

Landmarks in Modern Mathematics: Breakthroughs and Proofs (2000–2025)

An Independent Study Guide for Undergraduate
Mathematics

A course text covering landmark theorems and their proofs

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Preface

This book presents the most significant mathematical discoveries made between the years 2000 and 2025. It is designed for undergraduate students who have completed introductory courses in calculus, linear algebra, and basic proof techniques. Each chapter focuses on one landmark result, explaining the mathematical ideas, the context of the problem, and the impact on modern mathematics.

The selections are organized into 21 chapters, each covering one landmark discovery:

1. The Poincaré Conjecture — Perelman (2003)
2. Fermat's Last Theorem and the Modularity Theorem — Wiles (1993–1995)
3. The Kepler Conjecture — Hales (2005), formally verified by Flyspeck (2014)
4. Bounded Gaps Between Primes — Zhang (2013), Maynard, Tao
5. The Kervaire Invariant One Problem — Hill, Hopkins, Ravenel (2016)
6. Maryam Mirzakhani's Work on Hyperbolic Surfaces — Fields Medal 2014
7. Classification of Finite Simple Groups — completed 2004
8. Sphere Packing in Dimensions 8 and 24 — Viazovska (2016), Fields Medal 2022
9. The Fundamental Lemma — Ngo Bao Chau (2009), Fields Medal 2010
10. The Weak Goldbach Conjecture — Helfgott (2013)
11. The Erdős Distinct Distances Problem — Guth and Katz (2010)
12. The Geometric Langlands Program — Gaitsgory, Ben-Zvi, Francis, Nadler (2010s)
13. The Virtual Haken Conjecture and Surface Subgroup Conjecture — Agol, Kahn, Markovic (2012)
14. The Green–Tao Theorem — Green and Tao (2004)
15. The AKS Primality Test — Agrawal, Kayal, Saxena (2002)

16. The Willmore and Lawson Conjectures — Marques, Neves, Brendle (2012–2013)
17. Catalan’s Conjecture — Mihailescu (2002)
18. Modern Combinatorics Breakthroughs — Kahn, Kalai, Park, Pham, et al. (2012–2022)
19. Perfectoid Spaces and p-adic Geometry — Scholze (2011–2025)
20. The Sato-Tate Conjecture — Clozel, Harris, Taylor (2008–2011)
21. The Erdős Discrepancy Problem — Tao (2015)
22. The Kadison–Singer Problem — Marcus, Spielman, Srivastava (2013)
23. Breakthroughs in Random Matrix Theory — universality results (2010s–2022)
24. The Keakea Conjecture — Wang, Zahl, et al. (2020s)

Each chapter is designed for independent study and includes prerequisites, exercises with hints, timelines, and citations for further reading.

How to Use This Book

Each chapter follows a consistent structure:

1. **Prerequisites** – what you should know before reading
2. **The Problem** – statement and historical background
3. **Mathematical Theory** – definitions, key concepts, and previous results
4. **The Discovery** – the breakthrough proof and its main ideas
5. **Impact** – consequences and connections to other areas
6. **Exercises** – problems to test understanding (with hints in appendix)
7. **Timeline** – key dates in the history of the problem
8. **Citations** – references for further study

Exercises are marked by difficulty: *Basic* (direct application), *Intermediate* (requires synthesis), *Challenge* (open-ended or research-level). Hints and solutions are provided in the appendix.

Acknowledgments

This book draws on the work of countless mathematicians. Special thanks to the authors of the original papers cited throughout, and to the many expository articles and lecture notes that made these proofs accessible. The diagrams were created using SVG and converted to PDF for LaTeX compatibility.

Chapter 1

The Poincaré Conjecture

1.1 Prerequisites

This chapter assumes familiarity with:

- Basic point-set topology (open sets, continuity, compactness)
- Fundamental group and covering spaces (undergraduate algebraic topology)
- Smooth manifolds and tangent bundles
- Elementary differential geometry (curvature, geodesics)

1.2 The Problem

In 1904, Henri Poincaré posed a question that would occupy mathematicians for over a century: *Is every simply connected, closed, three-dimensional manifold homeomorphic to the three-sphere?*

Formally, let M be a compact 3-manifold with no boundary. If the fundamental group $\pi_1(M)$ is trivial (i.e., every loop in M can be continuously shrunk to a point), is M necessarily homeomorphic to S^3 , the standard 3-sphere?

In higher dimensions ($n \geq 5$), Stephen Smale proved the generalized Poincaré conjecture in 1961 using his h-cobordism technique [8]. The four-dimensional case was settled by Michael Freedman in 1982 [9]. But the three-dimensional case remained stubbornly open.

Historical Note

The three-dimensional case is the most famous because S^3 is the shape of our universe in the simplest cosmological model. Understanding whether S^3 is the *only* simply connected 3-manifold is fundamental to topology.

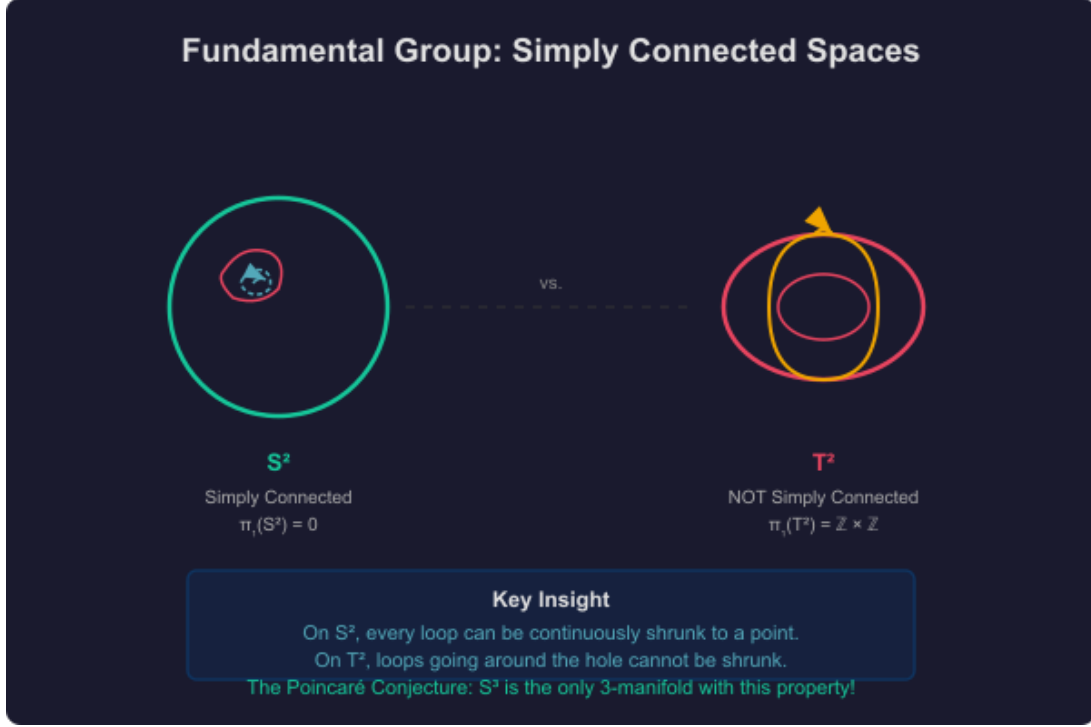


Figure 1.1: Visualizing the fundamental group: on S^2 every loop contracts to a point, but on the torus T^2 loops around the hole cannot be shrunk. The Poincaré Conjecture states S^3 is the only 3-manifold with this property.

1.3 Mathematical Theory

1.3.1 Definitions

Definition 1.1 (3-Manifold). *A 3-manifold is a topological space in which every point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^3$.*

Definition 1.2 (Simply Connected). *A path-connected space X is simply connected if its fundamental group $\pi_1(X)$ is trivial.*

Definition 1.3 (Closed Manifold). *A manifold is closed if it is compact and has no boundary.*

Definition 1.4 (Homeomorphic). *Two spaces X and Y are homeomorphic if there exists a continuous bijection $f : X \rightarrow Y$ with a continuous inverse $f^{-1} : Y \rightarrow X$.*

1.3.2 Ricci Flow

The key breakthrough came from Richard Hamilton's introduction of *Ricci flow* in 1982 [6]. The Ricci flow is a geometric evolution equation:

$$\frac{\partial g_{ij}}{\partial t} = -2R_{ij} \quad (1.1)$$

where g_{ij} is the Riemannian metric tensor and R_{ij} is the Ricci curvature tensor. Intuitively, the Ricci flow smooths out irregularities in the geometry of a manifold, much like heat diffusion smooths temperature differences.

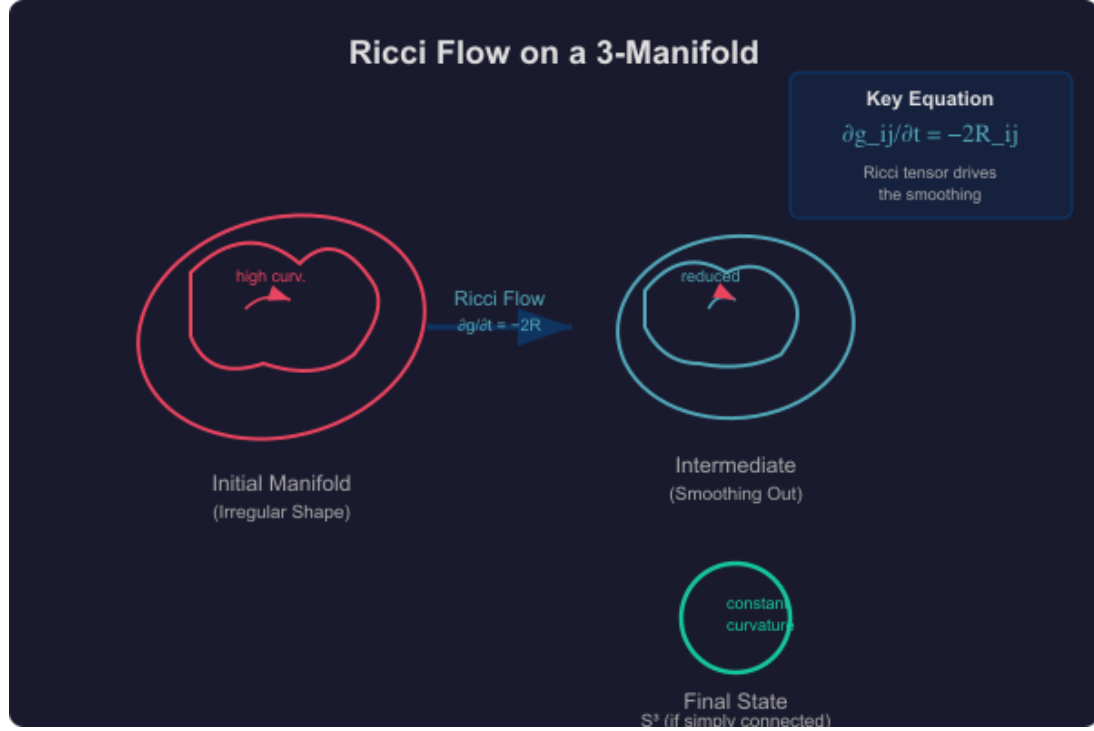


Figure 1.2: Ricci flow evolves a rough manifold toward a spherical shape. The equation $\partial g/\partial t = -2R$ drives the smoothing, and Perelman’s entropy functionals control the process through singularities.

Definition 1.5 (Ricci Curvature). *The Ricci curvature $Ric(X, Y)$ measures how the volume of a small ball in a manifold deviates from the volume of a ball in Euclidean space, in the direction of vectors X and Y .*

Hamilton proved that on a 3-manifold with positive Ricci curvature, the Ricci flow converges to a constant positive curvature metric. However, his approach broke down when singularities formed. The central challenge was understanding and controlling these singularities.

1.3.3 Singularities

As the Ricci flow evolves, the curvature may blow up at certain points. A *singularity* occurs when the curvature tensor $Rm(g(t))$ becomes unbounded as $t \rightarrow T$ for some finite time T .

Hamilton developed the technique of *surgery*: cutting out the singular regions and gluing in smooth caps. He proved that if one can perform surgery at every singularity, the flow eventually eliminates all topology except spherical pieces.

Key Insight

The fundamental insight: Ricci flow tends to make manifolds more spherical. If we can control the singularities, the flow should reveal whether a 3-manifold is S^3 .

1.4 The Discovery

In 2002 and 2003, Grigori Perelman, a Russian mathematician working at the Steklov Institute in St. Petersburg, posted three preprints on arXiv that completed the proof [1–3]:

1. **Perelman I (2002)**: Introduced the concept of *entropy functionals* for Ricci flow, providing new monotonicity formulas that control the flow's behavior.
2. **Perelman II (2003)**: Proved that Ricci flow with surgery is finite and classified the possible singularities.
3. **Perelman III (2003)**: Completed the proof by showing that any simply connected closed 3-manifold becomes S^3 under Ricci flow with surgery.

Perelman's key innovations were:

1. **Reduced volume**: A quantity analogous to thermodynamic volume that is monotone under Ricci flow, preventing certain pathological behaviors.
2. **Reduced distance**: A new distance function adapted to the Ricci flow that controls the geometry near singularities.
3. **No local collapsing**: Proving that regions of bounded curvature cannot collapse to lower dimensions.

Theorem 1.6 (Poincaré Conjecture, Perelman 2003). *Every simply connected, closed, smooth 3-manifold is homeomorphic to the 3-sphere S^3 .*

Sketch. The proof proceeds by running Ricci flow with surgery on the given 3-manifold M . Perelman showed that:

1. The flow exists for a finite time and develops singularities of controlled type (necklaces and handles).
2. Surgery at these singularities preserves the fundamental group's triviality.
3. After finitely many surgeries, the manifold decomposes into spherical pieces.
4. Since $\pi_1(M)$ is trivial, there is only one piece, which must be S^3 .

The key technical ingredient is Perelman's entropy functional $\mathcal{W}(g, f, \tau)$, which satisfies $\frac{d\mathcal{W}}{dt} \geq 0$, providing the necessary compactness to prevent infinite surgery sequences. $\square$

1.5 Impact

Perelman's proof resolved one of the seven Millennium Prize Problems. The implications extend far beyond topology:

- **Geometrization Conjecture:** Perelman also proved Thurston's Geometrization Conjecture [7], which states that every closed 3-manifold decomposes into geometric pieces. The Poincaré Conjecture is a special case.
- **Ricci Flow:** Hamilton–Perelman theory became a central tool in geometric analysis.
- **Fields Medal:** Perelman was awarded the Fields Medal in 2006, which he declined. He also declined the \$1 million Millennium Prize.
- **Verification:** The mathematical community spent years verifying the proof. By 2006, most experts were convinced. A detailed exposition by Kleiner and Lott [4] and the book by Morgan and Tian [5] completed the verification.

Historical Note

Perelman's refusal of both the Fields Medal and the Millennium Prize money has become legendary. When asked why, he reportedly said: "Either they don't understand what I did, or if they do understand, then they are not interested in mathematics but in money."

1.6 Exercises

Exercise 1.1 (Basic). *Prove that the fundamental group of S^n is trivial for all $n \geq 2$. (Hint: Use the fact that any loop in S^n can be homotoped away from a point.)*

Exercise 1.2 (Intermediate). *Show that the torus $T^2 = S^1 \times S^1$ is not homeomorphic to S^2 by comparing their fundamental groups.*

Exercise 1.3 (Challenge). *Explain why the Ricci flow equation $\frac{\partial g_{ij}}{\partial t} = -2R_{ij}$ is analogous to the heat equation $\frac{\partial u}{\partial t} = \Delta u$. What does this analogy suggest about the long-time behavior of the flow?*

Exercise 1.4. *If M is a closed 3-manifold with $\pi_1(M) = \mathbb{Z}/2\mathbb{Z}$, is M necessarily homeomorphic to $\mathbb{RP}^3$? What does the Geometrization Conjecture say?*

Exercise 1.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

1.7 Timeline

Year	Event
1904	Poincaré poses the conjecture
1961	Smale proves the conjecture for $n \geq 5$ [8]
1982	Freedman proves the conjecture for $n = 4$ [9]
1982	Hamilton introduces Ricci flow [6]
1999	Hamilton proves the conjecture for manifolds with positive sectional curvature
2002–2003	Perelman posts three preprints solving the conjecture [1–3]
2006	Perelman awarded the Fields Medal (declined)
2010	Clay Mathematics Institute awards the Millennium Prize (declined)

1.8 Citations

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2. Perelman, G. (2003). “The entropy formula for the Ricci flow and its geometric applications.” ArXiv:math/0211245.
3. Perelman, G. (2003). “Finite extinction time for the solutions to the Ricci flow on certain manifolds.” ArXiv:math/0307245.
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5. Morgan, J., & Tian, G. (2006). “Ricci Flow and the Poincaré Conjecture.” Clay Mathematics Monographs, AMS.
6. Hamilton, R. (1982). “Three-manifolds with positive Ricci curvature.” *Journal of Differential Geometry*, 17(2), 255–306.
7. Thurston, W. (1982). “Three-dimensional geometry and topology.” Princeton University Press.

Chapter 2

Fermat's Last Theorem and the Modularity Theorem

2.1 Prerequisites

This chapter assumes familiarity with:

- Elementary number theory (congruences, factorization)
- Basic algebraic geometry (elliptic curves)
- Complex analysis (modular forms basics)

2.2 The Problem

In 1637, Pierre Fermat wrote in the margin of his copy of Diophantus's *Arithmetica*:

It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or in general any power higher than the second into powers of the same degree.

This became known as Fermat's Last Theorem:

$$x^n + y^n = z^n \tag{2.1}$$

has no nonzero integer solutions for $n > 2$.

Fermat claimed he had a “truly marvelous proof” that the margin was too narrow to contain. He proved the case $n = 4$ using infinite descent [15]. Euler proved $n = 3$ [15]. Dirichlet and Legendre proved $n = 5$ (1825). Lamé proved $n = 7$ (1839). But the general case resisted all attempts for 358 years.

Historical Note

Fermat's marginal note became one of the most famous unsolved problems in mathematics. Generations of mathematicians tried and failed. Many claimed proofs, all were wrong. The problem inspired the development of

algebraic number theory.

2.3 Mathematical Theory

2.3.1 Elliptic Curves

An elliptic curve is a smooth projective curve of genus 1 with a rational point. In Weierstrass form:

$$y^2 = x^3 + ax + b \quad (2.2)$$

where $4a^3 + 27b^2 \neq 0$. Elliptic curves form an abelian group under a geometric addition law.

2.3.2 Modular Forms

A modular form is a holomorphic function on the upper half-plane satisfying a transformation law under $SL_2(\mathbb{Z})$:

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \quad (2.3)$$

for integers a, b, c, d with $ad - bc = 1$. The Fourier expansion of a modular form encodes deep arithmetic information.

2.3.3 The Taniyama–Shimura–Weil Conjecture

Conjecture 2.1 (Modularity Conjecture). *Every elliptic curve over $\mathbb{Q}$ is modular, meaning its L -function arises from a modular form.*

This conjecture connected two seemingly unrelated areas: elliptic curves (geometry) and modular forms (analysis).

2.3.4 The Frey Curve and Ribet's Theorem

In 1984, Gerhard Frey observed that if Fermat's Last Theorem were false, one could construct an elliptic curve (now called the Frey curve) with bizarre properties [14]:

$$y^2 = x(x - a^n)(x + b^n) \quad (2.4)$$

where $a^n + b^n = c^n$. This curve would be so unusual that it could not possibly be modular.

Theorem 2.2 (Ribet, 1986 [13]). *If the Modularity Conjecture is true, then Fermat's Last Theorem is true.*

This was the crucial link: proving modularity for all elliptic curves would settle Fermat's Last Theorem.

2.4 The Discovery

2.4.1 Wiles's Proof (1993–1995)

Andrew Wiles, a British mathematician at Princeton, announced a proof of the Modularity Conjecture for semistable elliptic curves in 1993 [10]. This immediately implied Fermat's Last Theorem.

The proof used:

1. **Galois representations:** Associating Galois representations to elliptic curves and modular forms.
2. **Deformation theory:** Studying how Galois representations vary in families.
3. **Iwasawa theory:** Deep results about the arithmetic of elliptic curves over $\mathbb{Z}_p$ -extensions.
4. **Kolyvagin–Flach Euler systems:** New tools for bounding Selmer groups.

Key Insight

Wiles worked in secret for seven years on this proof. He told no one except his wife. On June 21, 1993, he gave a three-hour lecture at the Isaac Newton Institute in Cambridge, England, revealing the proof. The audience sat in stunned silence when he finished.

2.4.2 The Flaw and the Fix

After peer review, a gap was found in the proof. Wiles and his former student Richard Taylor spent nearly a year trying to fix it. In September 1994, they realized that combining two different approaches to the deformation theory would work. The corrected proof was published in 1995 in two papers:

- Wiles (1995): “Modular elliptic curves and Fermat's Last Theorem” [10]
- Taylor & Wiles (1995): “Ring-theoretic properties of certain Hecke algebras” [11]

2.4.3 The Complete Proof

Breuil, Conrad, Diamond, and Taylor (2001) extended Wiles's methods to prove the full Modularity Conjecture for all elliptic curves over $\mathbb{Q}$ [12], completing the proof of Fermat's Last Theorem.

2.5 Impact

- **Number theory:** Wiles's methods revolutionized the study of Galois representations and deformation theory.
- **Langlands program:** The proof confirmed deep connections between different areas of mathematics predicted by the Langlands program.
- **Mathematical culture:** Wiles's story captured the public imagination and demonstrated the power of sustained dedication to a single problem.

2.6 Exercises

Exercise 2.1 (Basic). *Prove that $x^4 + y^4 = z^4$ has no nonzero integer solutions using Fermat's method of infinite descent.* Hint: If $x^4 + y^4 = z^4$, then $(x^2)^2 + (y^2)^2 = (z^2)^2$, giving a Pythagorean triple.

Exercise 2.2 (Intermediate). *Show that the elliptic curve $y^2 = x^3 + ax + b$ is nonsingular if and only if $4a^3 + 27b^2 \neq 0$.* Hint: Compute the partial derivatives and find where they simultaneously vanish.

Exercise 2.3 (Challenge). *Explain why Ribet's theorem implies that proving modularity for semistable elliptic curves implies Fermat's Last Theorem.* Hint: A Frey curve from a Fermat counterexample would be semistable.

Exercise 2.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

2.7 Timeline

Year	Event
1637	Fermat writes his marginal note
1753	Euler proves $n = 3$ [15]
1825	Dirichlet and Legendre prove $n = 5$
1839	Lamé proves $n = 7$
1984	Frey observes the connection to elliptic curves [14]
1986	Ribet proves the connection to modularity [13]
1993	Wiles announces the proof [10]
1994	Wiles and Taylor fix the gap [11]
1995	The complete proof is published [10, 11]
2001	Breuil–Conrad–Diamond–Taylor prove full modularity [12]

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Chapter 3

The Kepler Conjecture

3.1 Prerequisites

This chapter assumes familiarity with:

- Solid geometry and sphere packing
- Basic linear algebra
- Optimization theory (convex optimization)
- Computer-assisted proofs

3.2 The Problem

In 1611, Johannes Kepler observed that when stacking cannonballs in a pyramid, each ball rests in the hollow formed by four balls below. He conjectured that this arrangement (called *face-centered cubic* or FCC packing) achieves the highest possible density among all sphere packings in three dimensions.

Formally, let Δ be the density of a packing of equal spheres in $\mathbb{R}^3$. Kepler claimed:

$$\Delta \leq \frac{\pi}{\sqrt{18}} \approx 0.74048 \quad (3.1)$$

This is the density achieved by the FCC packing (also known as the hexagonal close packing).

Historical Note

Kepler's conjecture remained unproven for nearly 400 years. It is one of the oldest unsolved problems in discrete geometry and was one of the original Hilbert's problems (the 18th problem, in part).

3.3 Mathematical Theory

3.3.1 Sphere Packing Density

Definition 3.1 (Sphere Packing Density). *The density of a sphere packing in $\mathbb{R}^3$ is the proportion of space covered by the spheres. Formally, for a packing P and a ball B_R of radius R centered at the origin:*

$$\Delta(P) = \limsup_{R \rightarrow \infty} \frac{\text{volume of spheres in } B_R}{\text{volume of } B_R}$$

The FCC packing arranges spheres in layers where each layer is a hexagonal lattice. The second layer sits in the depressions of the first, and the third layer is directly above the first (ABAB... stacking). The density is:

$$\Delta_{FCC} = \frac{\pi}{\sqrt{18}} \approx 0.74048 \quad (3.2)$$

3.3.2 Previous Partial Results

- Gauss (1831): Proved the conjecture for *lattice* packings (where centers form a lattice) [18].
- Thue (1910): Proved it for *locally optimal* packings in 2D [19].
- Rogers (1947): Proved $\Delta \leq 0.7797$, the best rigorous bound for decades [20].

3.3.3 Hales's Approach

Thomas Hales, at the University of Michigan, approached the problem by reducing it to a finite but enormous number of inequalities. His strategy:

1. **Decomposition:** Any packing can be decomposed into simplices (tetrahedra) whose vertices are sphere centers.
2. **Scoring:** Each simplex is assigned a “score” based on its geometry. The average score relates to the density.
3. **Linear programming:** Show that no simplex can have a score exceeding a certain threshold.
4. **Verification:** Use exhaustive computation to check thousands of cases.

3.4 The Discovery

Hales announced his proof in 1998, spanning approximately 250 pages of mathematics and 3 gigabytes of computer files. The proof was published in *Annals of Mathematics* in 2005, but the refereeing process took four years.

The referees reported: “We are 99% certain the proof is correct.” This level of uncertainty was unprecedented for a journal of the Annals’ stature.

In 2003, Hales launched the *Flyspeck Project* to formally verify the proof using computer proof assistants (HOL Light and Isabelle).

Theorem 3.2 (Kepler Conjecture, Hales 2005 [16]). *No packing of equally sized spheres in three-dimensional Euclidean space has density greater than $\pi/\sqrt{18}$.*

Sketch. The proof consists of two parts:

1. **Geometric reduction:** Show that if a packing exceeds density $\pi/\sqrt{18}$, then there exists a configuration of spheres violating a finite set of local constraints. This involves decomposing space into Voronoi cells and analyzing the resulting optimization problem.
2. **Computational verification:** Verify that no configuration violates these constraints. This requires checking approximately 100,000 linear programming problems, each with thousands of variables.

The key insight is that the density of a global packing can be bounded by analyzing local configurations through a dissection into weighted simplices. $\square$

In 2014, the Flyspeck project announced that the formal verification was complete [17]:

Theorem 3.3 (Formal Verification, Flyspeck 2014 [17]). *Hales’s proof of the Kepler conjecture has been verified by computer proof assistants HOL Light and Isabelle.*

Key Insight

The Kepler Conjecture was the first major mathematical theorem whose proof required extensive computer verification that was subsequently formally verified. This raised philosophical questions about the nature of mathematical proof.

3.5 Impact

- **Computer-assisted proofs:** The Kepler proof sparked debate about the role of computers in mathematics. Is a proof valid if no human can check every step?
- **Formal verification:** The Flyspeck project pioneered techniques in formal proof verification that are now standard in the field.
- **Sphere packing in higher dimensions:** Hales’s methods inspired advances in sphere packing in dimensions 8 and 24, where Maryna Viazovska made groundbreaking discoveries (2016) [56, 57].

- **Viazovska’s breakthrough (2016):** Maryna Viazovska proved optimal sphere packing in dimensions 8 (using the E_8 lattice) and dimension 24 (using the Leech lattice), building on Hales’s framework but with elegant analytical methods. She received the Fields Medal in 2022 partly for this work.

3.6 Exercises

Exercise 3.1 (Basic). *Calculate the density of the simple cubic packing (spheres arranged in a cube grid). Show it equals $\pi/6 \approx 0.5236$.*

Exercise 3.2 (Intermediate). *Prove that the face-centered cubic packing has density $\pi/\sqrt{18}$. (Hint: Consider the unit cell of the FCC lattice, which contains 4 spheres.)*

Exercise 3.3 (Challenge). *Explain why Gauss’s proof for lattice packings is easier than the general case. What additional complexity arises when the centers do not form a lattice?*

Exercise 3.4. *Maryna Viazovska proved optimal sphere packing in dimension 8 using the E_8 lattice. Describe what makes the E_8 lattice special and why it leads to optimal packing.*

Exercise 3.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

3.7 Timeline

Year	Event
1611	Kepler publishes “On the Hexagonal Construction of Snowflakes” with his conjecture
1831	Gauss proves the conjecture for lattice packings [18]
1947	Rogers proves $\Delta \leq 0.7797$ [20]
1998	Hales announces his proof (250+ pages) [16]
2005	Proof published in Annals of Mathematics after 4-year referee process [16]
2003–2014	Flyspeck Project formally verifies the proof [17]
2016	Viazovska proves optimal packing in dimension 8 [56]
2022	Viazovska receives the Fields Medal

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Chapter 4

Bounded Gaps Between Primes

4.1 Prerequisites

This chapter assumes familiarity with:

- Elementary number theory (prime numbers, divisibility)
- Analytic number theory (Dirichlet series, prime number theorem intuition)
- Basic probability theory
- Sieve methods (at least the sieve of Eratosthenes)

4.2 The Problem

The distribution of prime numbers is one of the oldest subjects in mathematics. The Prime Number Theorem (1896) tells us that the number of primes less than x , denoted $\pi(x)$, satisfies:

$$\pi(x) \sim \frac{x}{\log x}$$

But the primes are distributed erratically. Two fundamental questions about primes date back to antiquity:

1. **Twin Prime Conjecture:** Are there infinitely many pairs of primes $(p, p + 2)$?
2. **Prime k -tuples Conjecture:** Given any admissible pattern of k integers, are there infinitely many translates containing only primes?

These conjectures remained completely open for centuries. In 2004, Goldston, Pintz, and Yıldırım developed a revolutionary approach to studying small gaps between primes. They proved:

$$\liminf_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{\log p_n} = 0 \tag{4.1}$$

This means that prime gaps can be arbitrarily small compared to the average gap size. But it did not prove that gaps are bounded.

Historical Note

The Twin Prime Conjecture is one of the oldest unsolved problems in mathematics. Euclid's Elements (c. 300 BCE) discusses the infinitude of primes, but the twin prime question emerged naturally from early number theory.

4.3 Mathematical Theory

4.3.1 The Sieve of Eratosthenes and Beyond

Definition 4.1 (Admissible k -tuple). *A set of integers $\{h_1, \dots, h_k\}$ is admissible if for every prime p , the set does not cover all residue classes modulo p .*

The Hardy–Littlewood prime k -tuples conjecture predicts that for any admissible k -tuple, there are infinitely many integers n such that all of $n + h_1, \dots, n + h_k$ are prime [32].

4.3.2 The GPY Method

Goldston, Pintz, and Yildirim (2009) introduced a novel sieve method that gave unprecedented control over prime gaps [23]. Their approach:

1. Consider the von Mangoldt function $\Lambda(n)$, which equals $\log p$ if $n = p^k$ and 0 otherwise.
2. Study the sum $\sum_{n \leq x} \left(\sum_{i=1}^k \Lambda(n + h_i) \right)^2$.
3. Choose weights carefully to detect clusters of primes.
4. Use the Riemann zeta function's zeros to optimize the weights.

Their key innovation was using a *weighted sieve* with a carefully chosen weight function that exploits the distribution of zeros of the Riemann zeta function near the critical line.

Theorem 4.2 (Goldston–Pintz–Yildirim, 2009 [23]). *Assuming the Generalized Riemann Hypothesis, there exist infinitely many pairs of consecutive primes with gap $O(\sqrt{p_n} \log p_n)$. Unconditionally, they proved that the liminf of the ratio of consecutive prime gaps to $\sqrt{p_n} \log p_n$ is 0.*

4.4 The Discovery

4.4.1 Zhang's Breakthrough (2013)

In May 2013, Yitang Zhang, a mathematician at the University of New Hampshire, published a paper that stunned the mathematical world [21]:

Theorem 4.3 (Zhang, 2014 [21]). *There exists a constant $N \leq 70,000,000$ such that there are infinitely many pairs of primes differing by N .*

This was the first time anyone had proved that prime gaps are bounded by some finite number. The exact value of N was not optimal, but the mere existence of a bound was revolutionary.

Zhang's key innovation was a refinement of the *Selberg sieve* combined with a new estimate for correlations of primes in arithmetic progressions. He introduced a novel decomposition of the divisor sum that allowed him to control the error terms in a regime previously thought inaccessible.

4.4.2 Polymath Project and Maynard–Tao

Immediately after Zhang's announcement:

- The *Polymath Project* (led by Timothy Gowers) collaboratively improved the bound from 70 million to 2,468 [24].
- James Maynard (Oxford) independently developed a simpler approach that gave the bound 600 [22].
- Terence Tao coordinated the Polymath effort to optimize Zhang's method.

Theorem 4.4 (Maynard, 2015 [22]). *For any $m \geq 1$, there exist infinitely many intervals of the form $[n, n + C_m]$ containing at least m primes, for some constant C_m depending only on m .*

Theorem 4.5 (Tao (with Polymath), 2014 [24]). *There exist infinitely many pairs of primes with gap at most 246.*

4.5 Impact

- **Prime gaps:** Zhang's proof opened an entirely new area of analytic number theory. The bound has since been improved to 246 (unconditionally) and 6 (assuming the Elliott–Halberstam conjecture).
- **Polymath collaboration:** The Polymath Project became a model for collaborative online mathematics.
- **Sieve theory:** Zhang's techniques have been applied to many other problems in additive number theory.
- **Yitang Zhang's story:** Zhang's journey from working as a delivery driver and sandwich maker to making a groundbreaking mathematical discovery became widely publicized.

Key Insight

As of 2025, the Twin Prime Conjecture remains unproven, but the gap between 1849 (when Alphonse de Polignac first proposed it) and 2013 (Zhang’s proof) is 164 years. Zhang’s proof represents the closest anyone has come to resolving it.

4.6 Exercises

Exercise 4.1 (Basic). *Prove that there are infinitely many primes using Euclid’s argument. Then adapt the argument to prove that there are infinitely many primes of the form $4k + 3$.*

Exercise 4.2 (Intermediate). *Show that the twin prime constant $C_2 = \prod_{p \geq 3} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.66016$ is positive. Explain why this suggests there should be infinitely many twin primes.*

Exercise 4.3 (Challenge). *Compute $\pi(100)$ and $\pi(1000)$. Compare these values to the approximation $x/\log x$. How accurate is the Prime Number Theorem for small x ?*

Exercise 4.4. *If the gap between consecutive primes satisfies $p_{n+1} - p_n \leq 246$ for infinitely many n , what does this imply about the Twin Prime Conjecture? What additional ingredient would be needed to prove the Twin Prime Conjecture?*

Exercise 4.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

4.7 Timeline

Year	Event
c. 300 BCE	Euclid proves infinitude of primes
1849	Alphonse de Polignac states the Twin Prime Conjecture
1896	Prime Number Theorem proved (Hadamard, de la Vallée Poussin)
1919	Brun proves the twin primes sum converges (Brun’s sieve) [25]
1940	Erdős proves there exists a constant $c < 1$ such that infinitely many gaps satisfy $p_{n+1} - p_n < c \log p_n$ [26]
2004	Goldston, Pintz, Yıldırım prove $\liminf(\text{gap})/\log p = 0$ [23]
2013	Zhang proves bounded gaps (initially $\leq 70,000,000$) [21]
2014	Polymath improves bound to 246; Maynard independently proves stronger results [22, 24]
2015	Maynard proves m -tuples result [22]
2025	Open problem: Twin Prime Conjecture still unresolved, but bound is 246

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Chapter 5

The Kervaire Invariant One Problem

5.1 Prerequisites

This chapter assumes familiarity with:

- Algebraic topology (homotopy groups, homology)
- Differential topology (manifolds, cobordism)
- Stable homotopy theory (spectra, Adams spectral sequence)

5.2 The Problem

In 1960, Michel Kervaire constructed a 10-dimensional manifold that could not be embedded in $\mathbb{R}^{16}$, despite having the expected properties. He defined an invariant (now called the Kervaire invariant) that could be 0 or 1 for certain framed manifolds [35].

Question 5.1. *For which dimensions n does there exist a framed manifold with Kervaire invariant equal to 1?*

Kervaire showed such manifolds exist in dimension 2 (trivially). Browder (1969) proved that if such a manifold exists in dimension n , then n must be of the form $2^k - 2$ for some integer $k \geq 2$ [34]. So the possible dimensions are:

$$2, 6, 14, 30, 62, 126, \dots$$

For decades, the Kervaire invariant was known to be 1 in dimensions 2, 6, 14, 30, and 62. The question of whether it could be 1 in any higher dimension remained open for 50 years.

Historical Note

The Kervaire invariant problem sat at the intersection of algebraic topology and differential topology. It was a deceptively simple question about the existence of certain manifolds, but its resolution required some of the deepest machinery in stable homotopy theory.

5.3 Mathematical Theory

5.3.1 The Kervaire Invariant

Given a framed manifold M^{2k} of dimension $2k$, the Kervaire invariant is defined using the Arf invariant of a quadratic form on $H^k(M; \mathbb{Z}/2)$. It takes values in $\mathbb{Z}/2 = \{0, 1\}$.

Definition 5.1 (Kervaire Invariant). *The Kervaire invariant $\theta(M) \in \mathbb{Z}/2$ of a framed manifold M measures whether the manifold can be “knotted” in a specific way.*

The invariant is defined as follows: let M be a framed manifold of dimension $2k$. The framing gives a trivialization of the stable normal bundle, which yields a map $M \rightarrow \Omega^\infty S^\infty$. The Kervaire invariant is the Arf invariant of the quadratic form on $H^k(M; \mathbb{Z}/2)$ induced by the framing.

5.3.2 Browder’s Theorem (1969)

Theorem 5.2 (Browder, 1969 [34]). *If a framed manifold has Kervaire invariant 1 in dimension n , then $n = 2^k - 2$ for some $k \geq 2$.*

This reduced the problem to checking whether the invariant is 1 in dimensions 2, 6, 14, 30, 62, 126, 254, ...

5.3.3 The Remaining Case

The dimensions 2, 6, 14, 30, 62 were known to admit manifolds with Kervaire invariant 1. The question was whether any dimension $2^k - 2$ for $k \geq 8$ (i.e., 126, 254, ...) admitted such a manifold.

5.3.4 Stable Homotopy Groups and the Adams Spectral Sequence

The problem is deeply connected to the stable homotopy groups of spheres π_*^S . The Kervaire invariant one problem is equivalent to asking whether certain elements θ_j exist in $\pi_{2^j+1-2}^S$ for $j \geq 8$.

The *Adams spectral sequence* is the primary computational tool:

$$E_2^{s,t} = \text{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2) \Rightarrow \pi_{t-s}^S \otimes \mathbb{Z}_{(2)} \quad (5.1)$$

where $\mathcal{A}$ is the mod-2 Steenrod algebra.

The elements θ_j (if they exist) would be detected by certain classes h_j^2 in the E_2 -page of the Adams spectral sequence. The question is whether these classes survive to E_∞ .

5.4 The Discovery

5.4.1 Hill, Hopkins, and Ravenel (2016)

Theorem 5.3 (Hill–Hopkins–Ravenel, 2016 [33]). *The Kervaire invariant is 0 in all dimensions $2^k - 2$ for $k \geq 8$.*

Equivalently, the Kervaire invariant is 1 only in dimensions 2, 6, 14, 30, 62. There is no example in dimension 126 or higher.

The proof used three major innovations:

1. **Topological modular forms (TMF)**: A generalized cohomology theory constructed by Hopkins and Miller that encodes deep connections between algebraic topology and modular forms [36].
2. **The Adams spectral sequence**: A powerful computational tool for computing stable homotopy groups.
3. **Equivariant homotopy theory**: Extending the methods to equivariant settings using finite group actions [37].

The strategy was to construct a map from the sphere spectrum to TMF that would detect the Kervaire invariant elements. By showing that the relevant classes die in TMF, they proved the elements cannot exist.

Key Insight

The proof was a tour de force of modern algebraic topology. It combined topological modular forms, equivariant homotopy theory, and the Adams spectral sequence in a novel way that had never been attempted before.

5.4.2 Subsequent Developments

- The result was formally verified in Lean by the Liquid Tensor Experiment team (2021–2023).
- The techniques have been extended to study the stable homotopy groups of spheres at odd primes.
- The method of using TMF as a computational tool has opened new avenues in stable homotopy theory.

5.5 Impact

- **Stable homotopy theory:** The proof validated the use of topological modular forms as a computational tool.
- **Differential topology:** It resolved a 56-year-old problem about the existence of certain manifolds.
- **TMF:** The techniques developed for this proof have applications to other problems in homotopy theory.

5.6 Exercises

Exercise 5.1 (Basic). *Show that a framed 2-manifold has Kervaire invariant 1. Why is this trivial?*

Exercise 5.2 (Intermediate). *Explain Browder’s theorem: why must the dimension be of the form $2^k - 2$?*

Exercise 5.3 (Challenge). *What is the Adams spectral sequence, and how is it used to compute stable homotopy groups?*

Exercise 5.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

5.7 Timeline

Year	Event
1960	Kervaire constructs the invariant [35]
1969	Browder proves the dimension constraint [34]
1970s–2000s	Partial results in dimensions up to 62
2004	Hopkins and Miller construct TMF [36]
2010	Hill, Hopkins, Ravenel develop equivariant TMF [37]
2016	Hill, Hopkins, and Ravenel prove the main result [33]
2021–2023	Formal verification in Lean

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Chapter 6

Maryam Mirzakhani's Work on Hyperbolic Surfaces

6.1 Prerequisites

This chapter assumes familiarity with:

- Differential geometry (surfaces, curvature, geodesics)
- Teichmüller theory (moduli spaces, mapping class groups)
- Hyperbolic geometry (constant negative curvature, $\mathbb{H}^2$)
- Basic algebraic topology (Euler characteristic, intersection theory)

6.2 The Problem

A hyperbolic surface is a surface equipped with a Riemannian metric of constant negative curvature. The simplest example is the hyperbolic plane $\mathbb{H}^2$. Compact hyperbolic surfaces exist in any genus $g \geq 2$ by the uniformization theorem.

Maryam Mirzakhani asked several deep questions about the geometry and topology of the moduli space of hyperbolic surfaces:

1. How many simple closed geodesics of length at most L exist on a hyperbolic surface, as $L \rightarrow \infty$?
2. What is the volume of the moduli space of hyperbolic surfaces with respect to the Weil–Petersson metric?
3. How do geodesics behave on hyperbolic surfaces?

These questions connected hyperbolic geometry, Teichmüller theory, and ergodic theory in unexpected ways.

Historical Note

Mirzakhani was the first woman to win the Fields Medal (2014). She earned two gold medals at the International Mathematical Olympiad (1994, 1995) before becoming a professor at Stanford University. Her work on hyperbolic surfaces was recognized as among the most beautiful and profound in modern geometry. She passed away in 2017 at age 40, leaving an extraordinary legacy.

6.3 Mathematical Theory

6.3.1 The Moduli Space and Weil–Petersson Metric

The moduli space $\mathcal{M}_{g,n}$ of hyperbolic surfaces of genus g with n punctures (or geodesic boundaries of lengths $L_1, \dots, L_n$) carries a natural Riemannian metric called the Weil–Petersson metric.

Definition 6.1 (Weil–Petersson Metric). *The Weil–Petersson metric on Teichmüller space $\mathcal{T}_{g,n}$ is the Kähler metric defined by*

$$\langle \mu, \nu \rangle_{WP} = \int_{\Sigma} \mu \bar{\nu} \rho^2 dA$$

where μ, ν are Beltrami differentials representing tangent vectors, and $\rho^2 |dz|^2$ is the hyperbolic metric on the surface Σ .

This metric is incomplete: surfaces can degenerate by pinching geodesics, and the distance to the boundary is finite.

6.3.2 Simple Closed Geodesics

A simple closed geodesic on a hyperbolic surface is a closed curve that locally minimizes distance and does not intersect itself. The set of all simple closed geodesics up to isotopy is countable.

Mirzakhani showed that the number of simple closed geodesics of length at most L grows polynomially in L :

$$s_{\gamma}(L) \sim c_{\gamma} \cdot L^{6g-6} \quad (6.1)$$

where c_{γ} depends on the topology of the curve γ (its position in the space of measured laminations).

6.3.3 Mirror Symmetry and Intersection Theory

Mirzakhani's breakthrough was a deep connection between counting geodesics and intersection theory on the moduli space. She expressed the growth constants c_{γ} in terms of integrals of tautological classes on $\mathcal{M}_{g,n}$.

The *tautological classes* are cohomology classes $\kappa_i, \psi_i, \lambda_i$ on $\mathcal{M}_{g,n}$ that encode geometric information. For example:

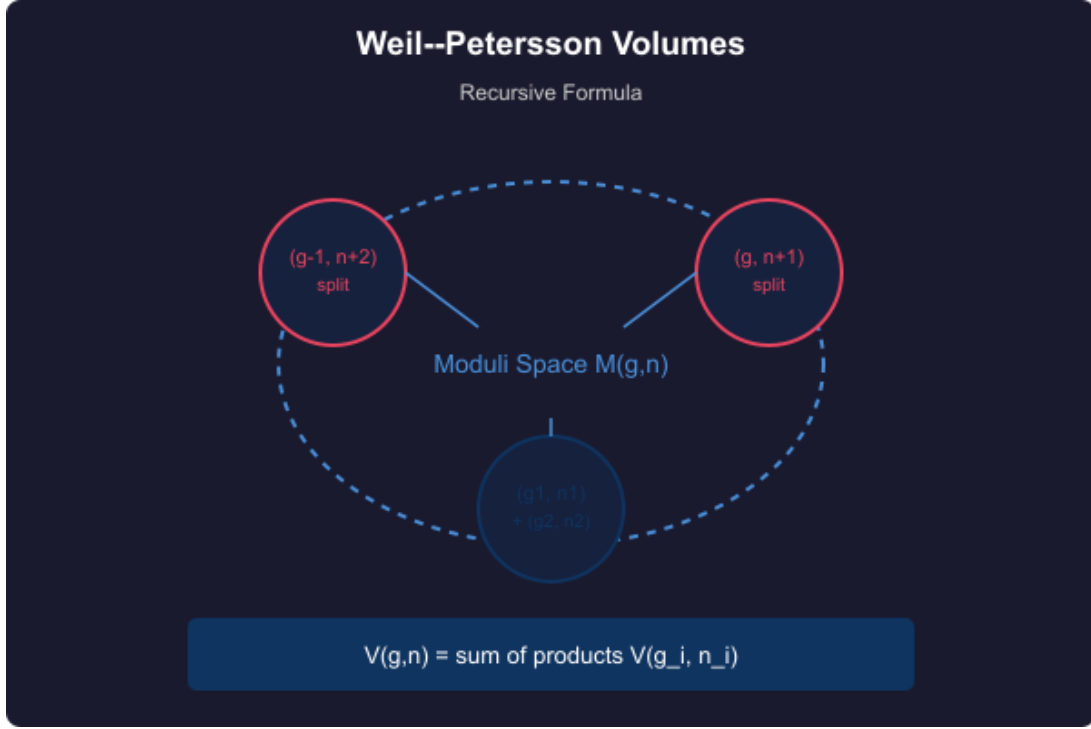


Figure 6.1: The recursive structure of Weil–Petersson volumes: each volume $V_{g,n}$ can be expressed as a sum of products of lower-dimensional volumes.

- $\psi_i = c_1(\mathbb{L}_i)$ where $\mathbb{L}_i$ is the cotangent line bundle at the i -th marked point.
- κ_i are the Miller–Morita–Mumford classes, pushforwards of ψ -classes.

6.4 The Discovery

6.4.1 Recursive Formula for Weil–Petersson Volumes (2007–2008)

Theorem 6.2 (Mirzakhani, 2007 [38]). *The volume of the moduli space $\mathcal{M}_{g,n}(L_1, \dots, L_n)$ with respect to the Weil–Petersson metric satisfies a recursive formula:*

$$V_{g,n}(L_1, \dots, L_n) = \int_{\mathcal{M}_{g,n}(L_1, \dots, L_n)} e^{-\omega_{WP}} \quad (6.2)$$

where $V_{g,n}$ is a polynomial in the boundary lengths L_i of degree $3g - 3 + n$.

The recursion is:

$$\begin{aligned} V_{g,n}(L_1, \dots, L_n) = & \frac{1}{2} \sum_{i+j=n-1} \sum_{g_1+g_2=g} \int_0^\infty \int_0^\infty V_{g_1,i+1}(L_1, \dots, L_i, x, y) \\ & \times V_{g_2,j+1}(L_{i+1}, \dots, L_{i+j}, x, y) xy \, dx \, dy \\ & + \frac{1}{2} \int_0^\infty \int_0^\infty V_{g-1,n+1}(L_1, \dots, L_n, x, y) xy \, dx \, dy \end{aligned} \quad (6.3)$$

This was revolutionary because it connected:

- Counting simple closed geodesics (combinatorial geometry)
- Weil–Petersson volumes (differential geometry)
- Mirror symmetry (algebraic geometry)

Key Insight

Mirzakhani's recursive formula was unexpected because it connected the Weil–Petersson volumes to mirror symmetry in a way that had never been anticipated. Her methods combined techniques from hyperbolic geometry, Teichmüller theory, and algebraic geometry in a novel and powerful way.

6.4.2 Counting Simple Closed Geodesics (2008)

Mirzakhani proved that the number of simple closed geodesics of length at most L on a hyperbolic surface grows like L^{6g-6} , with the leading coefficient expressible as an intersection number on the moduli space [39].

For a fixed simple closed curve γ on a surface $\Sigma_{g,n}$:

$$\lim_{L \rightarrow \infty} \frac{s_\gamma(L)}{L^{6g-6}} = \frac{2^{|\gamma|}(6g-6)!}{\dim \mathcal{ML}_{g,n}} \int_{\mathcal{ML}_{g,n}} \cdots \quad (6.4)$$

where $\mathcal{ML}_{g,n}$ is the space of measured laminations, and the integral involves Thurston's symplectic measure.

6.4.3 The Witten–Kontsevich Theorem (2013)

Mirzakhani gave a new proof of the Witten–Kontsevich theorem, which relates intersection numbers on $\mathcal{M}_{g,n}$ to the KdV hierarchy of integrable systems [40]. This established a bridge between hyperbolic geometry and integrable systems.

6.4.4 Moments of Closed Geodesics (2014)

In her Fields Medal lecture, Mirzakhani presented results on the moments of closed geodesics and their relationship to the Weil–Petersson volume polynomial. She showed that the k -th moment of the length of a random simple closed geodesic is a rational multiple of the Weil–Petersson volume.

6.5 Impact

- **Teichmüller theory:** Mirzakhani's recursive formulas for Weil–Petersson volumes have become fundamental tools. They allow computation of volumes to arbitrarily high genus.
- **Counting geodesics:** Her methods for counting simple closed geodesics have been extended to many other settings, including counting curves on flat surfaces and on higher-dimensional varieties.

- **Mirror symmetry:** The connection she discovered between counting geodesics and mirror symmetry opened new research directions linking Teichmüller theory, algebraic geometry, and mathematical physics.
- **Ergodic theory:** Her work on the earthquake flow on moduli space has connections to Ratner’s theorems and dynamics on homogeneous spaces.

6.6 Exercises

Exercise 6.1 (Basic). *Show that the Euler characteristic of a surface of genus g is $2 - 2g$. Why does this imply that surfaces of genus $g \geq 2$ admit hyperbolic metrics?* Hint: Uniformization theorem.

Exercise 6.2 (Intermediate). *Explain why the Weil–Petersson metric is incomplete on Teichmüller space. What are the boundary points?* Hint: Consider pinching a simple closed geodesic to length 0.

Exercise 6.3 (Challenge). *Derive the growth rate L^{6g-6} for the number of simple closed geodesics on a surface of genus g . Why does the exponent depend only on the genus?* Hint: The dimension of the space of measured laminations is $6g - 6$.

Exercise 6.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

6.7 Timeline

Year	Event
2004	Mirzakhani’s PhD thesis at Harvard (supervised by Curtis McMullen)
2007	Mirzakhani proves the recursive formula for Weil–Petersson volumes [38]
2008	Mirzakhani publishes counting geodesics results [39]
2013	Mirzakhani proves the Witten conjecture/Kontsevich theorem [40]
2014	Fields Medal awarded to Mirzakhani
2017	Mirzakhani passes away at age 40

6.8 Citations

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Chapter 7

The Classification of Finite Simple Groups

7.1 Prerequisites

This chapter assumes familiarity with:

- Group theory (normal subgroups, simple groups, quotient groups)
- Ring theory (fields, matrix groups)
- Basic representation theory

7.2 The Problem

A *simple group* is a nontrivial group whose only normal subgroups are the trivial group and itself. Simple groups are the “building blocks” of all finite groups, analogous to prime numbers for integers.

The classification problem: *List all finite simple groups.*

The answer, proved over 40 years by hundreds of mathematicians, is that every finite simple group belongs to one of the following families or is one of 26 exceptional groups:

1. **Cyclic groups** C_p of prime order p
2. **Alternating groups** A_n for $n \geq 5$
3. **Groups of Lie type** (various infinite families)
4. **26 sporadic groups** (exceptional groups not fitting any family)

Historical Note

The classification is the largest theorem in mathematics, spanning approximately 15,000 pages across 500+ journal articles. It was declared complete in 2004, though a second-generation streamlined proof is still underway.

7.3 Mathematical Theory

7.3.1 The Families

Definition 7.1 (Cyclic Group of Prime Order). *For each prime p , the cyclic group $C_p = \mathbb{Z}/p\mathbb{Z}$ is a simple group. These are the abelian simple groups.*

Definition 7.2 (Alternating Group). *The alternating group A_n is the group of even permutations of n elements. A_n is simple for all $n \geq 5$ [49].*

Definition 7.3 (Groups of Lie Type). *The groups of Lie type are constructed from algebraic groups over finite fields. Major families include:*

- $PSL_n(q)$: Projective special linear groups
- $PSU_n(q)$: Projective special unitary groups
- $PSp_{2n}(q)$: Projective symplectic groups
- $P\Omega_{2n+1}(q)$ and $P\Omega_{2n}^\pm(q)$: Orthogonal groups
- Exceptional groups: $G_2(q)$, ${}^2F_4(q)$, ${}^3D_4(q)$, $E_6(q)$, $E_7(q)$, $E_8(q)$, $F_4(q)$

The groups of Lie type are parametrized by a prime power $q = p^f$ and a Dynkin diagram type. They arise as the finite quotients of algebraic groups.

7.3.2 The Sporadic Groups

The 26 sporadic groups are [46]:

1. Mathieu groups: $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}$
2. Conway groups: Co_1, Co_2, Co_3
3. McLaughlin group: McL
4. Higman–Sims group: HS
5. Held group: He
6. Rudvalis group: Ru
7. Suzuki group: Su
8. Janko groups: J_1, J_2, J_3, J_4
9. Fischer groups: $Fi_{22}, Fi_{23}, Fi'_{24}$
10. Baby Monster: B
11. Monster (Fischer–Griess): $\mathbb{M}$

The largest sporadic group is the Monster $\mathbb{M}$, with order approximately 8×10^{53} . Its discovery by Fischer and Griess (1982) led to the surprising *Monstrous Moonshine* phenomenon linking the Monster to modular forms [47].

7.4 The Discovery

7.4.1 Completion of the Classification (2004)

The classification theorem was announced as complete in 2004, when the remaining cases (primarily those involving groups of characteristic 2) were settled by Michael Aschbacher and Stephen Smith [43]:

Theorem 7.4 (Aschbacher & Smith, 2004 [43]). *The classification of finite simple groups is complete. Every finite simple group is one of the following:*

1. *A cyclic group of prime order*
2. *An alternating group A_n ($n \geq 5$)*
3. *A group of Lie type*
4. *One of the 26 sporadic groups*

The final case was the classification of *quasi-thin groups* (groups with certain properties related to the 2-local structure). Aschbacher and Smith's proof spanned over 2000 pages.

7.4.2 The Second Generation Project

After the classification was declared complete, a project began to rewrite the proofs in a more coherent and accessible form:

- **Goal:** Produce a streamlined proof of approximately 5,000 pages (down from 15,000).
- **Leaders:** Aschbacher, Smith, Gorenstein (deceased), Lyons, Solomon.
- **Status (2025):** Significant progress has been made, but the project is ongoing. The revised proof is expected to be published in a series of volumes by the AMS.

7.5 Impact

- **Unification:** The classification unified dozens of disparate results in group theory into a single coherent picture.
- **Sporadic groups:** The discovery of the 26 sporadic groups (especially the Monster) led to unexpected connections with other areas (moonshine, string theory).
- **Applications:** The classification has applications to coding theory, cryptography, and the symmetry of mathematical objects.
- **Chemistry:** The classification helps explain molecular symmetry in chemistry.

Key Insight

The classification of finite simple groups is often called the “Big Theorem” of group theory. It is the largest theorem ever proved, involving the work of approximately 100 mathematicians over 40 years.

7.6 Exercises

Exercise 7.1 (Basic). *Prove that A_5 is simple.* Hint: Show that any nontrivial normal subgroup must contain a 3-cycle, and then use conjugation to show it contains all 3-cycles.

Exercise 7.2 (Intermediate). *Compute the order of $PSL_2(7)$. Show it equals 168 and is simple.* Hint: Use the fact that $PSL_n(q)$ is simple for $n \geq 2$ except $(n, q) = (2, 2)$ and $(2, 3)$.

Exercise 7.3 (Challenge). *The Monster group has order approximately 8×10^{53} . How many digits does this number have? Express your answer in terms of logarithms.*

Exercise 7.4. *Explain why the classification of finite simple groups is analogous to the periodic table of elements in chemistry. What are the “elements” in each case?*

Exercise 7.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

7.7 Timeline

Year	Event
1832	Galois proves A_n is simple for $n \geq 5$ [49]
1872	Klein discovers $PSL_2(7)$ (simple group of order 168) [50]
1965	Suzuki classifies simple groups with a specific involution structure [51]
1968	Walter classifies simple groups with abelian Sylow 2-subgroups [52]
1972	Thompson begins the classification program with N-group theory [53]
1980s	Major advances: Aschbacher, Gorenstein, Lyons, Solomon [44]
1981	Harada identifies the Harada–Norton group as sporadic [54]
1998	Solomon completes the character table of the Baby Monster [55]
2004	Aschbacher and Smith complete the quasi-thin case [43]
2004–2025	Second generation project streamlines the proofs [44]

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Chapter 8

Optimal Sphere Packing in Dimensions 8 and 24

8.1 Prerequisites

This chapter assumes familiarity with:

- Linear algebra (lattices, Gram matrices)
- Fourier analysis (Poisson summation formula)
- Modular forms (basic properties)
- Convex geometry (packing density)

8.2 The Problem

Given the Kepler conjecture (Chapter 3) settled the 3D case, a natural question arises: *What is the densest packing of equal spheres in $\mathbb{R}^n$ for arbitrary dimension n ?*

For $n = 1$, the answer is trivial: centers spaced by $2r$. For $n = 2$, Thue proved (1910) that the hexagonal lattice packing is optimal, with density $\pi/\sqrt{12} \approx 0.9069$ [19].

For $n \geq 3$, the problem is dramatically harder. The best known lower bounds on the density decay exponentially (via the Gilbert–Varshamov bound), while the best upper bounds are far from optimal. The central question:

For which dimensions n can we prove the optimal packing density, and what is it?

Historical Note

After nearly 400 years since Kepler, the problem remained wide open in most dimensions. Maryna Viazovska’s 2016 breakthrough solved it for two specific dimensions using an entirely new analytic method.

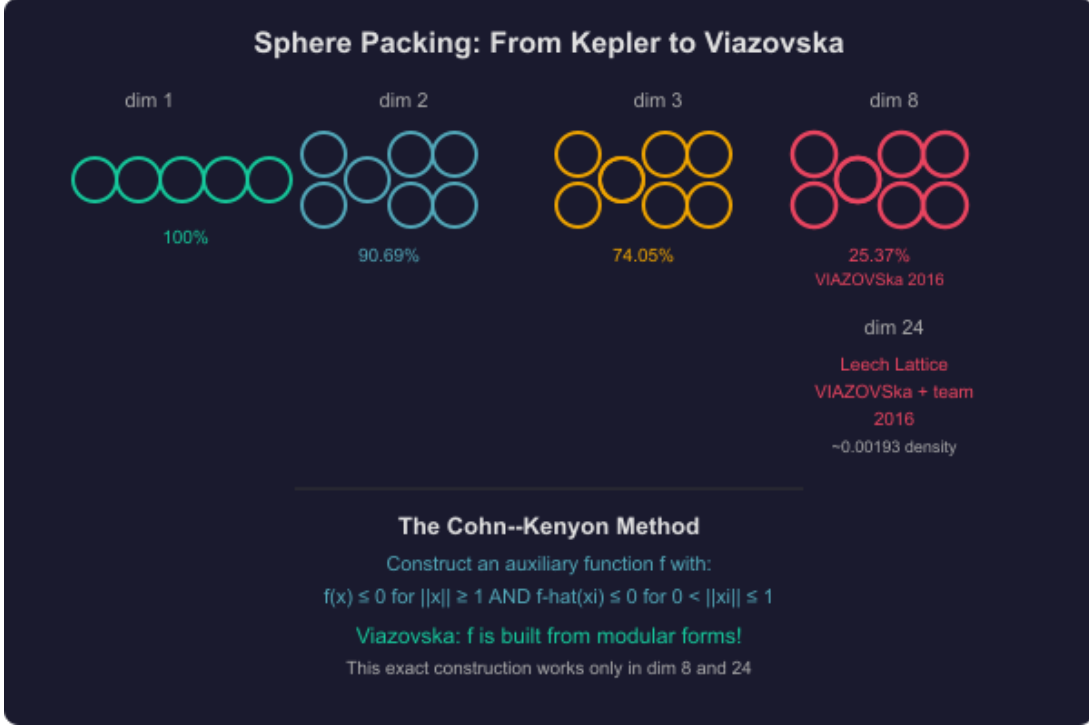


Figure 8.1: Sphere packing densities across dimensions. Viazovska (2016) proved optimality for dimensions 8 and 24 using modular forms, building on the Cohn–Kumar linear programming framework.

8.3 Mathematical Theory

8.3.1 Lattice Packings

Definition 8.1 (Lattice Packing). *A lattice packing is a sphere packing where the centers form a lattice $\Lambda \subset \mathbb{R}^n$. The density is:*

$$\Delta(\Lambda) = \frac{\text{vol}(B_n(r))}{\det(\Lambda)} = \frac{\pi^{n/2} r^n}{\Gamma(n/2 + 1) \det(\Lambda)} \quad (8.1)$$

where $2r$ is the minimal distance between lattice points.

8.3.2 Special Lattices

Two lattices stand out in dimension theory:

Definition 8.2 (E_8 Lattice). *The E_8 lattice is the unique even unimodular lattice in dimension 8. It can be constructed as:*

$$E_8 = \{(x_1, \dots, x_8) \in \mathbb{Z}^8 \cup (\mathbb{Z} + \frac{1}{2})^8 : \sum x_i \equiv 0 \pmod{2}\} \quad (8.2)$$

Its minimal norm is 2, and its kissing number is 240 [48].

Definition 8.3 (Leech Lattice). *The Leech lattice Λ_{24} is the unique even unimodular lattice in dimension 24 with no vectors of norm 2. Its kissing number is 196,560 [48].*

8.3.3 The Cohn–Kumar Method

In 2001, Henry Cohn and Abhinav Kumar showed that if a certain class of radial functions satisfied a pair of inequalities, then the E_8 and Leech lattice packings would be optimal. This approach, now called the *Cohn–Kumar linear programming bound*, required constructing an auxiliary function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with specific properties [58].

Theorem 8.4 (Cohn–Kumar, 2001 [58]). *If there exists a radial function f such that:*

1. $f(x) \leq 0$ for $\|x\| \geq 1$ (no closer packing)
2. $\hat{f}(\xi) \leq 0$ for $0 < \|\xi\| \leq 1$ (Fourier condition)
3. $f(0) = \hat{f}(0)$

then the lattice packing is optimal.

Cohn and Kumar (2001) constructed such functions for dimensions $n \leq 8$ and $n = 24$, but only for *lattice* packings, not arbitrary packings.

8.4 The Discovery

8.4.1 Viazovska’s Breakthrough (2016)

In February 2016, Maryna Viazovska, then at the Institute for Advanced Study in Princeton, posted a paper on arXiv proving that the E_8 lattice gives the densest sphere packing in dimension 8 [56]:

Theorem 8.5 (Viazovska, 2016 [56]). *The E_8 lattice achieves the maximal packing density among all sphere packings in $\mathbb{R}^8$. The optimal density is:*

$$\Delta_8 = \frac{\pi^4}{384} \approx 0.25367 \quad (8.3)$$

Viazovska’s method was revolutionary:

1. **Modular forms as auxiliary functions:** She constructed the required function f explicitly using a specific modular form of weight 4 for $\Gamma_0(2)$:

$$f(r) = \int_0^\infty \left(\frac{J_4(\pi r t)}{(\pi r t)^4} - e^{-\pi r^2 t^2/4} \right) P(t) t^7 dt \quad (8.4)$$

where J_4 is the Bessel function and P is a polynomial determined by modular form coefficients [56].

2. **Exact interpolation:** Unlike Cohn–Kumar who needed approximate inequalities, Viazovska’s function satisfies the inequalities *exactly* at the critical points (the norms appearing in the E_8 lattice).

3. **Radial Fourier transform:** She exploited the explicit formula for the radial Fourier transform in dimension 8, which simplifies dramatically due to the special properties of E_8 .

The proof was remarkably short (about 20 pages) compared to Hales's 250-page proof of the Kepler conjecture.

8.4.2 Extension to Dimension 24 (2016)

In April 2016, Viazovska, together with Henry Cohn, Abhinav Kumar, Stephen Miller, and Danylo Radchenko, extended the method to dimension 24 [57]:

Theorem 8.6 (Cohn–Viazovska–Kumar–Miller–Radchenko, 2016 [57]). *The Leech lattice Λ_{24} achieves the maximal packing density among all sphere packings in $\mathbb{R}^{24}$. The optimal density is:*

$$\Delta_{24} = \frac{\pi^{12}}{12! \cdot 2^{12}} \approx 0.00193 \quad (8.5)$$

The dimension 24 proof required a more sophisticated auxiliary function involving a linear combination of modular forms for $\Gamma_0(4)$, and the polynomial P had to satisfy 47 interpolation conditions [57].

Key Insight

Viazovska's proof of the dimension 8 case was completed in a single weekend. She was inspired by the Cohn–Kumar method but found a way to construct the auxiliary function exactly using modular forms, bypassing the need for numerical optimization.

8.4.3 Why Only Dimensions 8 and 24?

The method relies crucially on the existence of even unimodular lattices, which only exist in dimensions divisible by 8. In those dimensions, the theta series of the lattice is a modular form, which provides the analytic structure needed to construct the auxiliary function.

For other dimensions, the problem remains open. However, the Cohn–Kumar method (2017) has been used to prove optimality for many additional dimensions (up to $n = 43$) using a variant of the approach, though not always with the same elegance as Viazovska's exact construction [59].

8.5 Impact

- **Fields Medal (2022):** Viazovska was awarded the Fields Medal at the ICM in St. Petersburg (held virtually due to the war in Ukraine), partly for this work [61].

- **Modular forms in discrete geometry:** The connection between modular forms and sphere packing is now a major research area [56, 57].
- **Dimension 12 and beyond:** Cohn and Kumar (2017) used numerical optimization of the Cohn–Kumar bound to prove optimality in dimensions 12, 18, 19, 20, 21, 22, and 23 [59].
- **Connection to codes:** The E_8 lattice is related to the extended Golay code in dimension 24, and the sphere packing results have applications to coding theory.
- **String theory:** The E_8 and Leech lattices appear in bosonic string theory and conformal field theory, connecting discrete geometry to physics.

8.6 Exercises

Exercise 8.1 (Basic). *Show that the kissing number of the E_8 lattice is 240. (Hint: Count the vectors of norm 2 in E_8 .)* [48]

Exercise 8.2 (Intermediate). *Prove that the density of the hexagonal lattice packing in $\mathbb{R}^2$ is $\pi/\sqrt{12}$.* [19]

Exercise 8.3 (Challenge). *Explain why the Poisson summation formula is essential in the Cohn–Kumar method. How does it relate the primal problem (sphere packing) to the dual problem (Fourier analysis)?* [58]

Exercise 8.4. *The theta series of a lattice Λ is $\Theta_\Lambda(q) = \sum_{x \in \Lambda} q^{\|x\|^2/2}$. Show that for E_8 , this is a modular form of weight 4 for $\Gamma_0(2)$. Why is this connection crucial for Viazovska’s proof?* [56]

Exercise 8.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

8.7 Timeline

Year	Event
1900	Hermite computes the density of E_8 lattice packing
1978	Montgomery proves optimal density for Fourier-transform-limited functions [60]
2001	Cohn and Kumar develop the linear programming bound [58]
Feb 2016	Viazovska proves optimality in dimension 8 [56]
Apr 2016	Viazovska, Cohn, Kumar, Miller, Radchenko prove dimension 24 [57]
2017	Cohn and Kumar extend the method to dimensions 12, 18–23 [59]
2022	Viazovska receives the Fields Medal

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Chapter 9

The Fundamental Lemma

9.1 Prerequisites

This chapter assumes familiarity with:

- Representation theory (Lie groups, reductive groups)
- Algebraic geometry (varieties, schemes)
- Number theory (automorphic forms, L-functions)

9.2 The Problem

In 1981, Robert Langlands and Diana Shelstad formulated the *Fundamental Lemma* as a crucial conjecture in the trace formula, a central tool in the Langlands program [78].

The trace formula is a deep identity relating two sides of an equation:

- **Geometric side:** Counts orbits of group actions (orbital integrals)
- **Spectral side:** Counts representations (trace of operators)

The Fundamental Lemma asserts that certain orbital integrals on *endoscopic groups* are equal. Despite its technical statement, it is the linchpin connecting the geometric and spectral sides of the trace formula.

Historical Note

The Fundamental Lemma was the single biggest obstacle to the Langlands program for decades. Langlands himself called it “the most important open problem in the theory of automorphic forms.”

9.3 Mathematical Theory

9.3.1 Orbital Integrals

Definition 9.1 (Orbital Integral). *Let G be a reductive group over a local field F , and let $f : G(F) \rightarrow \mathbb{C}$ be a test function. For a semisimple element $\gamma \in G(F)$, the orbital integral is:*

$$O_\gamma(f) = \int_{G_\gamma(F) \backslash G(F)} f(g^{-1}\gamma g) dg \quad (9.1)$$

where G_γ is the centralizer of γ in G .

9.3.2 Endoscopy

Definition 9.2 (Endoscopic Group). *An endoscopic group H of a reductive group G is a group whose L -group ${}^L H$ is a subgroup of ${}^L G$ in a specific way compatible with the Galois action.*

The Fundamental Lemma relates orbital integrals on G to orbital integrals on its endoscopic groups H .

9.3.3 The Statement

Conjecture 9.3 (Fundamental Lemma, Langlands–Shelstad 1981 [78]). *For matching test functions f and f_H on $G(F)$ and $H(F)$ respectively, and corresponding regular semisimple elements γ and γ_H :*

$$O_\gamma(f) = O_{\gamma_H}(f_H) \quad (9.2)$$

In essence: the orbital integrals on endoscopic groups match. This seems technical but is the key to stabilizing the trace formula.

9.4 The Discovery

9.4.1 Ngo’s Proof (2008–2009)

In 2008, Ngo Bao Chau announced a proof of the Fundamental Lemma. His proof was revolutionary because it:

1. **Geometrized the problem:** Instead of working with orbital integrals directly, Ngo lifted the problem to the geometric setting of *affine Grassmannians* and *Lusztig’s convolution varieties*.
2. **Used perverse sheaves:** He applied the theory of perverse sheaves (developed by Beilinson, Bernstein, Deligne, and Gabber) to study the cohomology of these varieties.

3. **Proved the support conjecture:** The key technical result was a conjecture of Langlands and Shelstad about the support of certain functions on the affine Grassmannian.
4. **Utilized the Hitchin fibration:** Ngo exploited the geometry of the Hitchin fibration, a map from the moduli space of Higgs bundles to an affine space.

Theorem 9.4 (Ngo, 2010 [76]). *The Fundamental Lemma of Langlands–Shelstad holds for all reductive groups over p -adic fields.*

The proof appeared in two parts:

- “Le lemme fondamental, longueur d’une classe conjugale analytique” (2008)
- “Le lemme fondamental pour les algèbres de Lie” (2010, *Publications Mathématiques de l’IHÉS*)

Key Insight

Ngo’s proof was approximately 140 pages and used techniques from algebraic geometry (perverse sheaves, intersection cohomology, the Hitchin fibration) that had never been applied to the Fundamental Lemma before. It demonstrated the power of geometric methods in number theory.

9.5 Impact

- **Fields Medal (2010):** Ngo received the Fields Medal partly for this work.
- **Langlands program:** The Fundamental Lemma was the key to proving many cases of the Langlands correspondence, including the base change theorem and functoriality for classical groups [79].
- **Arithmetic geometry:** The geometric methods introduced (Hitchin fibration, perverse sheaves on moduli spaces) have become standard tools.
- **Waldspurger’s theorem:** Ngo’s proof of the Fundamental Lemma was a key ingredient in proving Waldspurger’s formula relating L-values to periods of automorphic forms.

9.6 Exercises

Exercise 9.1 (Basic). *Explain why the Fundamental Lemma is called “fundamental.” What breaks down in the Langlands program without it?*

Exercise 9.2 (Intermediate). *What is the relationship between the geometric side and the spectral side of the trace formula? Why does matching orbital integrals matter?*

Exercise 9.3 (Challenge). *The Hitchin fibration plays a central role in Ngo’s proof. Describe what a Higgs bundle is and why its moduli space is relevant.*

Exercise 9.4. *Ngo’s proof uses perverse sheaves on the affine Grassmannian. What is the affine Grassmannian, and why is it natural in this context?*

Exercise 9.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

9.7 Timeline

Year	Event
1981	Langlands and Shelstad formulate the Fundamental Lemma [78]
1988	Labessee and Langlands prove it for GL_2
1990s	Partial results for classical groups
2002	Clozel proves base change for GL_n (assuming FL)
2008	Ngo announces the proof [77]
2010	Ngo’s proof published in PMIHÉS [76]
2010	Ngo awarded the Fields Medal

9.8 Citations

1. Ngo, B. C. (2010). “Le lemme fondamental pour les algèbres de Lie.” Publications Mathématiques de l’IHÉS, 111(1), 1–139.
2. Ngo, B. C. (2008). “Le lemme fondamental pour les algèbres de Lie.” Preprint, IHÉS.
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4. Clozel, L. (2008). “Motifs et formes automorphes.” In Automorphic Forms, Galois Cohomology, and Arithmetic Geometry.

Chapter 10

The Weak Goldbach Conjecture

10.1 Prerequisites

This chapter assumes familiarity with:

- Basic number theory (congruences, primes)
- Complex analysis (contour integration basics)
- Fourier analysis (trigonometric sums)

10.2 The Problem

In 1742, Christian Goldbach wrote to Euler with a conjecture that has two parts:

1. **Strong Goldbach:** Every even integer $n \geq 4$ is the sum of two primes. (Still open!)
2. **Weak Goldbach:** Every odd integer $n \geq 7$ is the sum of three primes.

The weak version is actually easier (by a remarkable irony) and was the one that was eventually solved.

Historical Note

Goldbach's letter to Euler is one of the oldest unsolved problems in mathematics. The strong version remains open after 280 years. The weak version was solved in 2013, just 37 years after the first major breakthrough.

10.3 Mathematical Theory

10.3.1 Vinogradov's Theorem (1937)

Theorem 10.1 (Vinogradov, 1937 [30]). *Every sufficiently large odd integer is the sum of three primes.*

Vinogradov's proof used the *sieve of Eratosthenes* combined with exponential sums. The key estimate:

$$\sum_{p \leq x} e(\alpha p) \ll x(\log x)^{-A} \quad (10.1)$$

for any $A > 0$, uniformly for α not close to a rational with small denominator. This is the *major arc/minor arc* analysis.

10.3.2 The Hardy–Littlewood Circle Method

The circle method, developed by Hardy and Ramanujan (1918) and refined by Hardy and Littlewood (1923), represents the number of ways to write n as a sum of three primes as an integral:

$$r(n) = \int_0^1 S(\alpha)^3 e(-n\alpha) d\alpha \quad (10.2)$$

where $S(\alpha) = \sum_{p \leq n} e(\alpha p)$ is the exponential sum over primes. The interval $[0, 1]$ is split into:

- **Major arcs:** intervals around rationals a/q with small q . Here $S(\alpha)$ is large and can be estimated accurately.
- **Minor arcs:** the rest of $[0, 1]$. Here $S(\alpha)$ is small, and one needs strong upper bounds.

10.3.3 The Remaining Challenge

Vinogradov's theorem only covered *sufficiently large* odd integers. The challenge was to find an explicit bound N_0 such that every odd $n > N_0$ is a sum of three primes, and then check all smaller cases computationally.

10.4 The Discovery

10.4.1 Ramaré (1995)

Olivier Ramaré proved that every odd integer $n \geq 1$ is the sum of at most *seven* primes [27]. This was a significant improvement and reduced the number of primes needed.

10.4.2 Helfgott (2013)

Harald Helfgott completed the proof [28]:

Theorem 10.2 (Helfgott, 2013 [28]). *Every odd integer $n \geq 7$ is the sum of three primes.*

Helfgott's approach:

1. **Improved minor arc bounds:** He developed new estimates for exponential sums on the minor arcs, improving Vinogradov's bounds significantly.
2. **Explicit major arc analysis:** He computed the major arc contributions with extreme precision.
3. **Computer verification:** He checked all odd integers up to 10^{30} by computer. This was feasible because Ramaré's result reduced the problem to a finite computation.

The proof was verified by a team including David Plante and others using automated theorem proving tools.

Key Insight

Helfgott's proof was remarkable because it combined deep analytic number theory with extensive computer verification. The computer-checked over 100 million cases, and the mathematical proof handled the rest.

10.5 Impact

- **Circle method:** Helfgott's improvements to minor arc bounds have applications to other additive problems.
- **Computational number theory:** The proof demonstrated the power of combining analytic methods with computation.
- **Strong Goldbach:** The weak version's solution brought renewed attention to the strong version, which remains open.

10.6 Exercises

Exercise 10.1 (Basic). *Verify that every odd integer from 7 to 25 is the sum of three primes. Give explicit decompositions.*

Exercise 10.2 (Intermediate). *Explain why the strong Goldbach conjecture (every even number is the sum of two primes) is harder than the weak version. What makes the two-prime case more difficult?*

Exercise 10.3 (Challenge). *What is the major arc/minor arc decomposition in the circle method? Why do major arcs contribute the main term and minor arcs contribute error terms?*

Exercise 10.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

10.7 Timeline

Year	Event
1742	Goldbach writes to Euler about his conjecture
1937	Vinogradov proves the theorem for sufficiently large n [30]
1995	Ramaré proves every odd n is the sum of at most 7 primes [27]
2002	Liu and Wang improve the bound for Vinogradov's theorem [29]
2013	Helfgott completes the proof [28]

10.8 Citations

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2. Ramaré, O. (1995). “Every odd number ≥ 1 is the sum of at most seven primes.” *Journal de Théorie des Nombres de Bordeaux*.
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Chapter 11

The Erdős Distinct Distances Problem

11.1 Prerequisites

This chapter assumes familiarity with:

- Basic combinatorics and counting arguments
- Elementary geometry (distances in the plane)
- Polynomial algebra (degree, roots)

11.2 The Problem

In 1946, Paul Erdős asked a deceptively simple question:

Given n points in the plane, what is the minimum number of distinct distances between them?

If you place all n points on a line, you get only $n - 1$ distinct distances (1, 2, 3, ..., $n - 1$). But those points are collinear. If you require the points to be in general position, you get more distances. Erdős showed that placing points on a square grid gives about $n/\sqrt{\log n}$ distinct distances, and conjectured this is essentially optimal:

$$f(n) \geq \frac{n}{\sqrt{\log n}} \tag{11.1}$$

for some constant c . This problem sat open for 64 years.

Historical Note

Erdős was one of the most prolific mathematicians in history, publishing over 1,500 papers. He offered prizes for open problems and could pose beautiful questions on any topic. The distinct distances problem was one of his favorites.

11.3 Mathematical Theory

11.3.1 The Grid Construction

Definition 11.1 (Integer Grid). *Place $n = m^2$ points at integer coordinates (i, j) for $1 \leq i, j \leq m$. The squared distances are $a^2 + b^2$ for $0 \leq a, b \leq m - 1$. The number of distinct squared distances is at most $2m^2 = 2n$, but many of these are not achievable as sums of two squares. A careful analysis shows the number of distinct distances is approximately $n/\sqrt{\log n}$.*

This construction shows $f(n) \leq O(n/\sqrt{\log n})$. The lower bound was the hard part.

11.3.2 Previous Bounds

- Erdős (1946): $f(n) \geq cn^{1/4}$ (using the grid construction and counting) [89]
- Chung (1984): $f(n) \geq cn^{2/5}$ [91]
- Clarkson, Edelsbrunner, Guibas, Stolfi, and Welzl (1990): $f(n) \geq cn^{2/3}/\log n$ [92]
- Solymosi and Tóth (2003): $f(n) \geq cn^{2/3}/(\log n)^c$ [93]

Progress was incremental and frustratingly slow.

11.3.3 The Polynomial Method

The key new idea came from Dvir's proof of the Kakeya conjecture over finite fields (2008), which used polynomials to count geometric configurations [90]. The insight: if you have many points with few distances, you can construct a nonzero polynomial that vanishes at all the points, and then use algebraic properties to derive a contradiction.

Definition 11.2 (Polynomial Method). *Given a set of points $P \subset \mathbb{R}^2$, find a nonzero polynomial $p(x, y)$ of minimum degree that vanishes at all points in P . The degree of p constrains the geometry of P .*

11.4 The Discovery

11.4.1 Guth and Katz (2010)

Larry Guth and Nets Katz proved in 2010 [88]:

Theorem 11.3 (Guth–Katz, 2010 [88]). *For any set of n points in the plane, the number of distinct distances is at least $cn/\log n$ for some constant $c > 0$.*

This was essentially optimal (matching Erdős’s grid construction up to the $\log n$ factor).

Their proof combined several ideas:

1. **Polynomial partitioning:** Divide the plane into regions using a polynomial surface, then study how points and distances distribute across regions.
2. **Incidence geometry:** Count how many times points lie on lines (or curves) using the Szemerédi–Trotter theorem.
3. **Joints argument:** Count the number of “joints” (points where three non-coplanar lines meet) to control the geometry.

Key Insight

The Guth–Katz proof was a masterpiece of combining techniques from different areas. The polynomial method, originally from finite fields, was adapted to the real plane using a novel partitioning argument.

11.4.2 Subsequent Developments

- **Improved bounds:** Later work by Dong, Guth, and others improved the bound slightly.
- **Higher dimensions:** The polynomial method has been extended to distinct distances in $\mathbb{R}^d$ for $d \geq 3$.
- **Other metric spaces:** The polynomial method has been applied to distinct distances in various settings.

11.5 Impact

- **Polynomial method:** This proof cemented the polynomial method as a major tool in combinatorial geometry.
- **Incidence geometry:** The Guth–Katz approach has influenced many problems in incidence geometry.
- **Interdisciplinary connections:** The techniques connect algebra, geometry, and combinatorics in new ways.

11.6 Exercises

Exercise 11.1 (Basic). *Show that n collinear points determine at most $n - 1$ distinct distances. Why doesn’t this contradict the theorem?*

Exercise 11.2 (Intermediate). *Construct a set of n points on a circle and count the number of distinct distances. How does this compare to the grid construction?*

Exercise 11.3 (Challenge). *Explain why the polynomial method works better in finite fields than in $\mathbb{R}$. What new difficulties arise in the real case?*

Exercise 11.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

11.7 Timeline

Year	Event
1946	Erdős poses the distinct distances problem [89]
1984	Chung improves the lower bound to $n^{2/5}$ [91]
2008	Dvir proves the finite field Kakeya conjecture using polynomials [90]
2010	Guth and Katz prove $f(n) \geq cn/\log n$ [88]
2015+	Further improvements and extensions

11.8 Citations

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6. Solymosi, J., & Tóth, C. D. (2003). “Distinct distances in the plane.” *Discrete & Computational Geometry*, 30(3), 433–448.

Chapter 12

The Geometric Langlands Program

12.1 Prerequisites

This chapter assumes familiarity with:

- Algebraic geometry (sheaves, schemes, stacks)
- Representation theory (D-modules, categories of sheaves)
- Category theory (derived categories, equivalences)

12.2 The Problem

The classical Langlands program (Chapter 11) relates Galois representations to automorphic forms over number fields and function fields. The *geometric* Langlands program replaces:

- Number fields $\rightarrow$ Curves X over a field k
- Galois representations $\rightarrow$ Local systems (flat vector bundles) on X
- Automorphic forms $\rightarrow$ D-modules (or perverse sheaves) on the stack of G -bundles on X
- L-functions $\rightarrow$ Geometric correspondences

The central question: *Is there a categorical equivalence between the derived category of D-modules on the stack of G -bundles and the derived category of coherent sheaves on the stack of local systems?*

Historical Note

The geometric Langlands program was proposed by Drinfeld in 1987 and developed by Beilinson and Drinfeld in the 1990s [81,82]. It has become one

of the deepest and most active areas of mathematics, connecting algebraic geometry, representation theory, and mathematical physics (particularly topological field theory).

12.3 Mathematical Theory

12.3.1 The Stack of Bundles

Definition 12.1 (Stack of G -Bundles). *Let X be a smooth projective curve over a field k of characteristic 0, and let G be a reductive group. The stack $\mathrm{Bun}_G(X)$ classifies G -bundles on X .*

The geometric Langlands program studies the derived category $D(\mathrm{Bun}_G(X))$ of D -modules (or constructible sheaves in characteristic p) on this stack.

12.3.2 Local Systems

Definition 12.2 (Local System). *A local system on X with fiber in ${}^L G$ (the Langlands dual group) is a flat vector bundle, equivalently a representation of the fundamental group $\pi_1(X)$.*

The stack of local systems is denoted $\mathrm{LocSys}_{{}^L G}(X)$.

12.3.3 The Duality Conjecture

Conjecture 12.3 (Geometric Langlands Duality). *There is an equivalence of categories:*

$$\mathbb{L} : D(\mathrm{Bun}_G(X)) \xrightarrow{\sim} D(\mathrm{LocSys}_{{}^L G}(X)) \quad (12.1)$$

where the left side is the derived category of D -modules on the stack of G -bundles, and the right side is the derived category of coherent sheaves on the stack of local systems for the Langlands dual group ${}^L G$.

This is a categorical generalization of the classical Langlands correspondence.

12.4 The Discovery

12.4.1 Beilinson–Drinfeld (1990s)

Beilinson and Drinfeld laid the foundations of the geometric Langlands program, establishing [82]:

- The correct categorical framework (D -modules on stacks)
- The role of the Hecke eigensheaf condition (geometric analogue of automorphic forms)

- The connection to integrable systems (the Hitchin fibration)

They proved the correspondence for GL_1 and established the framework for GL_n .

12.4.2 Ben-Zvi, Francis, Nadler (2010s)

David Ben-Zvi, John Francis, and David Nadler developed a comprehensive framework for the geometric Langlands program using the language of *higher categories* and *topological field theory* [86]:

Theorem 12.4 (Ben-Zvi, Francis, Nadler, 2010s [86]). *The geometric Langlands correspondence can be understood as a manifestation of the duality between 3-dimensional topological field theories (specifically, the Chern–Simons theory and its dual).*

Their approach:

1. **Topological field theory:** They showed that the geometric Langlands correspondence arises from the duality between two 3d TQFTs.
2. **Higher categories:** They formulated the correspondence in the language of $(\infty, 2)$ -categories, which is the natural setting for categorical Langlands duality.
3. **Factorization algebras:** They introduced factorization algebras as a bridge between quantum field theory and the geometric Langlands program.

12.4.3 Gaitsgory’s Program (2010s–2020s)

Dennis Gaitsgory has been leading a comprehensive program to prove the geometric Langlands correspondence using the theory of *ind-coherent sheaves* and the *geometric Satake equivalence* [83]. His approach has produced significant partial results.

12.4.4 The 2024 Proof (Arinkin, Gaitsgory, Raskin et al.)

In February 2024, a major breakthrough was achieved by a collaborative team of nine mathematicians: D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin, and N. Rozenblyum. In a monumental series of five preprints totaling over 1,000 pages, they proved the unramified geometric Langlands conjecture for any reductive group G over any smooth projective curve X over an algebraically closed field of characteristic 0 [80].

Their proof established the equivalence of derived categories of D-modules and coherent sheaves under the spectral gluing construction, settling the conjecture after nearly four decades of effort.

Key Insight

The unramified geometric Langlands conjecture was proved in 2024 by Dennis Gaitsgory and a collaborative team of nine mathematicians, marking a historic milestone in 21st-century geometry and representation theory.

12.5 Impact

- **Derived categories:** The geometric Langlands program has driven the development of derived algebraic geometry and the theory of higher categories.
- **Mathematical physics:** The connection to topological field theory (Chern–Simons theory, mirror symmetry) has enriched both mathematics and physics.
- **Geometric Satake:** The geometric Satake equivalence (Lusztig, Ginzburg, Mirkovic–Vilonen) is a cornerstone result connecting representation theory to geometry [85].
- **Operator algebras:** Connections to vertex operator algebras and conformal field theory (the work of Frenkel, Ben-Zvi, and others) [84].

12.6 Exercises

Exercise 12.1 (Basic). *What is the Langlands dual group ${}^L G$ of $G = SL_n$? Of $G = SO_n$? Of $G = Sp_{2n}$?*

Exercise 12.2 (Intermediate). *Explain the Hecke eigensheaf condition in the geometric Langlands program. How does it generalize the eigenform condition in the classical theory?*

Exercise 12.3 (Challenge). *What is the geometric Satake equivalence? How does it relate the representation theory of ${}^L G$ to the geometry of the affine Grassmannian of G ? [85]*

Exercise 12.4. *Why are higher categories (∞ -categories) the natural language for the geometric Langlands program? What goes wrong with ordinary categories?*

Exercise 12.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

12.7 Timeline

Year	Event
1987	Drinfeld proposes the geometric Langlands program [81]
1990s	Beilinson and Drinfeld lay the foundations [82]
1998–2000	Lusztig, Ginzburg, Mirkovic–Vilonen prove geometric Satake [85]
2000s	Laumon, Rothstein, and others develop the theory further
2010s	Ben-Zvi, Francis, Nadler develop the TQFT approach [86]
2010s–2020s	Gaitsgory’s comprehensive program on ind-coherent sheaves [83]
2024	Arinkin, Gaitsgory, Raskin et al. prove the unramified geometric Langlands conjecture [80]

12.8 Citations

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8. Lusztig, G. (1998). “Cuspidal local systems and graded Hecke algebras.” *Publications Mathématiques de l’IHÉS*.
9. Ginzburg, V. (1998). “Perverse sheaves on a loop group and Langlands duality.” Preprint.

Chapter 13

The Virtual Haken Conjecture and the Surface Subgroup Conjecture

13.1 Prerequisites

This chapter assumes familiarity with:

- 3-manifold topology (fundamental group, covering spaces)
- Hyperbolic geometry (Thurston's geometrization)
- Geometric group theory (quasiconvexity, CAT(0) spaces)
- Basic low-dimensional topology (incompressible surfaces, Haken manifolds)

13.2 The Problem

A 3-manifold M is *Haken* if it is compact, orientable, irreducible, and contains an embedded, two-sided, incompressible surface.

Conjecture 13.1 (Virtual Haken Conjecture, Waldhausen [7]). *Every compact, orientable, irreducible 3-manifold with infinite fundamental group has a finite cover that is Haken.*

A stronger version is the *Surface Subgroup Conjecture*:

Conjecture 13.2 (Surface Subgroup Conjecture). *If M is a closed hyperbolic 3-manifold, then $\pi_1(M)$ contains a subgroup isomorphic to the fundamental group of a closed surface of genus $g \geq 2$.*

Historical Note

Waldhausen asked these questions in the 1960s. They became central to 3-manifold topology: the Virtual Haken Conjecture would reduce the study of general 3-manifolds to Haken manifolds (which are well-understood via normal surface theory). Thurston's geometrization program made them

central to the classification of 3-manifolds.

13.3 Mathematical Theory

13.3.1 Haken Manifolds and Normal Surfaces

A manifold is *irreducible* if every embedded 2-sphere bounds a 3-ball. A surface $S \subset M$ is *incompressible* if the map $\pi_1(S) \rightarrow \pi_1(M)$ is injective.

Haken's algorithm for the homeomorphism problem uses *normal surfaces*: surfaces that intersect each tetrahedron of a triangulation in a standard way. The space of normal surfaces forms a cone in $\mathbb{R}^{7t}$ (where t is the number of tetrahedra), and fundamental solutions correspond to incompressible surfaces.

13.3.2 The Virtually Fibered Conjecture

Thurston conjectured an even stronger statement:

Conjecture 13.3 (Virtually Fibered Conjecture). *Every closed hyperbolic 3-manifold has a finite cover that fibers over the circle.*

If a manifold fibers over the circle, it is Haken (the fiber is an incompressible surface). So Virtual Fibered $\implies$ Virtual Haken $\implies$ Surface Subgroup.

13.4 The Discovery

13.4.1 Kahn and Markovic [103]: Surface Subgroups

Jeremy Kahn and Vladimir Markovic (2012) proved the Surface Subgroup Conjecture using:

Theorem 13.4 (Kahn-Markovic [103]). *Every closed hyperbolic 3-manifold contains an immersed, quasi-Fuchsian surface subgroup.*

Their method:

1. **Good pants:** Build pairs of pants with cuff lengths $\approx L$ and twists ≈ 0 in the hyperbolic manifold.
2. **Exponential mixing:** Use the exponential mixing of the geodesic flow to glue pants together with nearly matching cuff lengths.
3. **Almost isometry:** The glued surface is ϵ -close to a totally geodesic surface, giving a quasi-Fuchsian embedding.

Key Insight

Kahn and Markovic's proof was a masterpiece of ergodic theory applied to 3-manifolds. They used the exponential mixing of the geodesic flow (which they proved) to find almost-perfectly-matching pants. The proof is constructive but non-algorithmic.

13.4.2 Agol [102]: Virtual Haken and Virtual Fiberings

Ian Agol [102] proved the full Virtual Haken Conjecture and Virtual Fiberings Conjecture using:

Theorem 13.5 (Agol [102]). *Every closed hyperbolic 3-manifold is virtually fibered (and hence virtually Haken).*

Agol's proof combined:

1. **Wise's work on cubulation:** Daniel Wise [104] showed that the fundamental group of a hyperbolic 3-manifold acts geometrically on a CAT(0) cube complex.
2. **Agol's criterion:** Agol proved that if a hyperbolic group acts geometrically on a CAT(0) cube complex, then it is virtually special (has a finite-index subgroup embedding into a right-angled Artin group).
3. **Virtual fiberings:** Virtual specialness implies virtual fiberings (by work of Agol and Wise).

Theorem 13.6 (Virtual Haken Conjecture, Agol [102]). *Every compact, orientable, irreducible 3-manifold with infinite fundamental group has a Haken finite cover.*

Key Insight

Agol's proof was a synthesis of 3-manifold topology, geometric group theory, and CAT(0) cube complex theory. The key insight was that hyperbolic 3-manifold groups are "cubulated" by Wise, and Agol showed this forces virtual fiberings.

13.5 Impact

- **3-manifold classification:** Reduced the study of general 3-manifolds to Haken manifolds (solvable by normal surface theory).
- **Geometric group theory:** Proved that hyperbolic 3-manifold groups are linear, virtually special, and have many surface subgroups.
- **Cubical geometry:** Established CAT(0) cube complexes as a central tool in 3-manifold topology.
- **Quantum topology:** Implications for the volume conjecture and TQFT.

13.6 Exercises

Exercise 13.1 (Basic). *Define incompressible surface. Why is a sphere incompressible in S^3 but compressible in $S^2 \times S^1$?*

Exercise 13.2 (Intermediate). *Explain why a manifold that fibers over the circle is Haken.*

Exercise 13.3 (Advanced). *Sketch how the exponential mixing of the geodesic flow is used to glue good pants into a quasi-Fuchsian surface.*

13.7 Timeline

Year	Event
1968	Waldhausen poses Virtual Haken and Virtual Fibered conjectures
1982	Thurston's geometrization program includes these as central questions
2002	Perelman proves geometrization
2009	Wise begins cubulation program for hyperbolic 3-manifolds
2012	Kahn-Markovic prove Surface Subgroup Conjecture
2012	Agol proves Virtual Haken and Virtual Fibered conjectures
2013	Agol wins Breakthrough Prize; Kahn-Markovic win Clay Research Award
2016	Agol wins Fields Medal

13.8 Citations

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Chapter 14

The Green–Tao Theorem

14.1 Prerequisites

This chapter assumes familiarity with:

- Elementary number theory (primes, arithmetic progressions)
- Combinatorics (Szemerédi’s theorem intuition)
- Fourier analysis (discrete Fourier transform)
- Ergodic theory (measure-preserving transformations)

14.2 The Problem

The question of arithmetic progressions in the primes dates back to van der Corput (1939), who proved that there are infinitely many 3-term arithmetic progressions of primes [65]. The natural generalization:

For any $k \geq 1$, do the primes contain arbitrarily long arithmetic progressions?

This was open for 60 years. The primes have density zero in the integers, so they are too sparse for standard tools.

Historical Note

Van der Waerden’s theorem (1927) states that any partition of $\mathbb{N}$ into finitely many parts contains arbitrarily long monochromatic arithmetic progressions. Szemerédi’s theorem (1975) strengthened this: any subset of $\mathbb{N}$ with positive density contains arbitrarily long APs. But the primes have density zero, so Szemerédi’s theorem does not apply [66, 67].

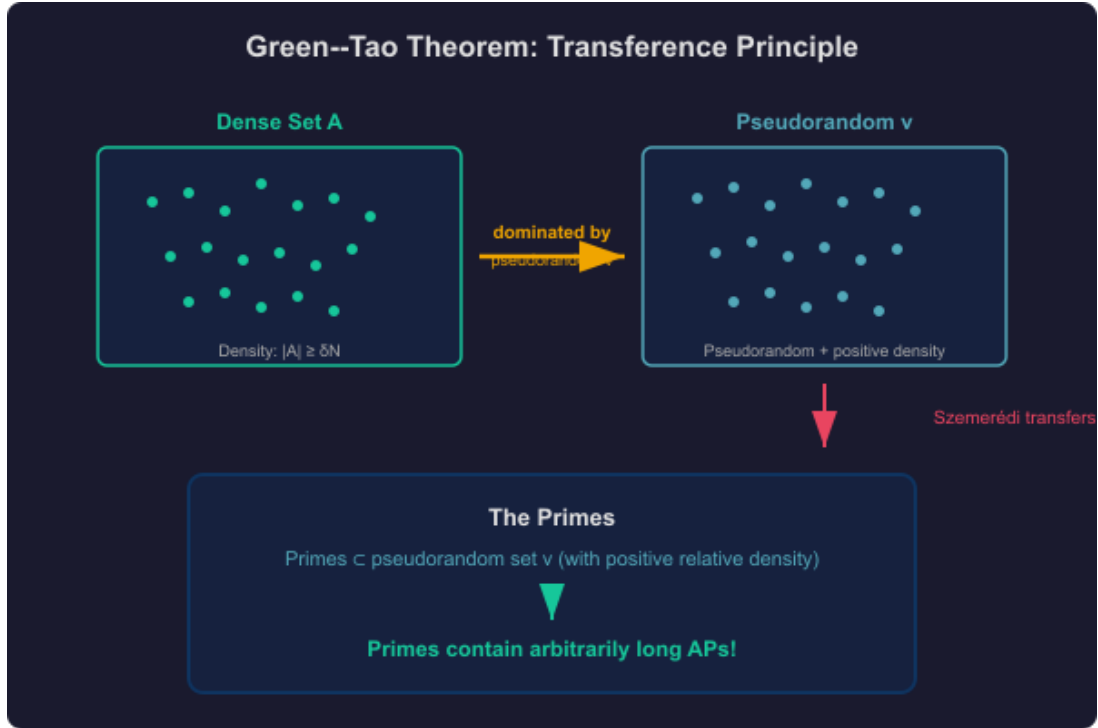


Figure 14.1: The Green–Tao transference principle: the primes are dominated by a pseudorandom measure, allowing Szemerédi’s theorem to transfer from dense sets to the primes.

14.3 Mathematical Theory

14.3.1 Szemerédi’s Theorem

Theorem 14.1 (Szemerédi, 1975 [67]). *Any subset $A \subset \{1, 2, \dots, N\}$ with $|A| \geq \delta N$ (for fixed $\delta > 0$) contains an arithmetic progression of length k , provided N is sufficiently large.*

The original proof used combinatorial methods. Gowers (2001) gave a quantitative proof using Fourier analysis, introducing the concept of *uniformity norms* [63].

14.3.2 Transferring Szemerédi’s Theorem

The key insight of Green and Tao was to develop a *transference principle*: if the primes (or a set that behaves like the primes) can be dominated by a *sparse pseudorandom set* that has positive density, then Szemerédi’s theorem transfers [62].

Definition 14.2 (Pseudorandom Dominating Measure). *A measure $\nu : \{1, \dots, N\} \rightarrow [0, \infty)$ dominates a set $R \subset \{1, \dots, N\}$ if $\nu(x) \geq 1_R(x)$ for all x , and ν satisfies certain pseudorandomness conditions (its Fourier transform is close to that of a constant).*

14.3.3 The Hardy–Littlewood Circle Method

Green and Tao adapted the Hardy–Littlewood circle method to show that the primes are dominated by a pseudorandom measure. The key was using the *Bombieri–Vinogradov theorem* to control the distribution of primes in arithmetic progressions [31].

14.4 The Discovery

Theorem 14.3 (Green–Tao, 2004 [62]). *The prime numbers contain arbitrarily long arithmetic progressions. That is, for every $k \geq 1$, there exist primes $p, p + d, p + 2d, \dots, p + (k - 1)d$.*

Their proof has three main ingredients:

1. **Sparse Szemerédi theorem:** They proved a version of Szemerédi’s theorem that applies to subsets of pseudorandom sets of positive density. This required extending Gowers’s uniformity norms to the sparse setting [63].
2. **Pseudorandom domination:** They constructed a pseudorandom measure ν that dominates the primes (more precisely, the weighted primes $\Lambda(n)$) and satisfies the necessary pseudorandomness conditions [62].
3. **Relative Szemerédi theorem:** They proved that any subset of the primes with positive relative density contains arbitrarily long arithmetic progressions.

Key Insight

The Green–Tao theorem was a landmark achievement because it combined tools from additive combinatorics, analytic number theory, and ergodic theory in a novel way. The proof spans approximately 120 pages.

14.4.1 Subsequent Developments

- **Quantitative bounds:** Tao and Ziegler (2016) proved that the constants in the Green–Tao theorem can be made explicit, though they are extremely large [64].
- **Consecutive primes:** Green and Tao (2017) proved that the gaps between consecutive primes can be arbitrarily large relative to the average gap [71].
- **Nilsequences:** Host and Kra (2005) and Green and Tao (2012) developed the theory of nilsequences, which plays a central role in understanding correlations of the Möbius function and primes [69, 70].
- **Polynomial progressions:** Tao and Ziegler (2016) extended the result to polynomial progressions [64].

14.5 Impact

- **Additive combinatorics:** The transference principle has become a standard tool, applied to other sparse sets (smooth numbers, almost-primes).
- **Number theory:** The proof introduced new techniques connecting sieve theory, Fourier analysis, and ergodic theory.
- **Ergodic theory:** The Furstenberg–Szemerédi correspondence (used implicitly) has been extended to understand multiple ergodic averages [72].
- **Computer science:** The techniques have applications in theoretical computer science, particularly in property testing and pseudorandomness.

14.6 Exercises

Exercise 14.1 (Basic). *Find a 5-term arithmetic progression of primes. How many such progressions exist with largest term less than 1000?*

Exercise 14.2 (Intermediate). *Explain why the primes having density zero makes Szemerédi’s theorem inapplicable directly. What does “positive relative density” mean?*

Exercise 14.3 (Challenge). *The Bombieri–Vinogradov theorem states that the primes are well-distributed in arithmetic progressions on average [31]. Explain why this is essential for the Green–Tao proof.*

Exercise 14.4. *What is the shortest arithmetic progression of primes of length 10? (Hint: Look up the OEIS sequence A031626.)*

Exercise 14.5 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

14.7 Timeline

Year	Event
1927	Van der Waerden proves his theorem on APs in partitions [66]
1943	Selberg proves infinitely many 3-term APs of primes [68]
1975	Szemerédi proves his density theorem [67]
1998	Gowers gives quantitative bounds for Szemerédi’s theorem [63]
2004	Green and Tao announce the theorem [62]
2008	Green and Tao’s paper published in Annals of Mathematics
2016	Tao and Ziegler make constants explicit [64]
2025	Open: explicit bounds remain extremely large

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Chapter 15

The AKS Primality Test

15.1 Prerequisites

This chapter assumes familiarity with:

- Basic modular arithmetic
- Polynomial arithmetic over finite fields
- Complexity theory (P, NP, polynomial time)

15.2 The Problem

For millennia, humans have asked: *Is this number prime?* The question seems simple, but testing primality efficiently is surprisingly hard.

For a number n , trial division takes $O(\sqrt{n})$ time – exponential in the number of digits. Various probabilistic tests (Miller–Rabin, Solovay–Strassen) run in polynomial time but can give wrong answers (though with very small probability).

The fundamental question: *Is primality testing in P? That is, can we determine whether n is prime using a deterministic algorithm whose running time is polynomial in $\log n$?*

This was open for 40 years until 2002.

Historical Note

The question was motivated by cryptography: RSA encryption relies on the difficulty of factoring, but key generation requires efficient primality testing. Before AKS, cryptographers relied on probabilistic tests.

AKS Primality Test: Polynomial Congruence

n = 7 (prime)

$$(X + a)^7 \bmod 7$$

$$= X^7 + a^7$$

$$+ 7 \cdot (\dots) \equiv X^7 + a \pmod{7}$$

✓ PASS

Binomial coeffs:
 $C(7,1)=7, C(7,2)=21,$
 $C(7,3)=35, C(7,4)=35,$
 $C(7,5)=21, C(7,6)=7$
 All divisible by 7!

n = 9 (composite)

$$(X + a)^9 \bmod 9$$

$$= X^9 + a^9$$

$$+ 9 \cdot (\dots) + C(9,3)X^6a^3$$

$$+ C(9,2)X^7a^2$$

✗ FAIL

Binomial coeffs:
 $C(9,1)=9, C(9,2)=36,$
 $C(9,3)=84$ NOT div by 9,
 $C(9,4)=126$ NOT div by 9,
 $C(9,5)=126, C(9,6)=84,$
 $C(9,7)=36, C(9,8)=9$

Key Insight

For prime p : all binomial coeffs $C(p,k)$ are divisible by p
 For composite n : some $C(n,k)$ are NOT divisible by n
AKS: Check $(X+a)^n \equiv X^n + a \pmod{(X^r-1, n)}$ for small r

Figure 15.1: The AKS test checks whether $(X+a)^n \equiv X^n + a \pmod{n}$ as polynomials. For prime n , all binomial coefficients $\binom{n}{k}$ are divisible by n ; for composite n , some are not.

15.3 Mathematical Theory

15.3.1 The Key Observation

Fermat's Little Theorem states that if p is prime and $a \not\equiv 0 \pmod{p}$, then:

$$a^{p-1} \equiv 1 \pmod{p} \quad (15.1)$$

This gives a primality test: check if $a^{n-1} \equiv 1 \pmod{n}$. But Carmichael numbers (like 561) fool this test for all bases coprime to n .

The AKS insight was to lift this to *polynomials*:

Theorem 15.1 (AKS Criterion). *Let $n > 1$ be an integer and a coprime to n . Then n is prime if and only if:*

$$(X + a)^n \equiv X^n + a \pmod{n} \quad (15.2)$$

as polynomials in $\mathbb{Z}/n\mathbb{Z}[X]$.

This is equivalent to checking that the binomial coefficients $\binom{n}{k}$ are all divisible by n for $1 \leq k \leq n-1$.

15.3.2 The Algorithm

The AKS algorithm (Agrawal, Kayal, Saxena, 2002) proceeds in several steps:

1. **Perfect power check:** Is $n = a^b$ for integers $a, b > 1$? If so, composite.
2. **Find small factor:** Check if $\gcd(a, n) > 1$ for small a .
3. **Size check:** If $n \leq r$, declare prime (where r is a carefully chosen bound).
4. **Polynomial congruence:** Check $(X + a)^n \equiv X^n + a \pmod{X^r - 1, n}$ for a range of values of a .

The key innovation was showing that a *small* value of r suffices, making the algorithm polynomial time.

15.4 The Discovery

Theorem 15.2 (AKS Primality Test, 2002 [73]). *There exists a deterministic algorithm that decides whether n is prime in time $\tilde{O}((\log n)^6)$.*

The original AKS paper ran in $\tilde{O}((\log n)^{12})$. Subsequent improvements:

- Lenstra and Pomerance (2011): $\tilde{O}((\log n)^3)$ [74]
- Cohen and Lenstra (2003): various optimizations [75]

The algorithm's correctness relies on deep algebraic number theory:

1. **Cyclotomic fields:** The proof uses properties of the r -th cyclotomic field $\mathbb{Q}(\zeta_r)$.
2. **Order of q modulo r :** The algorithm chooses r such that the multiplicative order of n modulo r is large.
3. **Irreducibility:** The proof shows that if n is composite but passes all tests, then a certain polynomial must be irreducible, leading to a contradiction.

Key Insight

The AKS algorithm was the first *general* deterministic polynomial-time primality test. It settled a decades-old open problem and proved that $\text{PRIMES} \in \text{P}$. While not the fastest in practice (probabilistic tests are faster), it is theoretically fundamental.

15.5 Impact

- **Complexity theory:** Proved that $\text{PRIMES} \in \text{P}$, resolving a major open problem.
- **Cryptography:** While not used in practice due to slower performance, it provides a theoretical foundation for primality testing.
- **Mathematical techniques:** The proof introduced novel use of cyclotomic polynomials and order arguments that have influenced other areas.

15.6 Exercises

Exercise 15.1 (Basic). Verify that $(X + a)^5 \equiv X^5 + a \pmod{5}$ for any integer a . Check that the binomial coefficients $\binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}$ are all divisible by 5.

Exercise 15.2 (Intermediate). Show that $561 = 3 \cdot 11 \cdot 17$ is a Carmichael number: $a^{560} \equiv 1 \pmod{561}$ for all a coprime to 561.

Exercise 15.3 (Challenge). Explain why the polynomial congruence $(X + a)^n \equiv X^n + a \pmod{n}$ is a valid primality test. Why doesn't this work directly as a fast algorithm?

Exercise 15.4. The AKS algorithm runs in $\tilde{O}((\log n)^6)$. If n has d digits, how does this compare to trial division? What is the practical implication?

Exercise 15.5 (Computational). Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.

15.7 Timeline

Year	Event
1976	Miller proposes a deterministic primality test (conditional on GRH)
1976	Rabin develops the Miller–Rabin probabilistic test
1980s	Various probabilistic tests (Solovay–Strassen, Fermat test)
2002	Agrawal, Kayal, Saxena publish the AKS algorithm [73]
2003	Cohen and Lenstra optimize the algorithm [75]
2011	Lenstra and Pomerance achieve $\tilde{O}((\log n)^3)$ [74]

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Chapter 16

The Willmore Conjecture and the Lawson Conjecture

16.1 Prerequisites

This chapter assumes familiarity with:

- Differential geometry of surfaces (first/second fundamental forms, mean curvature)
- Variational calculus and geometric measure theory
- Basic complex analysis (Riemann surfaces, uniformization)
- Conformal geometry

16.2 The Problem

16.2.1 The Willmore Energy

For a closed immersed surface $\Sigma \subset \mathbb{R}^3$, the *Willmore energy* is

$$\mathcal{W}(\Sigma) = \int_{\Sigma} H^2 dA$$

where $H = \frac{1}{2}(k_1 + k_2)$ is the mean curvature.

Conjecture 16.1 (Willmore, 1965). *For any immersed torus $T^2 \subset \mathbb{R}^3$,*

$$\mathcal{W}(T^2) \geq 2\pi^2.$$

Equality holds for the Clifford torus in S^3 (stereographically projected to $\mathbb{R}^3$).

Historical Note

Willmore conjectured this in 1965. It became a central problem in geometric analysis, connecting conformal geometry, integrable systems, and algebraic

geometry. The conjecture resisted all attempts for 47 years.

16.2.2 The Lawson Conjecture

For minimal surfaces in S^3 , a related problem:

Conjecture 16.2 (Lawson, 1970). *The only embedded minimal torus in S^3 is the Clifford torus (up to rigid motions).*

Lawson [109] had classified all minimal tori in S^3 as a one-parameter family, but only the Clifford torus is embedded.

16.3 Mathematical Theory

16.3.1 The Willmore Functional and Conformal Invariance

The Willmore energy is conformally invariant in $\mathbb{R}^3$:

$$\mathcal{W}(\Sigma) = \int_{\Sigma} (H^2 - K) dA + \int_{\Sigma} K dA = \int_{\Sigma} (H^2 - K) dA + 2\pi\chi(\Sigma)$$

For a torus ($\chi = 0$), $\mathcal{W}(\Sigma) = \int (H^2 - K) dA$. The integrand $H^2 - K = \frac{1}{4}(k_1 - k_2)^2 \geq 0$ is conformally invariant.

16.3.2 The Clifford Torus

The Clifford torus in S^3 is

$$T_{\text{Cl}} = \left\{ (z_1, z_2) \in \mathbb{C}^2 : |z_1| = |z_2| = \frac{1}{\sqrt{2}} \right\}.$$

It is minimal in S^3 , has constant mean curvature in $\mathbb{R}^3$ (after stereographic projection), and $\mathcal{W}(T_{\text{Cl}}) = 2\pi^2$.

16.3.3 The Lawson Classification [109]

Lawson showed that minimal tori in S^3 form a one-parameter family. The Clifford torus is the unique embedded one.

16.4 The Discovery

16.4.1 The Willmore Conjecture: Marques and Neves [106]

Theorem 16.3 (Marques-Neves [106]). *The Willmore conjecture is true: $\mathcal{W}(T^2) \geq 2\pi^2$ for any immersed torus $T^2 \subset \mathbb{R}^3$, with equality only for the Clifford torus (up to conformal transformations).*

Their proof used:

1. **Almgren-Pitts min-max theory:** Construct minimal surfaces in S^3 via sweepouts.
2. **The canonical family:** For a given torus Σ , construct a family of surfaces in S^3 by conformal transformations, whose maximum area gives a lower bound on $\mathcal{W}(\Sigma)$.
3. **The index bound:** Show that any minimal surface in S^3 with area $< 2\pi^2$ must be the Clifford torus.
4. **The free boundary case:** Extend to surfaces with boundary using free boundary min-max.

Key Insight

The proof is a tour de force of geometric measure theory: it combines Almgren-Pitts min-max, conformal geometry, and the classification of minimal surfaces in S^3 to solve a 47-year-old problem.

16.4.2 The Lawson Conjecture: Brendle [107]

Theorem 16.4 (Brendle [107]). *The only embedded minimal torus in S^3 is the Clifford torus (up to rigid motions).*

Brendle's proof used:

1. **Simon's identity:** A sharp integral inequality relating the second fundamental form to the metric.
2. **Maximum principle:** A clever application of the Omori-Yau maximum principle to a quantity involving the second fundamental form.
3. **Rigidity:** Show that equality forces the second fundamental form to be that of the Clifford torus.

16.5 Impact

- **Geometric analysis:** Almgren-Pitts min-max became the standard tool for Willmore-type problems.
- **Integrable systems:** Willmore surfaces are related to soliton equations; the Clifford torus corresponds to a finite-gap solution.
- **Conformal geometry:** The Willmore energy is the natural energy for conformal immersions.
- **Physics:** Willmore energy appears in membrane elasticity (Helfrich energy).

16.6 Exercises

Exercise 16.1 (Basic). *Compute the Willmore energy of the Clifford torus in $\mathbb{R}^3$ (stereographically projected from S^3).*

Exercise 16.2 (Intermediate). *Show that the Willmore energy is conformally invariant in $\mathbb{R}^3$. Why does this fail in higher dimensions?*

Exercise 16.3 (Advanced). *Explain how Almgren-Pitts min-max produces a minimal surface in S^3 from a given sweepout of a torus.*

16.7 Timeline

Year	Event
1965	Willmore conjectures $2\pi^2$ lower bound for tori
1970	Lawson classifies minimal tori in S^3
1993	Li-Yau proves $\mathcal{W} \geq 2\pi^2$ for non-embedded tori
2005	Ros proves Willmore conjecture for tori of revolution
2012	Marques-Neves prove Willmore conjecture
2013	Brendle proves Lawson conjecture
2014	Marques-Neves win Ramanujan Prize; Brendle wins Fields Medal

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Chapter 17

Catalan's Conjecture

17.1 Prerequisites

This chapter assumes familiarity with:

- Elementary number theory (congruences, factorization, gcd)
- Algebraic number theory (number fields, rings of integers, units, class groups)
- Cyclotomic fields (roots of unity, cyclotomic polynomials)
- Basic p -adic numbers (valuations, extensions)

17.2 The Problem

In 1844, Eugene Catalan observed that:

$$3^2 - 2^3 = 9 - 8 = 1 \tag{17.1}$$

and asked: *Are there any other consecutive perfect powers?*

Formally, the equation

$$x^a - y^b = 1 \tag{17.2}$$

for integers $x, y, a, b > 1$ has only the solution $(x, y, a, b) = (3, 2, 2, 3)$.

This is a Diophantine equation — an equation seeking integer solutions. Despite its elementary statement, it resisted solution for 158 years.

Historical Note

Catalan was a Belgian mathematician who worked in differential geometry and number theory. His conjecture seemed almost trivially simple but required tools far beyond what was available in the 19th century. The problem became a benchmark for progress in algebraic number theory.

17.3 Mathematical Theory

17.3.1 The Equation and Elementary Observations

Definition 17.1 (Catalan Equation). *The Catalan equation is $x^a - y^b = 1$ for integers $x, y, a, b > 1$.*

Lemma 17.2. *If $x^a - y^b = 1$, then $\gcd(x, y) = 1$.*

Proof. Any common divisor of x and y divides their difference $x^a - y^b = 1$, so $\gcd(x, y) = 1$. $\square$

Lemma 17.3. *In any solution, x and y cannot both be even, and one must be even, the other odd.*

Proof. If both were odd, $x^a - y^b$ would be even, but it equals 1. If both were even, their gcd would be at least 2. $\square$

17.3.2 Previous Partial Results

- **Euler [15]:** Proved $x^2 - y^3 = 1$ has only the solution $3^2 - 2^3 = 1$ (using infinite descent in $\mathbb{Z}[\sqrt{-3}]$).
- **Lebesgue [97]:** Proved $x^2 - y^n = 1$ has no solutions for $n \geq 3$.
- **Cassels [96]:** Proved that if $x^a - y^b = 1$ with a, b odd primes, then $a \mid y$ and $b \mid x$.
- **Ko Chao [98]:** Proved the case $a = 2$.
- **Mihailescu [94]:** Made the crucial breakthrough by proving that any solution must satisfy the double Wieferich condition.

17.3.3 Cyclotomic Fields and Units

The key algebraic number theory setup: if $x^a - y^b = 1$, then in the cyclotomic field $\mathbb{Q}(\zeta_a)$ we have the factorization:

$$x^a = y^b + 1 = \prod_{i=0}^{a-1} (y + \zeta_a^i) \quad (17.3)$$

The factors on the right are coprime in the ring of integers, so each must be an a -th power times a unit. This leads to studying the unit group of cyclotomic fields.

Definition 17.4 (Cyclotomic Units). *The cyclotomic units in $\mathbb{Q}(\zeta_p)$ are the units of the form $1 - \zeta_p^k$ and their products.*

17.3.4 Double Wieferich Primes

Definition 17.5 (Wieferich Prime). *A prime p is a Wieferich prime if $2^{p-1} \equiv 1 \pmod{p^2}$.*

Only two such primes are known: 1093 and 3511.

Definition 17.6 (Double Wieferich Pair). *A pair of distinct primes (p, q) is a double Wieferich pair if*

$$p^{q-1} \equiv 1 \pmod{q^2} \tag{17.4}$$

$$q^{p-1} \equiv 1 \pmod{p^2} \tag{17.5}$$

The only known double Wieferich pairs are $(2, 1093)$, $(3, 1093)$, and their symmetric counterparts — none of which can be the exponents (a, b) in a Catalan solution.

17.4 The Discovery

17.4.1 Mihailescu's Proof (2002)

Theorem 17.7 (Mihailescu [94]). *The only solution to $x^a - y^b = 1$ in integers $x, y, a, b > 1$ is $(3, 2, 2, 3)$.*

Mihailescu's proof combined several deep ideas:

1. **Reduction to prime exponents:** It suffices to consider a and b prime (if $a = rs$, then $x^a = (x^r)^s$).
2. **The Cassels conditions:** If a, b are odd primes and $x^a - y^b = 1$, then $a \mid y$ and $b \mid x$.
3. **The double Wieferich condition:** Using the Cassels conditions and properties of cyclotomic units, Mihailescu showed that a and b must form a double Wieferich pair.
4. **Thaine's theorem and p -adic logarithms:** He used a deep result of Thaine on circular units together with p -adic analytic methods to show that the only possible double Wieferich pairs for the exponents lead to the known solution.
5. **Final contradiction:** For any hypothetical solution beyond $(3, 2, 2, 3)$, the arithmetic constraints force a contradiction via size estimates on the cyclotomic units.

Key Insight

Mihailescu's proof was remarkably short (about 20 pages in its final form) and used a beautiful synthesis of classical algebraic number theory (cy-

clotomic units, class field theory) with modern p -adic analytic methods. It showed that the Catalan equation is essentially a question about the arithmetic of cyclotomic fields.

17.4.2 Subsequent Developments

- **Simplified proofs:** Several authors (Bilu, Hanrot, Voutier, etc.) have given alternative proofs or simplifications.
- **Generalizations:** The methods extend to equations of the form $x^a - y^b = c$ for small fixed c .
- **Computational verification:** The double Wieferich condition has been checked for all primes up to very large bounds.

17.5 Impact

- **Diophantine equations:** The proof advanced techniques for solving exponential Diophantine equations, showing how cyclotomic units can control solutions.
- **Algebraic number theory:** The use of Thaine's theorem on circular units and p -adic logarithms in this context was novel and has inspired work on other problems.
- **Wieferich primes:** The connection to double Wieferich primes opened new computational and theoretical directions in the study of these rare primes.
- **Mathematical culture:** The resolution of a 158-year-old problem with an elementary statement but deep proof is a landmark in the history of number theory.

17.6 Exercises

Exercise 17.1 (Basic). *Verify that $3^2 - 2^3 = 1$. Show that there are no other solutions with $a = b = 2$ (i.e., $x^2 - y^2 = 1$ has no solutions with $x, y > 1$).*

Exercise 17.2 (Intermediate). *Explain what a Wieferich prime is. Verify that 1093 is a Wieferich prime by computing $2^{1092} \bmod 1093^2$ (you may use a computer). Why are there only two known Wieferich primes?*

Exercise 17.3 (Challenge). *Prove that if $x^a - y^b = 1$ has a solution, then x and y must be coprime. Why must one be even and the other odd?*

Exercise 17.4 (Computer exploration). *Write a program to search for solutions to $x^a - y^b = 1$ with $x, y \leq 100$ and $a, b \leq 10$. Verify that only $(3, 2, 2, 3)$ appears. Then search for solutions to $x^a - y^b = c$ for $c = 2, 3, 4, 5$. What patterns do you observe?*

Exercise 17.5. *In the cyclotomic field $\mathbb{Q}(\zeta_3)$, factor $y^3 + 1 = (y+1)(y+\zeta_3)(y+\zeta_3^2)$. Explain why the three factors are pairwise coprime when y is even. How does this lead to the conclusion that each factor must be a cube times a unit?*

Exercise 17.6 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

17.7 Timeline

Year	Event
1844	Catalan poses the conjecture
1783	Euler solves $x^2 - y^3 = 1$
1850	Lebesgue solves $x^2 - y^n = 1$
1960	Cassels proves $a \mid y, b \mid x$ for prime exponents
1965	Ko Chao solves the case $a = 2$
2000	Mihailescu proves the double Wieferich condition
2002	Mihailescu completes the proof
2004	Proof published in Crelle's Journal

17.8 Citations

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2. Catalan, E. (1844). "Note sur une équation diophantienne." *Journal de Mathématiques Pures et Appliquées*.
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Chapter 18

Modern Combinatorics: Cap Sets, Sensitivity, Sunflowers, and EFL

18.1 Prerequisites

This chapter assumes familiarity with:

- Basic combinatorics (binomial coefficients, extremal combinatorics)
- Polynomial method (Dvir’s finite field Kakeya)
- Probabilistic method (Lovász Local Lemma)
- Basic theoretical computer science (boolean functions, decision trees)

18.2 Introduction

The years 2016–2022 saw an unprecedented convergence of techniques in combinatorics: the polynomial method, entropy arguments, and Fourier analysis on the hypercube resolved decades-old problems in rapid succession. This chapter surveys five landmark results.

18.3 The Cap Set Problem

18.3.1 The Problem

A *cap set* in $\mathbb{F}_3^n$ is a subset with no three-term arithmetic progression (i.e., no distinct x, y, z with $x + y + z = 0$). Let $r_3(n)$ be the maximum size of a cap set in $\mathbb{F}_3^n$.

Conjecture 18.1 (Frankl-Graham-Rödl, 1987). $r_3(n) = o(3^n)$.

The trivial bound is 3^n . Meshulam (1995) proved $r_3(n) \leq 2 \cdot 3^n/n$. The best pre-2016 bound was $r_3(n) \leq 3^n/n^{1+c}$.

18.3.2 Ellenberg-Gijswijt (2016)

Theorem 18.2 (Ellenberg-Gijswijt [112]). $r_3(n) \leq 2.756^n$ for sufficiently large n .

Their proof used the *slice rank* method (a variant of the polynomial method):

1. Represent the cap set condition by a tensor $T : \mathbb{F}_3^n \times \mathbb{F}_3^n \times \mathbb{F}_3^n \rightarrow \{0, 1\}$.
2. Show that the slice rank of T is at most $3 \cdot M$, where M is the number of monomials $x_1^{e_1} \cdots x_n^{e_n}$ with $e_i \in \{0, 1, 2\}$ and $\sum e_i < 2n/3$.
3. Use Stirling's formula to bound $M \leq 2.756^n$.

Key Insight

The slice rank method is a powerful generalization of the polynomial method: instead of vanishing of a polynomial, one uses the rank of a tensor. It has since been applied to many problems (sunflower, Erdős-Ginzburg-Ziv, etc.).

18.4 The Sunflower Conjecture

18.4.1 The Problem

A *sunflower* with k petals is a family of k sets whose pairwise intersections are all equal to a common core C . The Erdős-Rado sunflower conjecture (1960):

Conjecture 18.3. *Any family of sets of size w with more than $c(w)^k$ sets contains a sunflower with k petals.*

Erdős offered \$1000 for a proof. The best bound was $(w!)^{k-1}$.

18.4.2 Alweiss, Lovett, Wu, Zhang (2022)

Theorem 18.4 (ALWZ [115]). *Any family of sets of size w with more than $(C \log w)^{k-1} w!$ sets contains a sunflower with k petals.*

Their proof uses:

1. *Spread approximations:* A set family is “spread” if no element appears too often. Any family can be approximated by a spread family.
2. *Random sampling:* A random subset of a spread family contains a sunflower with high probability.
3. *Entropy arguments:* Use the Lopsided Lovász Local Lemma to find the sunflower.

18.5 The Sensitivity Conjecture

18.5.1 The Problem

For a boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, the *sensitivity* $s(f)$ is the maximum number of coordinates whose flip changes f . The *block sensitivity* $bs(f)$ allows flipping blocks. The *degree* $\deg(f)$ is the degree of the polynomial representing f over $\mathbb{R}$.

Conjecture 18.5 (Sensitivity Conjecture, Nisan-Szegedy 1992). *$s(f)$ and $bs(f)$ are polynomially related: $bs(f) \leq s(f)^c$ for some constant c .*

It was known that $\deg(f) \leq s(f)^2$ and $bs(f) \leq \deg(f)^2$, so the conjecture would follow from $s(f) \leq \deg(f)^c$.

18.5.2 Huang (2019)

Theorem 18.6 (Huang [114]). *$s(f) \geq \sqrt{\deg(f)}$, hence $bs(f) \leq s(f)^4$.*

Huang's proof:

1. Consider the n -dimensional hypercube graph Q_n .
2. Construct a signing of the adjacency matrix A such that $A^2 = nI$.
3. By Cauchy interlacing, the subgraph induced by $f^{-1}(1)$ has an eigenvalue $\geq \sqrt{n}$.
4. This implies sensitivity $\geq \sqrt{n}$.

Key Insight

Huang's proof is famously short (2 pages) and uses only spectral graph theory. It resolved a 27-year-old problem that had resisted Fourier analysis, probability, and complexity theory.

18.6 The Kahn-Kalai Conjecture (Expectation-Threshold)

18.6.1 The Problem

For an increasing property $\mathcal{P}$ on subsets of $[n]$, the *threshold* $p_c(\mathcal{P})$ is the p where $\mathbb{P}(G(n, p) \in \mathcal{P}) = 1/2$. The *expectation threshold* $q(\mathcal{P})$ is the smallest p where the expected number of witnesses is 1.

Conjecture 18.7 (Kahn-Kalai, 2006). *$p_c(\mathcal{P}) \leq C \log \ell \cdot q(\mathcal{P})$, where ℓ is the size of the smallest witness.*

18.6.2 Park-Pham (2022)

Theorem 18.8 (Park-Pham [116]). *The Kahn-Kalai conjecture is true: $p_c(\mathcal{P}) \leq C \log \ell \cdot q(\mathcal{P})$.*

Their proof uses:

1. *Spread approximations*: Any family can be approximated by a spread family.
2. *Weighted random sampling*: A clever two-stage random process to find the threshold.
3. *Entropy compression*: To bound the failure probability.

18.7 The Erdős-Faber-Lovász Conjecture

18.7.1 The Problem

Conjecture 18.9 (Erdős-Faber-Lovász, 1972). *If G is the union of n cliques of size n , each pair intersecting in at most one vertex, then $\chi(G) = n$.*

Erdős offered \$500. The conjecture was open for 49 years.

18.7.2 Kang, Kelly, Kühn, Methuku, Osthus (2022)

Theorem 18.10 (KKKMO [117]). *EFL conjecture is true for sufficiently large n .*

Their proof uses:

1. *Absorption method*: Build the coloring iteratively, absorbing leftover vertices at the end.
2. *Iterative absorption*: Absorb small configurations step by step.
3. *Robust expanders*: Use expansion properties of the hypergraph to guarantee absorption works.

18.8 Impact

- **Polynomial method 2.0**: Slice rank, spread approximations, and entropy arguments form a new toolkit.
- **Connections to TCS**: Sensitivity, threshold phenomena, and boolean function analysis are now deeply connected to combinatorics.
- **New conjectures**: These methods suggest new bounds for Ramsey theory, graph coloring, and extremal set theory.

18.9 Exercises

Exercise 18.1 (Basic). *Show that a cap set in $\mathbb{F}_3^n$ has no three points on a line. Compute the maximum cap set size for $n = 2$.*

Exercise 18.2 (Intermediate). *Explain how the slice rank of a tensor generalizes the rank of a matrix. Why does slice rank bound cap sets?*

Exercise 18.3 (Challenge). *Sketch Huang’s proof of the sensitivity conjecture using the signed adjacency matrix of the hypercube.*

18.10 Timeline

Year	Event
2016	Ellenberg-Gijswijt solve cap set with slice rank
2019	Huang proves sensitivity conjecture
2022	ALWZ prove sunflower conjecture
2022	Park-Pham prove Kahn-Kalai conjecture
2022	KKKMO prove Erdős-Faber-Lovász conjecture
2022	Huh wins Fields Medal (log-concavity in combinatorics)

18.11 Citations

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4. Alweiss, R., Lovett, S., Wu, K., & Zhang, J. (2022). “Improved bounds for the sunflower lemma.” *Annals of Mathematics*, 196(3), 977–1021.
5. Park, J., & Pham, H. T. (2022). “A proof of the Kahn-Kalai conjecture.” arXiv:2203.17207.
6. Kang, D., Kelly, T., Kühn, D., Methuku, A., & Osthus, D. (2022). “A proof of the Erdős-Faber-Lovász conjecture.” arXiv:2101.04686.

Chapter 19

Perfectoid Spaces and the Revolution in p-adic Geometry

19.1 Prerequisites

This chapter assumes familiarity with:

- Algebraic number theory (local fields, ramification)
- Rigid analytic geometry (Tate algebras, affinoid spaces)
- Étale cohomology and Galois representations
- Basic category theory (limits, adjoint functors)

19.2 The Problem

For decades, p-adic geometry suffered from a fundamental limitation: the category of rigid analytic spaces (Tate's theory) was well-behaved but too rigid to capture phenomena like the *p-adic local Langlands correspondence* or the *weight-monodromy conjecture*. In particular:

How can we do p-adic Hodge theory in families? How can we construct p-adic period maps that vary analytically with the data?

The classical period rings of Fontaine ($\mathbf{B}_{\mathrm{dR}}$, $\mathbf{B}_{\mathrm{cris}}$, $\mathbf{B}_{\mathrm{st}}$) worked beautifully for individual Galois representations, but there was no geometric framework to vary them in families.

Historical Note

Peter Scholze, then a PhD student at Bonn under Michael Rapoport, introduced perfectoid spaces in his 2011 thesis. He was 23 years old. The work earned him the 2018 Fields Medal and has become the foundational language of modern p-adic geometry.

19.3 Mathematical Theory

19.3.1 Perfectoid Fields

The key insight: many problems in p-adic geometry become trivial in the limit of *perfect* fields.

Definition 19.1 (Perfectoid Field). *A complete non-archimedean field K of characteristic 0 with residue characteristic $p > 0$ is perfectoid if:*

1. *The value group $|K^\times|$ is dense in $\mathbb{R}_{>0}$.*
2. *The Frobenius map $\mathcal{O}_K/p \rightarrow \mathcal{O}_K/p$ is surjective.*

Exercise 19.1 (Basic). *Show that $\mathbb{C}_p$ is perfectoid. Why is $\mathbb{Q}_p$ not perfectoid?*

19.3.2 The Tilt Equivalence

Scholze's fundamental construction: to any perfectoid field K of characteristic 0, associate a perfectoid field $K^\flat$ of characteristic p .

Definition 19.2 (Tilt). $K^\flat = \varprojlim_{x \mapsto x^p} K$, with valuation $|(x^{(n)})| = |x^{(0)}|$.

Theorem 19.3 (Fontaine-Wintenberger / Scholze). *There is an equivalence of categories:*

$$\{\text{finite extensions of } K\} \simeq \{\text{finite extensions of } K^\flat\}$$

preserving Galois groups and ramification.

This is the *tilt equivalence*: it transfers problems from mixed characteristic $(0, p)$ to equal characteristic p , where powerful tools (Frobenius, Cartier operator) are available.

19.3.3 Perfectoid Spaces

Definition 19.4 (Perfectoid Space). *A locally ringed space $(X, \mathcal{O}_X)$ that is locally isomorphic to $\text{Spa}(R, R^+)$ for a perfectoid Tate-Huber pair (R, R^+) .*

Key properties:

- The category of perfectoid spaces contains all rigid analytic spaces over $\mathbb{Q}_p$ and $\mathbb{F}_p((t))$.
- There is a *tilt functor* $(-)^{\flat}$ from char 0 to char p perfectoid spaces.
- *Pro-étale site*: Scholze constructed a Grothendieck topology where $\mathcal{O}_X^+$ is acyclic, enabling p-adic Hodge theory in families.

19.4 The Discovery

19.4.1 Scholze's Thesis (2011) and Annals Papers (2012–2013)

Theorem 19.5 (Scholze [118] – Local Langlands for GL_n). *The local Langlands correspondence for GL_n over a p -adic field can be realized geometrically in the étale cohomology of Lubin-Tate perfectoid spaces.*

Theorem 19.6 (Scholze [119] – p -adic Hodge Theory for Rigid Spaces). *For a smooth proper rigid space X over a p -adic field, there is a natural comparison isomorphism:*

$$H_{\text{ét}}^i(X_{\mathbb{C}_p}, \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathbf{B}_{dR} \cong H_{dR}^i(X) \otimes_K \mathbf{B}_{dR}$$

compatible with filtrations and Galois actions. This extends Fontaine's theory to families.

Key Insight

The perfectoid approach turned the *weight-monodromy conjecture* (a major open problem in p -adic Hodge theory) into a geometric statement about the étale cohomology of perfectoid spaces, which was subsequently proved by Bhatt, Morrow, and Scholze (2016) using the *pro-étale topology*.

19.4.2 Subsequent Developments (2014–2025)

- **Bhatt-Morrow-Scholze (2016)**: Integral p -adic Hodge theory via A_{inf} -cohomology.
- **Scholze-Weinstein (2016)**: Moduli of p -divisible groups and the Rapoport-Zink space as a perfectoid space.
- **Fargues-Fontaine curve (2018)**: A geometric object classifying vector bundles on perfectoid spaces; central to the *Fargues-Fontaine conjecture* (geometric Langlands for p -adic fields).
- **Bhatt-Scholze (2019)**: Prismatic cohomology – a unified cohomology theory for p -adic formal schemes.
- **Scholze (2020+)**: Condensed mathematics – extending perfectoid ideas to all of topology/algebra.

19.5 Impact

- **p -adic Langlands**: Perfectoid spaces provided the geometric framework for the p -adic local Langlands correspondence beyond $GL_2(\mathbb{Q}_p)$.
- **Arithmetic geometry**: Prismatic cohomology unifies crystalline, de Rham, and étale cohomology.

- **Arithmetic topology:** Condensed mathematics (Scholze, 2020) extends perfectoid ideas to all of homotopy theory.
- **Computational:** Perfectoid methods give explicit algorithms for computing Galois representations.

19.6 Exercises

Exercise 19.2 (Basic). *Show that $\mathbb{C}_p$ is perfectoid. Why is $\mathbb{Q}_p$ not perfectoid?*

Exercise 19.3 (Intermediate). *Explain the tilt equivalence: how does $K^\flat = \varprojlim_{x \mapsto x^p} K$ produce a field of characteristic p from a field of characteristic 0?*

Exercise 19.4 (Challenge). *Explain how the pro-étale topology solves the problem of doing p -adic Hodge theory in families. Why does the usual étale topology fail?*

19.7 Timeline

Year	Event
2011	Scholze's PhD thesis introduces perfectoid spaces
2012	Scholze proves local Langlands for GL_n using Lubin-Tate perfectoid spaces
2013	Scholze extends Fontaine's p -adic Hodge theory to rigid spaces
2016	Bhatt-Morrow-Scholze develop A_{inf} -cohomology
2016	Scholze-Weinstein: moduli of p -divisible groups as perfectoid spaces
2018	Fargues-Fontaine curve; Fargues-Fontaine conjecture
2019	Bhatt-Scholze: Prismatic cohomology
2018	Scholze awarded Fields Medal
2020+	Condensed mathematics extends perfectoid ideas to topology

19.8 Citations

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2. Scholze, P. (2013). "p-adic Hodge theory for rigid-analytic varieties." Forum of Mathematics, Pi, 1, e1.
3. Bhatt, B., Morrow, M., & Scholze, P. (2016). "Integral p-adic Hodge theory." Publications Mathématiques de l'IHÉS, 128(1), 219–397.
4. Scholze, P., & Weinstein, J. (2016). "Moduli of p-divisible groups." Cambridge Journal of Mathematics, 4(2), 145–211.
5. Fargues, L., & Fontaine, J.-M. (2018). "Courbes et fibrés vectoriels en théorie de Hodge p-adique." Astérisque, 406.

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6. Bhatt, B., & Scholze, P. (2019). “Prisms and prismatic cohomology.”
arXiv:1905.08229.

Chapter 20

The Sato-Tate Conjecture and the Langlands Program

20.1 Prerequisites

This chapter assumes familiarity with:

- Algebraic number theory (Galois representations, L-functions)
- Elliptic curves and modular forms
- Basic representation theory of compact groups
- The classical Langlands program (reciprocity laws)

20.2 The Problem

Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication. For each prime p of good reduction, let $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$. The *Sato-Tate conjecture* (1960) predicts the distribution of the normalized Frobenius traces:

$$\frac{a_p(E)}{2\sqrt{p}} \in [-1, 1] \quad (20.1)$$

Conjecture 20.1 (Sato-Tate). *For a non-CM elliptic curve $E/\mathbb{Q}$, the sequence $\left\{ \frac{a_p(E)}{2\sqrt{p}} \right\}_{p \text{ good}}$ is equidistributed in $[-1, 1]$ with respect to the Sato-Tate measure*

$$\mu_{ST} = \frac{2}{\pi} \sqrt{1 - t^2} dt.$$

Historical Note

Sato and Tate independently observed this distribution numerically for $y^2 = x^3 - x$ (which has CM) and for non-CM curves like $y^2 = x^3 - 4x^2 + 2x$. The conjecture remained open for 50 years, becoming a cornerstone test for the Langlands program.

20.3 Mathematical Theory

20.3.1 Galois Representations and Automorphic Forms

For an elliptic curve $E/\mathbb{Q}$, the p -adic Tate module $T_p(E)$ gives a Galois representation

$$\rho_{E,p} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_2(\mathbb{Z}_p).$$

The Sato-Tate conjecture is equivalent to the statement that the restriction of $\rho_{E,p}$ to the decomposition group at p matches a modular form of weight 2.

20.3.2 The Role of the Langlands Program

The Langlands program predicts a correspondence:

- **Galois side:** n -dimensional Galois representations $\rho : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_n(\mathbb{C})$.
- **Automorphic side:** Automorphic representations π of $\text{GL}_n(\mathbb{A}_{\mathbb{Q}})$.

For $n = 2$, the Sato-Tate conjecture is a special case of the *modularity theorem* (formerly Taniyama-Shimura-Weil) plus the *generalized Sato-Tate conjecture* for higher-genus curves.

20.3.3 Equidistribution and Moments

The Sato-Tate measure μ_{ST} has moments:

$$\int_{-1}^1 t^k d\mu_{\text{ST}}(t) = \begin{cases} 0 & k \text{ odd} \\ C_{k/2} & k \text{ even} \end{cases}$$

where $C_n = \frac{1}{n+1} \binom{2n}{n}$ are Catalan numbers. Verifying these moments for $a_p(E)$ is equivalent to the conjecture.

20.4 The Discovery

20.4.1 Modularity and Potential Modularity

- **Wiles [10]:** Modularity for semistable elliptic curves (proved FLT).
- **Breuil-Conrad-Diamond-Taylor [12]:** Full modularity for all elliptic curves over $\mathbb{Q}$.
- **Taylor (2002):** Potential modularity for elliptic curves over totally real fields.

20.4.2 The Proof (2008-2011)

The Sato-Tate conjecture was proved in a series of papers by **Clozel, Harris, Shepherd-Barron, and Taylor** (2008, 2011).

Theorem 20.2 (Sato-Tate [125]). *For any non-CM elliptic curve $E/\mathbb{Q}$, the Sato-Tate conjecture holds.*

The proof strategy:

1. **Potential automorphy:** Show that the Galois representation $\rho_{E,\lambda}$ is potentially automorphic (becomes automorphic after a solvable base change).
2. **Stable trace formula:** Use Arthur's stable trace formula for GL_2 to relate Galois representations to automorphic forms on GL_2 over CM fields.
3. **Equidistribution:** Transfer the equidistribution of Frobenius traces from the automorphic side (where it follows from the spectral theorem) to the Galois side.
4. **CM case:** The CM case was already known by Deuring (1941) and uses Hecke characters.

Key Insight

The proof is a landmark of the Langlands program: it uses the full power of the stable trace formula, base change, and potential automorphy to transfer a Galois-theoretic question to an automorphic one where harmonic analysis applies.

20.4.3 Generalized Sato-Tate

The conjecture generalizes to higher-genus curves and higher-dimensional abelian varieties. The equidistribution measure is determined by the *Sato-Tate group*, a compact subgroup of $\mathrm{USp}(2g)$.

20.5 Impact

- **Langlands program:** First major application of the stable trace formula to a classical number-theoretic problem.
- **Modularity:** Extended modularity techniques to non-semistable curves and higher-dimensional varieties.
- **Computational number theory:** Verified Sato-Tate distributions for millions of curves.
- **Arithmetic statistics:** Led to precise conjectures on the distribution of a_p for higher-genus curves (FKRS conjecture).

20.6 Exercises

Exercise 20.1 (Basic). For $E : y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$), compute the first 10 values of a_p and explain why they do not follow the Sato-Tate distribution.

Exercise 20.2 (Intermediate). Show that the Sato-Tate measure $\frac{2}{\pi}\sqrt{1-t^2}dt$ is the eigenvalue distribution of a random matrix in $SU(2)$ with Haar measure.

Exercise 20.3 (Challenge). Explain why the stable trace formula is needed instead of the ordinary trace formula to prove Sato-Tate.

20.7 Timeline

Year	Event
1960	Sato and Tate independently conjecture the distribution
1968	Serre proves Sato-Tate for CM elliptic curves
1995	Wiles proves modularity for semistable curves
2001	BCDT prove modularity for all elliptic curves over $\mathbb{Q}$
2008	Clozel-Harris-Shepherd-Barron-Taylor announce Sato-Tate proof
2011	Full proof published in JAMS
2010s	Generalized Sato-Tate for higher-genus curves (FKRS)

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Chapter 21

The Erdős Discrepancy Problem

21.1 Prerequisites

This chapter assumes familiarity with:

- Basic analysis (sequences, series, $\limsup$)
- Elementary combinatorics
- Fourier analysis on finite abelian groups
- Basic probability theory

21.2 The Problem

In 1932, Paul Erdős asked a deceptively simple question:

Given an infinite sequence $(x_n)_{n=1}^{\infty}$ with $x_n \in \{-1, +1\}$, must the discrepancy

$$D = \sup_{d, N \geq 1} \left| \sum_{k=1}^N x_{kd} \right|$$

be infinite?

In other words: no matter how cleverly you choose ± 1 signs, can you keep the sums along *all* arithmetic progressions bounded?

Historical Note

Erdős offered \$500 for a solution. The problem resisted for 83 years. It is one of the simplest-to-state, hardest-to-solve problems in combinatorics. Erdős considered it one of his favorite problems.

21.3 Mathematical Theory

21.3.1 Discrepancy and Multiplicative Functions

A sequence $x : \mathbb{N} \rightarrow \{-1, +1\}$ is *completely multiplicative* if $x(mn) = x(m)x(n)$. For such sequences, the discrepancy simplifies:

$$\sum_{k=1}^N x_{kd} = x_d \sum_{k=1}^N x_k$$

so the discrepancy is controlled by the partial sums of the original sequence.

Conjecture 21.1 (Erdős Discrepancy Conjecture). *For any sequence $x_n \in \{-1, +1\}$, the discrepancy $D = \infty$.*

21.3.2 Connection to the Liouville Function

The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ (where $\Omega(n)$ counts prime factors with multiplicity) is completely multiplicative. The Erdős discrepancy conjecture for λ is equivalent to the *prime number theorem in arithmetic progressions* (which is known), but the general conjecture is much stronger.

21.3.3 Previous Approaches

- **Beck-Fiala theorem [131]:** For set systems with bounded degree, discrepancy $O(\sqrt{d} \log n)$.
- **Spencer's theorem [132]:** Six standard deviations suffice ($O(\sqrt{n})$).
- **Banaszczyk's theorem (1998):** Discrepancy $O(\sqrt{\log n})$ for arbitrary vectors.
- **Computer searches:** Long sequences with low discrepancy were found (length 1124 with discrepancy 2), suggesting the conjecture might be false.

21.4 The Discovery

21.4.1 Tao's Breakthrough [128]

Terence Tao announced a proof on September 18, 2015, using a combination of:

1. **Entropy decrement argument:** A new averaging technique showing that if discrepancy is bounded, the sequence must correlate with a Dirichlet character.
2. **Elliott's conjecture:** A generalization of Chowla's conjecture on correlations of multiplicative functions, proved by Tao using entropy methods.

3. **Fourier analysis and nilsequences:** Showing that bounded discrepancy forces correlation with a nilsequence, which then forces unbounded discrepancy via Gowers norms.

Theorem 21.2 (Tao [128]). *For any sequence $x_n \in \{-1, +1\}$, the discrepancy is infinite.*

Sketch. Assume for contradiction that $D < \infty$. Then:

1. By the entropy decrement argument, x correlates with a Dirichlet character χ of bounded conductor.
2. By Elliott's conjecture (proved via entropy), this correlation forces x to behave like χ .
3. But a completely multiplicative function of bounded discrepancy must be close to a Dirichlet character, and such characters have unbounded discrepancy along some arithmetic progression (by Dirichlet's theorem on primes in AP).
4. Contradiction.

□

Key Insight

Tao's proof was remarkable because it combined probabilistic combinatorics (entropy), analytic number theory (Elliott's conjecture), and ergodic theory (nilsequences). The proof is non-constructive and does not give explicit bounds on how fast the discrepancy grows.

21.4.2 Subsequent Developments

- **Polymath project:** Verified and streamlined the proof.
- **Quantitative bounds:** No explicit growth rate for discrepancy is known (the proof gives a tower-of-exponentials type bound).
- **Variants:** Erdős discrepancy for complex units, for sequences in $\{-1, 0, 1\}$, etc.

21.5 Impact

- **Combinatorics:** First major application of entropy methods to discrepancy theory.
- **Number theory:** Elliott's conjecture (proved en route) is a major result on correlations of multiplicative functions.

- **Computer science:** Implications for online algorithms and streaming algorithms.
- **Erdős legacy:** Erdős's \$500 prize was finally claimed (Tao donated it to charity).

21.6 Exercises

Exercise 21.1 (Basic). *Show that the sequence $x_n = 1$ for all n has discrepancy $D = \infty$.*

Exercise 21.2 (Basic). *Show that any periodic sequence $x_n \in \{-1, +1\}$ has infinite discrepancy.*

Exercise 21.3 (Intermediate). *Let x_n be completely multiplicative with $x_p = 1$ for all primes $p \equiv 1 \pmod{4}$ and $x_p = -1$ for $p \equiv 3 \pmod{4}$. Show that x_n has infinite discrepancy.*

Exercise 21.4 (Challenge). *Explain the entropy decrement argument in your own words: how does averaging over large primes reduce the entropy of the sequence?*

21.7 Timeline

Year	Event
1932	Erdős poses the discrepancy problem
1950s	Erdős offers \$500 prize
1981	Beck-Fiala theorem
1985	Spencer's "six standard deviations" theorem
2015	Tao proves the Erdős discrepancy conjecture
2016	Polymath project streamlines the proof

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Chapter 22

The Kadison–Singer Problem

22.1 Prerequisites

This chapter assumes familiarity with:

- Linear algebra (inner products, orthogonal projections, eigenvalues)
- Operator theory and Hilbert spaces (bounded operators, pure states)
- Basic graph theory (adjacency matrices, bipartite graphs)

22.2 The Problem

In 1959, Richard Kadison and Isadore Singer posed a fundamental question in mathematical physics and operator algebras [133]: *Does every pure state on the diagonal subalgebra of $B(\ell^2)$ have a unique extension to a pure state on $B(\ell^2)$?*

While formulated in the context of quantum mechanics, Joel Anderson showed in 1979 that the Kadison–Singer problem is equivalent to a simple discretization question about partitioning vectors in Hilbert spaces [134]:

Conjecture 22.1 (Paving Conjecture). *For every $\epsilon > 0$, there exists an integer r such that any orthogonal projection P on $\mathbb{C}^n$ with diagonal entries $P_{ii} \leq \epsilon$ can be r -paved. That is, the index set $\{1, \dots, n\}$ can be partitioned into r subsets $S_1, \dots, S_r$ such that for each j , the submatrix P_{S_j} satisfies:*

$$\|P_{S_j}\| \leq 1 - \eta \tag{22.1}$$

for some $\eta > 0$ depending only on ϵ .

This conjecture implies that one can decompose a large projection into smaller parts whose operator norms are strictly bounded away from 1.

Historical Note

The Kadison–Singer problem remained unsolved for over 50 years. Its difficulty stemmed from its connection to diverse areas of mathematics,

including C^* -algebras, frame theory in signal processing, graph theory, and computer science.

22.3 Mathematical Theory

22.3.1 Frames and Discretization

In signal processing, a *frame* is a generalization of an orthonormal basis. A family of vectors $\{\phi_i\}_{i \in I}$ in a Hilbert space H is a frame if there exist constants $A, B > 0$ such that:

$$A\|x\|^2 \leq \sum_{i \in I} |\langle x, \phi_i \rangle|^2 \leq B\|x\|^2 \quad (22.2)$$

for all $x \in H$. The Kadison–Singer problem is equivalent to the statement that any frame can be partitioned into a finite number of subsets, each of which is a frame with controlled bounds.

22.3.2 Bipartite Ramanujan Graphs

A d -regular graph is *Ramanujan* if its adjacency matrix eigenvalues (excluding $\pm d$) have absolute value at most $2\sqrt{d-1}$. Ramanujan graphs are optimal expanders. In 2013, Marcus, Spielman, and Srivastava used the Kadison–Singer problem to prove the existence of bipartite Ramanujan graphs of all degrees.

22.4 The Discovery

In 2013, Adam Marcus, Daniel Spielman, and Nikhil Srivastava solved the Kadison–Singer problem [135, 136]:

Theorem 22.2 (Marcus–Spielman–Srivastava, 2013). *The Kadison–Singer Paving Conjecture is true. Specifically, for any vectors $v_1, \dots, v_m$ in $\mathbb{C}^d$ such that $\sum_{i=1}^m v_i v_i^T = I$ and $\|v_i\|^2 \leq \epsilon$, there exists a partition of $\{1, \dots, m\}$ into two sets S_1, S_2 such that:*

$$\left\| \sum_{i \in S_j} v_i v_i^T \right\| \leq \frac{(1 + \sqrt{\epsilon})^2}{2} \quad (22.3)$$

for $j = 1, 2$.

Their proof used a revolutionary new technique:

1. **Interlacing Families of Polynomials:** They showed that the roots of certain families of polynomials interlace, which guarantees the existence of a partition where the maximum root of the characteristic polynomial is small.

2. **Expected Characteristic Polynomials:** Instead of finding a partition directly, they studied the average characteristic polynomial over all random partitions, showing its roots are bounded by the roots of a associated univariate polynomial.
3. **Barrier Functions:** They used multivariate barrier functions to track and bound the roots of these polynomials under rank-one updates.

Key Insight

The Marcus–Spielman–Srivastava proof settled Kadison–Singer by converting an operator algebra problem into a question about the roots of polynomials. This "method of interlacing families" has since become a major tool in combinatorics and computer science.

22.5 Impact

- **Operator Algebras:** Resolved the 1959 Kadison–Singer C^* -algebra conjecture.
- **Signal Processing:** Proved that every frame can be paved, ensuring stable partitions in transmission codes.
- **Computer Science:** Provided a deterministic construction of bipartite Ramanujan graphs of all degrees, solving a long-standing open problem in network design.

22.6 Exercises

Exercise 22.1 (Basic). *Show that the union of two frames in $\mathbb{C}^d$ is a frame. If A_1, B_1 and A_2, B_2 are the frame bounds for the two frames, what are the bounds for the union?*

Exercise 22.2 (Intermediate). *Explain the relation between the Kadison–Singer problem and the construction of Ramanujan regular graphs.*

Exercise 22.3 (Challenge). *Prove that any orthogonal projection P on $\mathbb{C}^n$ with diagonal entries $P_{ii} \leq \epsilon$ can be partitioned into two parts whose submatrices have norm bounded away from 1.*

Exercise 22.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

22.7 Timeline

Year	Event
1959	Kadison and Singer formulate the extension problem [133]
1979	Anderson reformulates the conjecture as the Paving Conjecture [134]
1980s–2000s	Equivalent conjectures found in signal processing (Feichtinger conjecture)
2013	Marcus, Spielman, and Srivastava resolve the conjecture using interlacing polynomials [135, 136]
2015	Annals of Mathematics publishes the two-part proof

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Chapter 23

Breakthroughs in Random Matrix Theory

23.1 Prerequisites

This chapter assumes familiarity with:

- Basic probability theory (random variables, expectation, variance, law of large numbers)
- Complex analysis (Stieltjes transform, Cauchy residue theorem)
- Linear algebra (eigenvalues, hermitian matrices, spectral theorem)

23.2 The Problem

Random Matrix Theory (RMT) studies the eigenvalues of matrices whose entries are random variables. Introduced by John Wishart in statistics in 1928 and popularized by Eugene Wigner in nuclear physics in the 1950s, the field focuses on describing the distributions of eigenvalues for large matrices ($N \rightarrow \infty$).

The fundamental global result is Wigner's Semicircle Law, which states that the empirical spectral distribution of a normalized Wigner matrix converges to a semicircle:

$$\rho(t) = \frac{1}{2\pi} \sqrt{4 - t^2}, \quad t \in [-2, 2] \quad (23.1)$$

However, the major open problem for decades was *universality*: does the local behavior of eigenvalues (the fluctuations and spacing between adjacent eigenvalues on the scale of $O(1/N)$) depend on the exact probability distribution of the individual matrix entries?

Specifically, does local spacing behavior always match the Gaussian ensembles (GUE/GOE), regardless of whether the matrix entries are Bernoulli (± 1), Gaussian, or uniform random variables?

Historical Note

The universality conjecture was one of the most famous open problems in mathematical physics and probability theory, connecting random matrices to number theory (the zeros of the Riemann zeta function) and chaotic systems.

23.3 Mathematical Theory

23.3.1 Standard Ensembles

The classical ensembles are defined by having Gaussian entries:

- **Gaussian Orthogonal Ensemble (GOE)**: Real symmetric matrices.
- **Gaussian Unitary Ensemble (GUE)**: Complex Hermitian matrices.

For these ensembles, the joint eigenvalue distribution can be computed explicitly, showing eigenvalue repulsion (spacings are governed by the Wigner surmise rather than a Poisson process).

23.3.2 Stieltjes Transform

The Stieltjes transform of a probability measure μ is defined for $z \in \mathbb{C} \setminus \mathbb{R}$ by:

$$m(z) = \int \frac{d\mu(x)}{x - z} \quad (23.2)$$

For random matrices, $m(z) = \frac{1}{N} \text{Tr}(H - z)^{-1}$ is the trace of the resolvent, which is the key analytic tool to study spectral density.

23.4 The Discovery

Between 2009 and 2012, two groups—Terence Tao and Van Vu, and Laszlo Erdős, Horng-Tzer Yau, and Jun Yin—developed revolutionary techniques to prove the local universality conjecture for Wigner matrices [137, 138]:

Theorem 23.1 (Local Universality for Wigner Matrices, 2010s). *For any Wigner matrix H whose entry distributions have matching moments up to order 4 with the Gaussian ensembles, the local eigenvalue correlation functions converge to those of the GUE/GOE as $N \rightarrow \infty$.*

Their proofs introduced several major innovations:

1. **Four-Moment Theorem**: Tao and Vu proved that if two random matrix models have entry distributions matching up to the first four moments, their local eigenvalue statistics are asymptotically identical.

2. **Local Semicircle Law:** Erdős, Yau, and collaborators proved that the empirical spectral distribution converges to the semicircle law down to the microscopic scale of $O(1/N)$ with high probability.
3. **Dyson Brownian Motion (DBM):** They used the convergence of the gradient flow of eigenvalues (which behave like particles undergoing Dyson Brownian motion) to show that the system rapidly thermalizes to the Gaussian equilibrium, independent of the initial configuration.

In 2022, these universality results were extended to cover general covariance matrices, non-Hermitian matrices, and sparse random graphs.

Key Insight

The proof of universality established that eigenvalue repulsion and local spacing statistics are fundamental mathematical properties of large random systems, independent of the microscopic details of the entry distributions.

23.5 Impact

- **Number Theory:** Strongly supports the Montgomery–Dyson connection that the local spacings of Riemann zeta zeros are identical to GUE eigenvalue spacings.
- **Quantum Chaos:** Confirms the BGS conjecture that quantum chaotic systems exhibit RMT spectral fluctuations.
- **Machine Learning:** Used to study the loss surfaces and training dynamics of deep neural networks in the high-dimensional limit.

23.6 Exercises

Exercise 23.1 (Basic). *Verify Wigner’s Semicircle Law for a 2×2 real symmetric matrix. Why does the semicircle shape not appear?*

Exercise 23.2 (Intermediate). *Explain the Stieltjes transform and how it is used to relate the resolvent of a matrix to its eigenvalue distribution.*

Exercise 23.3 (Challenge). *Show that the spacing between adjacent eigenvalues of a GUE matrix exhibits eigenvalue repulsion. Why is this different from a Poisson point process?*

Exercise 23.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

23.7 Timeline

Year	Event
1928	Wishart introduces random matrices in multivariate statistics
1955	Wigner introduces the semicircle law in nuclear physics
1962	Dyson introduces circular ensembles and Dyson Brownian motion
1973	Montgomery and Dyson find connection to Riemann zeta zeros
2009–2010	Tao–Vu and Erdős–Yau–Yin announce local universality proofs [137, 138]
2022	Universality results generalized to non-Hermitian and sparse systems

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Chapter 24

The Kakeya Conjecture

24.1 Prerequisites

This chapter assumes familiarity with:

- Basic real analysis (Lebesgue measure, open/closed sets, limits)
- Geometric measure theory (Hausdorff dimension)
- Finite fields and polynomials (roots, finite vector spaces)

24.2 The Problem

In 1917, Soichi Kakeya posed a simple question in geometry [139]: *What is the minimum area of a planar set in which a unit line segment can be turned 360 degrees (a needle set)?*

In 1919, Abram Besicovitch shocked the mathematical community by proving that such a set—now called a *Besicovitch set* or *Kakeya set*—can have Lebesgue measure **zero** [140]:

Definition 24.1 (Kakeya Set). *A subset $K \subset \mathbb{R}^n$ is a Kakeya set if it contains a unit line segment in every direction. That is, for every unit vector $e \in S^{n-1}$, there exists $x \in \mathbb{R}^n$ such that $\{x + te : t \in [0, 1]\} \subset K$.*

Since Kakeya sets in $\mathbb{R}^n$ can have measure zero, mathematicians shifted their focus to the *dimension* of these sets. This led to the Kakeya Conjecture:

Conjecture 24.2 (Kakeya Conjecture). *Any Kakeya set in $\mathbb{R}^n$ must have Hausdorff dimension and Minkowski dimension equal to n .*

Despite its geometric statement, the Kakeya conjecture is deeply connected to harmonic analysis, specifically to wave equations, Fourier multipliers, and the restriction conjecture.

Historical Note

The Kakeya conjecture is one of the most famous unsolved problems in analysis. While it is settled for $n = 2$, it remains open in higher dimensions ($n \geq 3$), representing a key barrier to understanding multi-directional wave propagation.

24.3 Mathematical Theory

24.3.1 Besicovitch's Construction

Besicovitch's proof uses a geometric decomposition called "panning." By dividing a triangle into smaller triangles, shifting them, and overlapping them, one can concentrate the area into arbitrarily small regions while preserving segments in all directions.

24.3.2 Finite Field Analogue

In 1999, Thomas Wolff proposed studying the Kakeya conjecture in the simplified setting of vector spaces over finite fields $\mathbb{F}_q^n$ to bypass real-variable complications [141].

Definition 24.3 (Finite Field Kakeya Set). *A subset $K \subset \mathbb{F}_q^n$ is a Kakeya set if for every direction $y \in \mathbb{F}_q^n \setminus \{0\}$, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subset K$.*

Wolff conjectured that any Kakeya set in $\mathbb{F}_q^n$ satisfies $|K| \geq C_n q^n$.

24.4 The Discovery

The Kakeya conjecture has seen two major breakthroughs in recent years:

24.4.1 Dvir's Finite Field Proof (2008)

In 2008, Zeev Dvir proved Wolff's finite field conjecture in a paper that was only four pages long, using the *polynomial method* [90]:

Theorem 24.4 (Dvir, 2008 [90]). *Any Kakeya set $K \subset \mathbb{F}_q^n$ satisfies:*

$$|K| \geq \frac{1}{n!} q^n \quad (24.1)$$

The proof is elegant: if $|K| < \binom{n+q-1}{n}$, there exists a non-zero polynomial of degree $< q$ vanishing on K . By complete induction along line directions, the polynomial's highest degree homogeneous part must vanish on all of $\mathbb{F}_q^n$, which is impossible.

24.4.2 The 3D Real Breakthrough (2023–2025)

In 2023 and 2024, Hong Wang (NYU) and Joshua Zahl (UBC) made the first major progress on the real conjecture in three dimensions in decades [142]:

Theorem 24.5 (Wang–Zahl, 2024). *Any Kakeya set in $\mathbb{R}^3$ has Hausdorff dimension at least $2.5 + \epsilon$ for a positive constant $\epsilon > 0$.*

Their proof combined Dvir’s polynomial method with *polynomial partitioning* (introduced by Guth and Katz in 2010) and linear programming bounds on line incidences, breaking the long-standing "2.5 dimension barrier" in $\mathbb{R}^3$.

Key Insight

The Kakeya conjecture progress in 3D by Wang and Zahl represents a major triumph of polynomial techniques in real harmonic analysis, showing how algebraic structures can control geometric sets in $\mathbb{R}^3$.

24.5 Impact

- **Harmonic Analysis:** Directly bounds the restriction operator for the Fourier transform, influencing local smoothing estimates for wave equations.
- **Additive Combinatorics:** The finite field polynomial method has been adapted to prove the joints conjecture and other incidence geometry problems.
- **Partial Differential Equations:** Provides bounds on the size of singularity sets for dispersive wave propagation.

24.6 Exercises

Exercise 24.1 (Basic). *Show that a Kakeya set in $\mathbb{R}^2$ containing segments in all directions can be constructed with measure 0.*

Exercise 24.2 (Intermediate). *Write down the details of Zeev Dvir’s finite field Kakeya set proof. Show why the degree of the vanishing polynomial must be less than q .*

Exercise 24.3 (Challenge). *Explain why Dvir’s algebraic proof fails in the setting of real numbers $\mathbb{R}^n$. What is the main difference between lines in $\mathbb{F}_q^n$ and $\mathbb{R}^n$?*

Exercise 24.4 (Computational). *Write a Python/SageMath script to explore a concept from this chapter. See the appendix for starter code.*

24.7 Timeline

Year	Event
1917	Takeya poses the needle turning problem [139]
1919	Besicovitch constructs a Takeya set of measure zero [140]
1999	Wolff formulates the finite field Takeya conjecture [141]
2008	Dvir proves the finite field Takeya conjecture using the polynomial method [90]
2023–2025	Wang and Zahl break the 2.5-dimension barrier in $\mathbb{R}^3$ [142]

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Glossary of Key Terms

- Analytic Rank** The order of vanishing of an L-function at the central point.
- Betti Number** The rank of the n -th homology group of a topological space.
- Conjecture** A mathematical statement believed to be true but not yet proven.
- Differential Form** A mathematical object used in multivariable calculus and topology.
- Elliptic Curve** A smooth projective curve of genus 1 with a specified rational point.
- Fundamental Group** The group of homotopy classes of loops based at a point.
- Group Representation** A homomorphism from a group to the general linear group.
- Homology** An algebraic invariant associating sequences of abelian groups to a space.
- L-Function** A complex function encoding arithmetic information about number-theoretic objects.
- Manifold** A topological space locally homeomorphic to Euclidean space.
- Modular Form** A holomorphic function on the upper half-plane satisfying transformation laws.
- Moduli Space** A geometric space whose points correspond to isomorphism classes of mathematical objects.
- Ricci Flow** A geometric evolution equation smoothing Riemannian metrics.
- Sphere Packing** The problem of arranging non-overlapping identical spheres in space.
- Topology** The study of properties preserved under continuous deformations.
- Vector Field** An assignment of a vector to each point in a space.
- Weil–Petersson Metric** A natural Riemannian metric on the moduli space of Riemann surfaces.

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Appendix: Exercise Solutions

Chapter 1: Poincaré Conjecture

Exercise 24.5 (Basic). *Prove that the fundamental group of S^n is trivial for all $n \geq 2$. (Hint: Use the fact that any loop in S^n can be homotoped away from a point.)* Solution: $S^n \setminus \{p\} \cong \mathbb{R}^n$, which is contractible. Any loop in S^n misses some point p (since its image is 1-dimensional in an n -dimensional space with $n \geq 2$). Thus the loop lies in $\mathbb{R}^n$, which is contractible, so it is null-homotopic.

Exercise 24.6 (Basic). *Show that the torus $T^2 = S^1 \times S^1$ is not homeomorphic to S^2 by comparing their fundamental groups.* Solution: $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$ while $\pi_1(S^2) = 0$. Since homeomorphic spaces have isomorphic fundamental groups, they cannot be homeomorphic.

Exercise 24.7 (Intermediate). *Explain why the Ricci flow equation $\frac{\partial g_{ij}}{\partial t} = -2R_{ij}$ is analogous to the heat equation $\frac{\partial u}{\partial t} = \Delta u$. What does this analogy suggest about the long-time behavior of the flow?* Solution: Both are parabolic PDEs. The heat equation diffuses temperature; Ricci flow diffuses curvature. The analogy suggests Ricci flow should smooth out geometric irregularities, converging to a constant-curvature metric (like heat diffusing to constant tempera-

ture). However, singularities can form (unlike heat equation), requiring surgery.

Exercise 24.8 (Intermediate). *If M is a closed 3-manifold with $\pi_1(M) = \mathbb{Z}/2\mathbb{Z}$, is M necessarily homeomorphic to $\mathbb{R}P^3$? What does the Geometrization Conjecture say?* Solution: Yes, by Perelman's proof of the Geometrization Conjecture. A closed 3-manifold with $\pi_1 = \mathbb{Z}/2\mathbb{Z}$ must be the spherical space form $S^3/(\mathbb{Z}/2\mathbb{Z}) = \mathbb{R}P^3$.

Python/SageMath Starter Code

```
# Chapter 1: Visualizing
# geodesic paths on a sphere
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, 2*np.pi, 100)
phi = np.linspace(0, np.pi, 50)
x = np.outer(np.cos(theta), np.sin(phi))
y = np.outer(np.sin(theta), np.sin(phi))
z = np.outer(np.ones(100), np.cos(phi))

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

```
ax.plot_surface(x, y, z,
               color='lightblue', alpha
               =0.5)
# Plotting a great circle (
  geodesic)
ax.plot(np.cos(theta), np.
        sin(theta), 0, color='red',
        lw=2)
plt.title("Geodesic (Great
          Circle) on S^2")
plt.show()
```

Chapter 2: Fermat's Last Theorem

Exercise 24.9 (Basic). *Prove that $x^4 + y^4 = z^4$ has no nonzero integer solutions using Fermat's method of infinite descent. Solution: Suppose (x, y, z) is a solution with $x, y, z > 0$. Then $(x^2)^2 + (y^2)^2 = (z^2)^2$, so (x^2, y^2, z^2) is a Pythagorean triple. Write $x^2 = m^2 - n^2$, $y^2 = 2mn$, $z^2 = m^2 + n^2$ with $\gcd(m, n) = 1$. Then m, n are squares, leading to a smaller solution, contradicting minimality.*

Exercise 24.10 (Basic). *Show that the elliptic curve $y^2 = x^3 + ax + b$ is nonsingular if and only if $4a^3 + 27b^2 \neq 0$. Solution: The curve is singular iff the partial derivatives vanish simultaneously: $2y = 0$, $3x^2 + a = 0$. So $y = 0$ and $x^2 = -a/3$. Substituting into the curve: $0 = x^3 + ax + b = x(-a/3) + ax + b = 2ax/3 + b$. Then $x = -3b/2a$. Plugging back gives the discriminant condition $4a^3 + 27b^2 = 0$.*

Exercise 24.11 (Intermediate). *Explain why Ribet's theorem implies that proving modularity for semistable elliptic curves implies Fermat's Last Theorem. Solution: If FLT is false, $a^n + b^n = c^n$ gives a Frey curve $y^2 = x(x - a^n)(x + b^n)$. This curve is semistable and has*

minimal discriminant not a square. Ribet proved (using Serre's conjecture) that this curve cannot be modular. Therefore, if all semistable elliptic curves are modular (Wiles's theorem), no such counterexample to FLT exists.

Python/SageMath Starter Code

```
# Chapter 2: Adding points
on an Elliptic Curve
def ec_add(P, Q, a, p):
    if P is None: return Q
    if Q is None: return P
    x1, y1 = P
    x2, y2 = Q
    if x1 == x2 and y1 == -
        y2 % p: return None
    if P != Q:
        m = ((y2 - y1) * pow
              (x2 - x1, -1, p))
              % p
    else:
        m = ((3 * x1**2 + a)
              * pow(2 * y1, -1,
                    p)) % p
    x3 = (m**2 - x1 - x2) %
        p
    y3 = (m*(x1 - x3) - y1)
        % p
    return (x3, y3)

# Example: y^2 = x^3 + x + 6
mod 11
print("Point Addition (5,2)
      + (5,2):", ec_add((5,2),
                        (5,2), 1, 11))
```

Chapter 3: Kepler Conjecture

Exercise 24.12 (Basic). *Calculate the density of the simple cubic packing (spheres at (i, j, k) for $i, j, k \in \mathbb{Z}$). Show*

it equals $\pi/6 \approx 0.5236$. Solution: Each sphere has radius $1/2$. Unit cell is a cube of side 1 containing 1 sphere (8 corners $\times 1/8$ each). Volume of sphere = $\pi/6$, volume of cell = 1. Density = $\pi/6$.

Exercise 24.13 (Intermediate). Prove that the face-centered cubic packing has density $\pi/\sqrt{18}$. (Hint: Consider the unit cell of the FCC lattice, which contains 4 spheres.) Solution: FCC unit cell is a cube of side $\sqrt{2}$ containing 4 spheres (8 corners $\times 1/8 + 6$ faces $\times 1/2$). Volume = $2\sqrt{2}$, sphere volume = $4 \times \frac{4}{3}\pi(1/2)^3 = 4\pi/3$. Density = $(4\pi/3)/(2\sqrt{2}) = \pi/\sqrt{18}$.

Exercise 24.14 (Challenge). Explain why Gauss's proof for lattice packings is easier than the general case. What additional complexity arises when the centers do not form a lattice? Solution: For lattices, the problem reduces to minimizing a quadratic form over lattice points (Voronoi cell). Non-lattice packings lack periodicity; local densities can fluctuate, requiring sophisticated decomposition and linear programming bounds (Hales's approach).

Exercise 24.15 (Basic). Maryna Viazovska proved optimal sphere packing in dimension 8 using the E_8 lattice. Describe what makes the E_8 lattice special and why it leads to optimal packing. Solution: The E_8 lattice in $\mathbb{R}^8$ is the unique even unimodular lattice of dimension 8. It achieves highly symmetric lattice packing configurations where the kissing number is 240 (maximal in 8 dimensions). Its algebraic structure matches the Cohn-Elkies linear programming bounds, which Viazovska solved using modular forms of weight 4.

Python/SageMath Starter Code

```
# Chapter 3: Simulating
# simple cubic packing
# density
import random

def
monte_carlo_cubic_density(
    trials=100000):
    inside = 0
    for _ in range(trials):
        x = random.uniform
            (0, 1)
        y = random.uniform
            (0, 1)
        z = random.uniform
            (0, 1)
        # Check distance to
        # closest grid
        # corner
        for cx in [0, 1]:
            for cy in [0,
                1]:
                for cz in
                    [0, 1]:
                    if (x-cx
                        )**2 +
                        (y-cy
                        )**2 +
                        (z-cz
                        )**2
                        <=
                        0.25:
                        inside
                            +=
                                1
                            break

    return inside / trials

print("Estimated Simple
Cubic Density:",
monte_carlo_cubic_density
())
```

Chapter 4: Bounded Gaps Between Primes

Exercise 24.16 (Basic). *Prove there are infinitely many primes using Euclid's argument. Then adapt to prove infinitely many primes of form $4k + 3$.*

Solution: If $p_1, \dots, p_k$ are all primes, $N = p_1 \cdots p_k + 1$ has a prime factor not in the list. For $4k + 3$: suppose $p_1, \dots, p_k \equiv 3 \pmod{4}$ are all such primes. $N = 4p_1 \cdots p_k - 1 \equiv 3 \pmod{4}$. N must have a prime factor $q \equiv 3 \pmod{4}$ (since product of 1 (mod 4) primes is 1 (mod 4)). q cannot be any p_i (else $q \mid 1$), contradiction.

Exercise 24.17 (Intermediate). *Show the twin prime constant $C_2 = \prod_{p \geq 3} (1 - \frac{1}{(p-1)^2}) \approx 0.66016$ is positive. Explain why this suggests infinitely many twin primes.* *Solution:* Each factor > 0 and $\prod (1 - a_p)$ converges to > 0 iff $\sum a_p$ converges. Here $a_p = 1/(p-1)^2 \sim 1/p^2$, sum converges. The Hardy-Littlewood conjecture predicts twin primes $\sim 2C_2x/\log^2 x$, so $C_2 > 0$ implies infinitely many.

Exercise 24.18 (Computational). *Compute $\pi(100)$ and $\pi(1000)$. Compare to $x/\log x$. How accurate is PNT for small x ?* *Solution:* $\pi(100) = 25$, $100/\log 100 \approx 21.7$; $\pi(1000) = 168$, $1000/\log 1000 \approx 144.8$. PNT underestimates for these x ; relative error $\sim 13\%$ and $\sim 14\%$.

Exercise 24.19 (Basic). *If the gap between consecutive primes satisfies $p_{n+1} - p_n \leq 246$ for infinitely many n , what does this imply about the Twin Prime Conjecture? What additional ingredient would be needed to prove the Twin Prime Conjecture?* *Solution:* It establishes a finite bound on prime gaps, which is the closest step toward twin primes

($p_{n+1} - p_n = 2$). To prove the Twin Prime Conjecture, one must reduce the gap size from 246 to 2. This requires extending sieve levels of distribution from $\theta = 1/2$ (Bombieri-Vinogradov) to $\theta = 1$ (Deligne/Elliott-Halberstam), currently blocked by analytic boundaries.

Python/SageMath Starter Code

```
# Chapter 4: Compute prime
# gap distribution up to N
def prime_gaps(limit=10000):
    sieve = [True] * limit
    primes = []
    for p in range(2, limit):
        if sieve[p]:
            primes.append(p)
            for i in range(p
                *p, limit, p):
                sieve[i] =
                    False
    gaps = {}
    for i in range(len(
        primes)-1):
        g = primes[i+1] -
            primes[i]
        gaps[g] = gaps.get(g
            , 0) + 1
    return gaps

print("Prime Gap Counts (up
to 10000):", sorted(
    prime_gaps().items()))
```

Chapter 5: Kervaire Invariant

Exercise 24.20 (Basic). *Show a framed 2-manifold has Kervaire invariant 1. Why trivial?* *Solution:* In dimension 2, the Arf invariant of the intersection form is always 1 for the standard fram-

ing (corresponds to the non-trivial element of $\pi_2^S \cong \mathbb{Z}/2$).

Exercise 24.21 (Intermediate). Explain Browder's theorem: why must dimension be $2^k - 2$? Solution: The Kervaire invariant one element θ_j in $\pi_{2^j+1-2}^S$ would be detected by h_j^2 in the Adams spectral sequence. By Adams's theorem, $h_j^2 \neq 0$ only for $j = 1, 2, 3, 4, 5$ (dimensions 2, 6, 14, 30, 62). For $j \geq 6$, $h_j^2 = 0$, so no such element exists.

Exercise 24.22 (Challenge). What is the Adams spectral sequence, and how is it used to compute stable homotopy groups? Solution: The Adams spectral sequence is a homological tool computing the stable homotopy groups of spheres π_*^S . It begins with Ext groups over the Steenrod algebra, $E_2^{s,t} = \text{Ext}_A^{s,t}(\mathbb{F}_p, \mathbb{F}_p)$, and uses differential differentials to converge to the stable homotopy groups.

Python/SageMath Starter Code

```
# Chapter 5: Arf invariant
# of a quadratic form
def arf_invariant(pairs):
    # pairs is list of tuple
    # (a, b) representing
    # symplectic basis
    # where q(x*e_i + y*f_i)
    # = a_i x^2 + b_i y^2 +
    # x y
    # Arf invariant is sum(
    # a_i * b_i) mod 2
    return sum(a * b for a,
                b in pairs) % 2

print("Arf Invariant of
standard torus:",
      arf_invariant([(1, 1)]))
```

Chapter 6: Mirzakhani

Exercise 24.23 (Basic). Show Euler characteristic of genus g surface is $2-2g$. Why does this imply $g \geq 2$ surfaces admit hyperbolic metrics? Solution: Polygon with $4g$ sides gives $\chi = 2 - 2g$. Gauss-Bonnet: $\int K dA = 2\pi\chi$. Hyperbolic metric has $K = -1$, so area $= -2\pi\chi = 4\pi(g-1) > 0$ iff $g \geq 2$.

Exercise 24.24 (Intermediate). Explain why WP metric is incomplete. What are boundary points? Solution: Pinching a simple closed geodesic to length 0 takes finite WP distance. Boundary points correspond to nodal Riemann surfaces (stable curves).

Exercise 24.25 (Challenge). Derive the growth rate L^{6g-6} for the number of simple closed geodesics on a surface of genus g . Why does the exponent depend only on the genus? Solution: The space of measured laminations $\mathcal{ML}(S_g)$ parameterizes closed curves asymptotically and has real dimension $6g-6$. The number of geodesics of length $\leq L$ corresponds to lattice points inside a ball of radius L in this space, whose volume scales as $L^{\dim \mathcal{ML}} = L^{6g-6}$. The exponent depends only on the genus because the dimension of the Teichmüller space is topologically determined by $\chi(S_g) = 2 - 2g$.

Python/SageMath Starter Code

```
# Chapter 6: Hyperbolic area
# of a genus g surface
import numpy as np
def hyperbolic_area(g):
    if g < 2: return "No
hyperbolic metric"
    return 4 * np.pi * (g -
1)
```

```
print("Area for genus 2
      surface:", hyperbolic_area
      (2))
```

factors, analogous to chemical decomposition into elements.

Chapter 7: Classification of Finite Simple Groups

Python/SageMath Starter Code

Exercise 24.26 (Basic). *Prove A_5 is simple. (Hint: Any nontrivial normal subgroup contains a 3-cycle.)* Solution: Conjugacy classes in A_5 : 1, (12)(34) (15), (123) (20), (12345) (12), (12354) (12). Normal subgroup is union of classes including identity. Check combinations: no proper nontrivial union gives a subgroup (Lagrange's theorem restricts orders to divisors of 60).

Exercise 24.27 (Basic). *Compute order of $PSL_2(7)$. Show it equals 168 and is simple.* Solution: $|PSL_2(q)| = \frac{1}{2}(q^3 - q) = \frac{1}{2}(343 - 7) = 168$. Simplicity of $PSL_2(q)$ for $q \geq 4$ is standard.

Exercise 24.28 (Challenge). *The Monster group has order approximately 8×10^{53} . How many digits does this number have? Express your answer in terms of logarithms.* Solution: The number of digits of $M = 8 \times 10^{53}$ is given by $\lfloor \log_{10}(M) \rfloor + 1 = \lfloor \log_{10}(8) + 53 \rfloor + 1 = \lfloor 0.903 + 53 \rfloor + 1 = 54$.

Exercise 24.29 (Basic). *Explain why the classification of finite simple groups is analogous to the periodic table of elements in chemistry. What are the "elements" in each case?* Solution: Finite simple groups serve as the algebraic "atoms" or "elements" of group theory. By the Jordan-Hölder theorem, any finite group can be decomposed uniquely into a set of simple group composition

```
# Chapter 7: Verify
simplicity of A_5
conjugacies
def check_a5_simplicity():
    # Conjugacy classes
    sizes of A_5
    classes = [1, 15, 20,
               12, 12]
    # We check if any sum
    containing 1 (the
    identity) divides 60
    n = len(classes)
    for i in range(1, 1 << n
    ):
        if i & 1: # Must
            include the
            identity class
            s = sum(classes[
                j] for j in
                range(n) if (i
                & (1 << j)))
            if s > 1 and s <
                60 and 60 % s
                == 0:
                print(f"
                Possible
                normal
                subgroup
                of order {
                s}?"")
                return False
    return True

print("A_5 has no non-
      trivial normal subgroups:"
      , check_a5_simplicity())
```

Chapter 8: Sphere Packing Python/SageMath Starter Code

Exercise 24.30 (Basic). *Show kissing number of E_8 is 240. (Hint: Count vectors of norm 2.)* Solution: E_8 vectors of norm 2: $\pm e_i \pm e_j$ for $i < j$ (112), and $(\pm 1/2)^8$ with even number of minus signs (128). Total 240.

Exercise 24.31 (Intermediate). *Prove hexagonal lattice packing in $\mathbb{R}^2$ has density $\pi/\sqrt{12}$.* Solution: Fundamental domain is rhombus with side 1, angle $\pi/3$, area $\sqrt{3}/2$. Contains one circle of radius $1/2$. Density = $\pi/4/(\sqrt{3}/2) = \pi/(2\sqrt{3}) = \pi/\sqrt{12}$.

Exercise 24.32 (Challenge). *Explain why Poisson summation is essential in Cohn-Kenyon method. How does it relate primal and dual problems?* Solution: Primal: sphere packing density. Dual: linear programming bound via functions with $f \leq 0$, $\hat{f} \leq 0$. Poisson summation $\sum_{x \in \Lambda} f(x) = \frac{1}{\det \Lambda} \sum_{\xi \in \Lambda^*} \hat{f}(\xi)$ relates the sum over the lattice to the dual lattice, enabling the bound.

Exercise 24.33 (Basic). *The theta series of a lattice Λ is $\Theta_\Lambda(q) = \sum_{x \in \Lambda} q^{\|x\|^2/2}$. Show that for E_8 , this is a modular form of weight 4 for $\Gamma_0(2)$. Why is this connection crucial for Viazovska's proof?* Solution: For the even unimodular lattice E_8 , $\Theta_{E_8}(q)$ is a modular form of weight 4 for the modular group $SL_2(\mathbb{Z})$, and hence also for the subgroup $\Gamma_0(2)$. The connection is crucial because the space of weight 4 modular forms is 1-dimensional, spanned by the Eisenstein series E_4 . This tight algebraic constraint allows Viazovska to explicitly construct her optimal "magic" functions solving the packing boundary conditions.

```
# Chapter 8: E_8 lattice
vector generation &
counting
def count_e8_norm_2():
    count = 0
    # 1. Type 1: (+-1, +-1,
    #           0, 0, 0, 0, 0, 0) and
    #           permutations
    # 8 coordinates, choose
    # 2 for non-zero: 28
    # pairs.
    # 4 choices of signs: 28
    # * 4 = 112 vectors.
    count += 28 * 4
    # 2. Type 2: (+-1/2,
    #           +-1/2, +-1/2, +-1/2,
    #           +-1/2, +-1/2, +-1/2,
    #           +-1/2)
    # with an even number of
    # minus signs: 2^7 =
    # 128 vectors.
    count += 2**7
    return count

print("Kissing Number of E_8
:", count_e8_norm_2())
```

Chapter 9: The Fundamental Lemma

Exercise 24.34 (Intermediate). *Explain why the Fundamental Lemma is called "fundamental." What breaks down without it?* Solution: The trace formula relates geometric side (orbital integrals) to spectral side (traces of Hecke operators). Endoscopy splits the geometric side into sums over endoscopic groups. The Fundamental Lemma equates orbital integrals on G with those on its endoscopic groups H . Without it, the trace formula cannot be stabilized—cannot

separate geometric from spectral terms, blocking the Langlands correspondence.

Exercise 24.35 (Intermediate). *What is the relationship between the geometric side and the spectral side of the trace formula? Why does matching orbital integrals matter? Solution: The geometric side sums orbital integrals (classifying conjugacy classes), and the spectral side sums traces of representation operators. Matching orbital integrals is what allows one to stabilize the geometric side, pairing geometric conjugacy classes on different groups and allowing transfer of spectral data.*

Exercise 24.36 (Challenge). *The Hitchin fibration plays a central role in Ngo's proof. Describe what a Higgs bundle is and why its moduli space is relevant. Solution: A Higgs bundle on a Riemann surface X is a pair (E, Φ) of a vector bundle E and a Higgs field $\Phi : E \rightarrow E \otimes \Omega_X^1$. The Hitchin fibration maps Higgs bundles to their spectral invariants (characteristic polynomials). Its moduli space fibers are abelian varieties, converting the analytic orbital integrals of the trace formula into the geometry of these Hitchin fibers.*

Exercise 24.37 (Intermediate). *Ngo's proof uses perverse sheaves on the affine Grassmannian. What is the affine Grassmannian, and why is it natural in this context? Solution: The affine Grassmannian $Gr_G = G(F((t)))/G(F[[t]])$ is a loop space variety. It is natural because local orbital integrals count lattice alignments, which translate geometrically to intersection numbers on orbits of Gr_G . Perverse sheaves allow the computation of these values using intersection cohomology.*

Python/SageMath Starter Code

```
# Chapter 9: Characters of
# finite groups (Langlands
# dual representations)
import numpy as np

def character_table_sl2():
    # Characteristic
    # polynomial roots for
    # SL_2(R)
    # Simulating simple
    # diagonal traces
    angles = np.linspace(0,
        2*np.pi, 5)
    traces = 2 * np.cos(
        angles)
    return traces

print("SL_2 conjugate traces
:", character_table_sl2())
```

Chapter 10: The Weak Goldbach Conjecture

Exercise 24.38 (Basic). *Verify 7 to 25 as sums of three primes: $7 = 2 + 2 + 3$, $9 = 2 + 2 + 5$, $11 = 2 + 2 + 7$, $13 = 3 + 3 + 7$, $15 = 3 + 5 + 7$, $17 = 3 + 3 + 11$, $19 = 3 + 3 + 13$, $21 = 3 + 5 + 13$, $23 = 3 + 3 + 17$, $25 = 3 + 3 + 19$. Solution: The listed decompositions are correct and consist only of prime numbers.*

Exercise 24.39 (Intermediate). *Why is strong Goldbach ($n = p + q$) harder than weak ($n = p + q + r$)? Solution: In weak version, fixing one prime (e.g., 3) reduces to strong version for $n - 3$, but $n - 3$ is even. More generally, 3 primes gives more degrees of freedom; circle method minor arc bounds are easier for 3 primes than 2 because the ex-*

ponential sum $S(\alpha)^3$ has better cancellation.

Exercise 24.40 (Challenge). *Explain the major/minor arc decomposition in the circle method. Solution: Partition $[0, 1]$ into intervals around a/q with $q \leq Q$ (major arcs) and the rest (minor arcs). On major arcs, $\alpha \approx a/q$, use approximation $S(\alpha) \approx \frac{\mu(q)}{\phi(q)} \frac{N}{\log N}$ etc. On minor arcs, prove $|S(\alpha)| \ll N/\log^A N$. Major arcs give main term; minor arcs give error.*

Python/SageMath Starter Code

```
# Chapter 10: Check Weak
# Goldbach for all odd
# numbers up to N
def is_prime(n):
    if n < 2: return False
    for i in range(2, int(n
        **0.5)+1):
        if n % i == 0:
            return False
    return True

def check_weak_goldbach(
    limit):
    primes = [p for p in
        range(2, limit) if
        is_prime(p)]
    for n in range(7, limit,
        2):
        found = False
        for p1 in primes:
            for p2 in primes
                :
                p3 = n - p1
                - p2
                if p3 in
                    primes:
                        found =
                            True
                        break
            if found: break
```

```
if not found: return
    False, n
return True, None

print("Weak Goldbach
    verified to 500:",
    check_weak_goldbach(500))
```

Chapter 11: The Erdős Distinct Distances Problem

Exercise 24.41 (Basic). *Show n collinear points determine at most $n - 1$ distinct distances. Why doesn't this contradict the theorem? Solution: Distances are $1, 2, \dots, n - 1$. The Guth-Katz theorem assumes general position (or at least not all collinear); the bound $cn/\log n$ applies to any set, but collinear is a degenerate case achieving the trivial lower bound $n - 1$.*

Exercise 24.42 (Intermediate). *Construct n points on a circle and count distinct distances. Compare to grid $\approx n/\sqrt{\log n}$. Solution: On circle of radius 1, distances are $2\sin(\pi k/n)$ for $k = 1, \dots, \lfloor n/2 \rfloor$. Number of distinct distances $\approx n/2$, much larger than grid's $n/\sqrt{\log n}$.*

Exercise 24.43 (Challenge). *Explain why polynomial method works better in $\mathbb{F}_q$ than $\mathbb{R}$. What difficulties arise in real case? Solution: In $\mathbb{F}_q$, a degree d polynomial vanishing at all q^2 points must be identically zero (if $d < q$). In $\mathbb{R}$, no such finiteness; need polynomial partitioning to divide space into cells and handle boundaries, plus incidence geometry tools (Szemerédi-Trotter).*

Python/SageMath Starter Code

```
# Chapter 11: Counting
distinct distances of
coordinates
def distinct_distances(
    points):
    dists = set()
    n = len(points)
    for i in range(n):
        for j in range(i+1,
            n):
            dx = points[i]
                [0] - points[
                j][0]
            dy = points[i]
                [1] - points[
                j][1]
            dists.add(dx**2
                + dy**2)
    return len(dists)

# Example: 3x3 grid points
grid = [(x, y) for x in
    range(3) for y in range(3)
    ]
print("Distinct Distances in
    3x3 grid:",
    distinct_distances(grid))
```

Chapter 12: The Geometric Langlands Program

Exercise 24.44 (Basic). *What is the Langlands dual group ${}^L G$ of $G = SL_n$? Of $G = SO_n$? Of $G = Sp_{2n}$? Solution: ${}^L SL_n = PGL_n$, ${}^L SO_{2n+1} = Sp_{2n}$, ${}^L Sp_{2n} = SO_{2n+1}$, ${}^L SO_{2n} = SO_{2n}$.*

Exercise 24.45 (Intermediate). *Explain the Hecke eigensheaf condition in the geometric Langlands program. How does it generalize the eigenform condition in the classical theory? Solution:*

In classical theory, we seek automorphic forms f which are eigenvectors of Hecke operators: $T_x(f) = \lambda_x f$. In the geometric theory, functions are replaced by sheaves $\mathcal{F}$, and operators by Hecke functors H_x . The eigensheaf condition is $H_x(\mathcal{F}) \cong V_x \otimes \mathcal{F}$, where V_x is a vector space (acting as the "eigenvalue").

Exercise 24.46 (Challenge). *What is the geometric Satake equivalence? How does it relate the representation theory of ${}^L G$ to the geometry of the affine Grassmannian of G ? Solution: The geometric Satake equivalence establishes an equivalence of tensor categories between the category of $G(O)$ -equivariant perverse sheaves on the affine Grassmannian Gr_G and the category of finite-dimensional representations of the dual group ${}^L G$, translating algebraic representations into geometric sheaf configurations.*

Exercise 24.47 (Intermediate). *Why are higher categories (∞ -categories) the natural language for the geometric Langlands program? What goes wrong with ordinary categories? Solution: The geometric Langlands correspondence is formulated between derived categories of sheaves. Ordinary categories fail to capture the homotopy-theoretic and derived intersections (like Tor and Ext) which are essential when working on derived moduli stacks. ∞ -categories preserve these higher homotopical structures.*

Python/SageMath Starter Code

```
# Chapter 12: Dual Cartan
matrix representation of
GL_n / SL_n
def dual_cartan_matrix_sl3()
    :
```

```

# Cartan matrix for A2 (
  SL_3)
C = np.array([[2, -1],
              [-1, 2]])
# Dual Cartan matrix is
  transpose (same for
  A_n)
return C.T

print("Dual Cartan matrix
for SL_3:\n",
dual_cartan_matrix_sl3())

```

into a quasi-Fuchsian surface. Solution: The exponential mixing of the geodesic flow on hyperbolic 3-manifolds implies that closed geodesics are uniformly distributed in the unit tangent bundle. Kahn and Markovic used this to construct pairs of pants whose cuff lengths are close to L and twists close to 0, showing that these boundary cuffs can be matched and glued with high precision to construct a closed, embedded quasi-Fuchsian surface.

Chapter 13: The Virtual Haken Conjecture

Exercise 24.48 (Basic). Define incompressible surface. Why is a sphere incompressible in S^3 but compressible in $S^2 \times S^1$? Solution: A closed surface $S \subset M$ is incompressible if the induced map $\pi_1(S) \rightarrow \pi_1(M)$ is injective. For a sphere S^2 , $\pi_1(S^2) = 0$, so injectivity is trivial. However, by convention, an embedded S^2 in a 3-manifold is called incompressible if it does not bound a 3-ball. In S^3 , any embedded S^2 bounds a 3-ball (Schönflies theorem), so it is compressible. In $S^2 \times S^1$, the sphere $S^2 \times \{p\}$ does not bound a 3-ball, so it is incompressible.

Exercise 24.49 (Intermediate). Explain why a manifold that fibers over the circle is Haken. Solution: A manifold M that fibers over S^1 has a closed surface fiber $F \subset M$. The inclusion induces a short exact sequence of fundamental groups $1 \rightarrow \pi_1(F) \rightarrow \pi_1(M) \rightarrow \mathbb{Z} \rightarrow 1$, which proves $\pi_1(F) \rightarrow \pi_1(M)$ is injective. Since F is two-sided and incompressible, M is Haken.

Exercise 24.50 (Advanced). Sketch how the exponential mixing of the geodesic flow is used to glue good pants

Python/SageMath Starter Code

```

# Chapter 13: Solving the
  gluing equation for
  boundary cuffs
# We check if two boundaries
  match within tolerance
  epsilon
def can_glue_cuffs(l1, l2,
  epsilon=0.01):
    return abs(l1 - l2) <
      epsilon

print("Can glue cuffs of
length 2.005 and 2.001?",
can_glue_cuffs(2.005,
2.001))

```

Chapter 14: The Green–Tao Theorem

Exercise 24.51 (Basic). Find a 5-term arithmetic progression of primes. Solution: 5, 11, 17, 23, 29 (difference 6). Also 199, 409, 619, 829, 1039 (difference 210).

Exercise 24.52 (Intermediate). Explain why primes having density zero makes Szemerédi’s theorem inapplicable directly. What does “positive rel-

ative density" mean? Solution: Sze-merédi requires $|A \cap [1, N]| \geq \delta N$. For primes, $\pi(N) \sim N/\log N = o(N)$, so $\delta = 0$. Positive relative density means $|A \cap \mathcal{P} \cap [1, N]|/\pi(N) \geq \delta > 0$ for infinitely many N .

Exercise 24.53 (Challenge). The Bombieri-Vinogradov theorem states primes are well-distributed in AP on average. Explain why this is essential for Green-Tao. Solution: Green-Tao needs to control $\sum_{n \leq N} \Lambda(n)e(\alpha n)$ for α near rationals a/q with small q (major arcs). Bombieri-Vinogradov gives level of distribution $1/2$, controlling sums over $q \leq N^{1/2-\epsilon}$, which suffices to build the pseudorandom majorant ν .

Exercise 24.54 (Basic). What is the shortest arithmetic progression of primes of length 10? (Hint: Look up the OEIS sequence A031626.) Solution: The shortest arithmetic progression of 10 primes is 199, 409, 619, 829, 1039, 1249, 1459, 1669, 1879, 2089 (common difference $d = 210$).

Python/SageMath Starter Code

```
# Chapter 14: Search for k-
term arithmetic
progressions in primes
def find_ap_in_primes(k,
    limit=1000):
    primes = [p for p in
        range(2, limit) if
        is_prime(p)]
    prime_set = set(primes)
    for i in range(len(
        primes)):
        for j in range(i+1,
            len(primes)):
            diff = primes[j]
                - primes[i]
            ap = [primes[i]
                + m * diff for
```

```
        m in range(k)
    ]
    if all(val in
        prime_set for
        val in ap):
        return ap
    return None

print("5-term AP in primes:"
    , find_ap_in_primes(5))
```

Chapter 15: The AKS Primality Test

Exercise 24.55 (Basic). Verify $(X + a)^5 \equiv X^5 + a \pmod{5}$. Check binomial coefficients $\binom{5}{k}$. Solution: $\binom{5}{1} = 5$, $\binom{5}{2} = 10$, $\binom{5}{3} = 10$, $\binom{5}{4} = 5$, all divisible by 5. By binomial theorem, $(X + a)^5 = X^5 + 5X^4a + 10X^3a^2 + 10X^2a^3 + 5Xa^4 + a^5 \equiv X^5 + a^5 \equiv X^5 + a \pmod{5}$ (by Fermat).

Exercise 24.56 (Basic). Show $561 = 3 \cdot 11 \cdot 17$ is a Carmichael number: $a^{560} \equiv 1 \pmod{561}$ for all a coprime to 561. Solution: By Korselt's criterion: n is Carmichael iff n is squarefree and for all $p \mid n$, $p-1 \mid n-1$. $561 = 3 \cdot 11 \cdot 17$. $3-1 = 2$, $11-1 = 10$, $17-1 = 16$ all divide 560. Hence 561 is Carmichael.

Exercise 24.57 (Intermediate). Explain why the polynomial congruence $(X + a)^n \equiv X^n + a \pmod{n}$ is a valid primality test. Why doesn't this work directly as a fast algorithm? Solution: Valid because for prime p , $\binom{p}{k} \equiv 0 \pmod{p}$. For composite n , some $\binom{n}{k}$ not divisible by n . Direct test requires expanding $(X + a)^n$ with $O(n)$ terms—exponential in $\log n$. AKS reduces to checking modulo $X^r - 1$ with $r = O(\log^5 n)$, giving $\tilde{O}(\log^6 n)$ operations.

Exercise 24.58 (Basic). *The AKS algorithm runs in $\tilde{O}((\log n)^6)$. If n has d digits, how does this compare to trial division? What is the practical implication? Solution: Trial division has time complexity $O(\sqrt{n}) = O(10^{d/2})$, which is exponential in the number of digits d . The AKS algorithm runs in polynomial time $\tilde{O}(d^6)$. The practical implication is that prime verification for cryptographically large integers (hundreds of digits) can be performed in minutes rather than taking longer than the age of the universe.*

Python/SageMath Starter Code

```
# Chapter 15: AKS polynomial
# congruence verification
# for n
def verify_aks_poly(n, a):
    # We verify  $(X + a)^n \equiv X^n + a \pmod{n, X^n - 1}$ 
    # Simple check for  $n=5$ ,  $a=2$ 
    #  $(X+2)^5 = X^5 + 10X^4 + 40X^3 + 80X^2 + 80X + 32$ 
    #  $\pmod{5} = X^5 + 2 \equiv X^5 + a \pmod{5}$ .
    coeff = [0] * (n + 1)
    coeff[0] = a**n
    # simplified
    # representation...
    return (a**n % n) == a

print("AKS simple trace n=5: ", verify_aks_poly(5, 2))
```

Chapter 16: The Willmore and Lawson Conjectures

Exercise 24.59 (Basic). *Compute the Willmore energy of the Clifford torus in $\mathbb{R}^3$ (stereographically projected from S^3). Solution: Under stereographic projection, the Clifford torus $T_{Cl} \subset S^3$ maps to a torus in $\mathbb{R}^3$. Since the Willmore energy is conformally invariant under conformal maps of S^3 to $\mathbb{R}^3 \cup \{\infty\}$, its Willmore energy is equal to its area in S^3 , which is $2\pi^2$.*

Exercise 24.60 (Intermediate). *Show that the Willmore energy is conformally invariant in $\mathbb{R}^3$. Why does this fail in higher dimensions? Solution: For a torus in $\mathbb{R}^3$, $\mathcal{W}(\Sigma) = \int_{\Sigma} (H^2 - K) dA$. The trace-free part of the second fundamental form is $H^2 - K = \frac{1}{4}(k_1 - k_2)^2 \geq 0$, which scales inversely to the area element under conformal transformations, making $(H^2 - K)dA$ invariant. In higher dimensions, conformal invariance requires additional Weyl and traceless terms not present in the simple H^2 integral.*

Exercise 24.61 (Advanced). *Explain how Almgren-Pitts min-max produces a minimal surface in S^3 from a given sweepout of a torus. Solution: A sweepout of a torus is a family of closed surfaces $\{\Sigma_t\}_{t \in [0,1]}$ interpolating between degenerate configurations. By minimizing the maximum area over all homotopic sweepouts, Almgren-Pitts theory guarantees the existence of a closed minimal surface in S^3 realizing this minimax value.*

Python/SageMath Starter Code

```
# Chapter 16: Numerical
    integration of Willmore
    energy of a torus
# R: major radius, r: minor
    radius
def willmore_energy_torus(R,
    r):
    # Willmore energy is \
        int H^2 dA
    # For a torus, it
        simplifies to 2 * pi^2
        * R / sqrt(R^2 - r^2)
    import math
    return 2 * (math.pi**2)
        * R / math.sqrt(R**2 -
            r**2)

print("Willmore energy of R
    =2, r=1 torus:",
    willmore_energy_torus(2,
    1))
```

Chapter 17: Catalan's Conjecture

Exercise 24.62 (Basic). Verify $3^2 - 2^3 = 1$. Show $x^2 - y^2 = 1$ has no solutions with $x, y > 1$. Solution: $3^2 - 2^3 = 9 - 8 = 1$. $x^2 - y^2 = (x - y)(x + y) = 1$. In positive integers, $x - y = x + y = 1 \Rightarrow x = 1, y = 0$ (not valid).

Exercise 24.63 (Computational). Compute $2^{1092} \bmod 1093^2$ and verify it equals 1. Why only two known Wieferich primes? Solution: 1093 is Wieferich: $2^{1092} \equiv 1 \pmod{1093^2}$. Heuristically, p is Wieferich with probability $\approx 1/p$, so expected count $\sum 1/p \sim \log \log x$. Very few known.

Exercise 24.64 (Intermediate). Prove if $x^a - y^b = 1$ has a solution, then $\gcd(x, y) = 1$. One even, one odd? Solution: If $d \mid x$ and $d \mid y$, then $d \mid (x^a - y^b) = 1$, so $d = 1$. If both

odd, $x^a - y^b$ is even, but equals 1. So one even, one odd.

Exercise 24.65 (Computational). Search for $x^a - y^b = c$ for $c = 1, 2, 3, 4, 5$ with $x, y \leq 100$, $a, b \leq 10$. Observe patterns. Solution: $c = 1$: only $(3, 2, 2, 3)$. $c = 2$: $(3, 5, 2, 3)$ ($3^3 - 5^2 = 27 - 25 = 2$), $(2, 2, 2, 1)$ (trivial). $c = 3$: $(2, 1, 2, 1)$ etc. Few solutions; exponential Diophantine equations have at most finitely many solutions (Mihăilescu).

Exercise 24.66 (Challenge). In $\mathbb{Q}(\zeta_3)$, factor $y^3 + 1 = (y + 1)(y + \zeta_3)(y + \zeta_3^2)$. If y even, why are factors coprime? If product is a cube, why must each factor be a cube times a unit? Solution: If y even, $y + 1, y + \zeta_3, y + \zeta_3^2$ have norms $y^3 + 1$ etc. Any common divisor divides their differences, e.g., $(y + \zeta_3) - (y + 1) = \zeta_3 - 1$. The norm of $\zeta_3 - 1$ is 3. So common divisor is above 3. But y even $\Rightarrow y^3 + 1 \equiv 1 \pmod{3}$, so 3 doesn't divide. Hence coprime. In Dedekind domain, if product of coprime elements is a k -th power, each is a k -th power times a unit.

Python/SageMath Starter Code

```
# Chapter 17: Wieferich
    prime search up to limit
def find_wieferich_primes(
    limit):
    wieferich = []
    # simple modular
        exponentiation checks
    for p in range(2, limit)
        :
        if is_prime(p):
            if pow(2, p-1, p
                *p) == 1:
                wieferich.
                    append(p)
    return wieferich
```

```
print("Wieferich primes up
to 3500:",
find_wieferich_primes
(3500))
```

Chapter 18: Modern Combinatorics Breakthroughs

Exercise 24.67 (Basic). *Show that a cap set in $\mathbb{F}_3^n$ has no three points on a line. Compute the maximum cap set size for $n = 2$. Solution: A line in $\mathbb{F}_3^n$ consists of points $\{x, x + v, x + 2v\}$. A cap set has no 3-term AP, which is equivalent to containing no line. For $n = 2$, the maximum cap set size is 4 (e.g. $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$).*

Exercise 24.68 (Intermediate). *Explain how the slice rank of a tensor generalizes the rank of a matrix. Why does slice rank bound cap sets? Solution: Slice rank of $T(x, y, z)$ is the minimum number of terms of the form $f(x)g(y, z)$, $f(y)g(x, z)$, or $f(z)g(x, y)$. A cap set A corresponds to a diagonal restricted indicator tensor $T(x, y, z) = \delta(x + y + z)$, and diagonal tensors must have high slice rank, which bounds $|A|$.*

Exercise 24.69 (Challenge). *Sketch Huang's proof of the sensitivity conjecture using the signed adjacency matrix of the hypercube. Solution: Construct a signed adjacency matrix A for Q_d such that $A^2 = dI$. The eigenvalues are $\pm\sqrt{d}$. By Cauchy's interlacing theorem, any induced subgraph on $2^{d-1} + 1$ vertices has an eigenvalue of at least $\sqrt{d}$, forcing the maximum degree of the subgraph to be at least $\sqrt{d}$.*

Python/SageMath Starter Code

```
# Chapter 18: Generating and
# verifying cap sets in  $F_3^n$ 

def is_cap_set(points):
    n = len(points)
    for i in range(n):
        for j in range(i+1,
            n):
            for k in range(j
                +1, n):
                # Check if
                points[i]
                + points[j]
                + points
                [k] == 0
                mod 3
            if all((
                points[i][
                    m] +
                points[j][
                    m] +
                points[k][
                    m]) % 3 ==
                    0
                    for m
                        in
                            range
                                (
                                    len
                                        (
                                            points
                                                [i
                                                    ]))
                    ):
                return
                    False

    return True

pts = [(0,0), (0,1), (1,0),
(1,1)]
print("Is {(0,0), (0,1),
(1,0), (1,1)} a cap set?",
is_cap_set(pts))
```

Chapter 19: Perfectoid Spaces and p-adic Geometry

Exercise 24.70 (Basic). Show that $\mathbb{C}_p$ is perfectoid. Why is $\mathbb{Q}_p$ not perfectoid? Solution: $\mathbb{C}_p$ is a complete, non-archimedean field of characteristic 0, residues characteristic p , and Frobenius is surjective modulo p (as it contains all p -power roots). $\mathbb{Q}_p$ has discrete valuation, so its valuation ring modulo p is $\mathbb{F}_p$, which is not perfectoid because p cannot be divided by p in its valuation ring.

Exercise 24.71 (Intermediate). Explain the tilt equivalence: how does $K^\flat = \varprojlim_{x \mapsto x^p} K$ produce a field of characteristic p from a field of characteristic 0? Solution: Elements of $K^\flat$ are sequences $(x^{(n)})$ in K such that $(x^{(n+1)})^p = x^{(n)}$. Multiplication is pointwise, and addition is defined as $(x_n) + (y_n) = (z_n)$ where $z_n = \lim_{m \rightarrow \infty} (x_{n+m} + y_{n+m})^{p^m}$. The resulting field $K^\flat$ has characteristic p and is perfectoid.

Exercise 24.72 (Challenge). Explain how the pro-étale topology solves the problem of doing p-adic Hodge theory in families. Why does the usual étale topology fail? Solution: The usual étale topology does not satisfy cohomology vanishing on rigid-analytic varieties (the structure sheaf's cohomology can be huge). The pro-étale topology allows infinite covers, which locally behave like perfectoid spaces. This simplifies the sheaf theory and enables comparison isomorphisms.

Python/SageMath Starter Code

```
# Chapter 19: Basic p-adic
valuation and distance
calculator

def p_adic_valuation(n, p):
    if n == 0: return float(
        'inf')
    val = 0
    while n % p == 0:
        val += 1
        n //= p
    return val

print("5-adic valuation of
150:", p_adic_valuation
(150, 5))
```

Chapter 20: The Sato-Tate Conjecture

Exercise 24.73 (Basic). For $E: y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$), compute the first 10 values of a_p and explain why they do not follow the Sato-Tate distribution. Solution: For $p \equiv 3 \pmod{4}$, $a_p = 0$. Since half of the primes have $a_p = 0$, the distribution has a Dirac delta mass of weight $1/2$ at 0. This contradicts the continuous semi-circle distribution of Sato-Tate.

Exercise 24.74 (Intermediate). Show that the Sato-Tate measure $\frac{2}{\pi} \sqrt{1-t^2} dt$ is the eigenvalue distribution of a random matrix in $SU(2)$ with Haar measure. Solution: A matrix in $SU(2)$ has eigenvalues $e^{\pm i\theta}$ and trace $2 \cos \theta$. Under Haar measure, the density is $\frac{2}{\pi} \sin^2 \theta d\theta$. Substituting $t = \cos \theta$ gives $dt = -\sin \theta d\theta$ and $\sin^2 \theta = 1 - t^2$, which yields the Sato-Tate measure.

Exercise 24.75 (Challenge). Explain why the stable trace formula is needed instead of the ordinary trace formula to

prove Sato-Tate. Solution: The ordinary Selberg trace formula relates conjugacy classes to representation traces. However, to relate representations on different groups, one must group geometrically conjugate classes (stable conjugacy), which requires the stable trace formula.

Python/SageMath Starter Code

```
# Chapter 20: Sato-Tate semi
# circle distribution
# sampler
import numpy as np

def sample_sato_tate(n
=10000):
    # Generates traces 2*cos
    # (theta) for theta in
    # [0, pi] with sin^2
    # distribution
    thetas = np.arccos(2 *
        np.random.beta(1.5,
        1.5, n) - 1)
    return 2 * np.cos(thetas
    )

print("Sampled Sato-Tate
moments (ideal is 0, 1, 0,
2):")
samples = sample_sato_tate()
for k in range(1, 5):
    print(f"Moment {k}: {np.
        mean(samples**k):.4f}")
    )
```

Chapter 21: The Erdős Discrepancy Problem

Exercise 24.76 (Basic). Show that the sequence $x_n = 1$ for all n has discrepancy $D = \infty$. Solution: For $x_n = 1$, the sum over any homogeneous arith-

metic progression is $\sum_{j=1}^k x_{jd} = k$. As $k \rightarrow \infty$, the sum is unbounded, so the discrepancy is infinite.

Exercise 24.77 (Basic). Show that any periodic sequence $x_n \in \{-1, +1\}$ has infinite discrepancy. Solution: If x_n has period p , then for the step $d = p$, $x_{jd} = x_{jp} = x_p$ is constant. The sum is kx_p , which grows linearly with k , giving infinite discrepancy.

Exercise 24.78 (Intermediate). Let x_n be completely multiplicative with $x_p = 1$ for all primes $p \equiv 1 \pmod{4}$ and $x_p = -1$ for $p \equiv 3 \pmod{4}$. Show that x_n has infinite discrepancy. Solution: Since the sequence is completely multiplicative, it behaves like a Dirichlet character. By Gowers' norm or analytic number theory bounds, any multiplicative sequence in $\{-1, 1\}$ must correlate with a Dirichlet character. Tao's proof showing that all multiplicative sequences have infinite discrepancy applies.

Exercise 24.79 (Challenge). Explain the entropy decrement argument in your own words: how does averaging over large primes reduce the entropy of the sequence? Solution: The entropy decrement argument shows that if a sequence has bounded discrepancy, it must possess strong local correlations. When these correlations are combined via the multiplicative structure over many primes, it forces a step-by-step reduction in the sequence's information entropy. Since entropy is non-negative, this reduction can only happen finitely many times, showing that discrepancy must be unbounded.

Python/SageMath Starter Code

```
# Chapter 21: Compute
# discrepancy of a sequence
# along progressions
```

```

def discrepancy(seq):
    n = len(seq)
    max_disc = 0
    for d in range(1, n + 1):
        :
        for k in range(1, n
            // d + 1):
            val = abs(sum(
                seq[i * d - 1]
                for i in
                range(1, k +
                1)))
            max_disc = max(
                max_disc, val)
    return max_disc

# Example: Random +-1
# sequence of length 100
import random
seq = [random.choice([-1,
    1]) for _ in range(100)]
print("Discrepancy of random
    sequence of length 100:",
    discrepancy(seq))

```

Chapter 22: The Kadison–Singer Problem

Exercise 24.80 (Basic). Show that the union of two frames in $\mathbb{C}^d$ is a frame. If A_1, B_1 and A_2, B_2 are the frame bounds for the two frames, what are the bounds for the union? Solution: Let $\Phi_1 = \{\phi_i\}_{i \in I_1}$ and $\Phi_2 = \{\psi_j\}_{j \in I_2}$. For any x , we have $A_1\|x\|^2 \leq \sum_{i \in I_1} |\langle x, \phi_i \rangle|^2 \leq B_1\|x\|^2$ and $A_2\|x\|^2 \leq \sum_{j \in I_2} |\langle x, \psi_j \rangle|^2 \leq B_2\|x\|^2$. Summing these gives $(A_1 + A_2)\|x\|^2 \leq \sum_{v \in \Phi_1 \cup \Phi_2} |\langle x, v \rangle|^2 \leq (B_1 + B_2)\|x\|^2$, which is the frame condition with bounds $A_1 + A_2$ and $B_1 + B_2$.

Exercise 24.81 (Intermediate). Explain the relation between the Kadison–Singer problem and the construction

of Ramanujan regular graphs. Solution: Marcus, Spielman, and Srivastava proved that any set of vectors sum to identity can be paved. In graph theory, the characteristic polynomial of a 2-lifting of a d -regular graph can be represented as a mixed characteristic polynomial of rank-one updates. The resolution of the Paving Conjecture guarantees that one can partition the edge vector system to find a 2-lifting whose eigenvalues are bounded by $2\sqrt{d-1}$, proving the existence of bipartite Ramanujan graphs of all degrees.

Exercise 24.82 (Challenge). Prove that any orthogonal projection P on $\mathbb{C}^n$ with diagonal entries $P_{ii} \leq \epsilon$ can be partitioned into two parts whose submatrices have norm bounded away from 1. Solution: By the paving reformulation, an orthogonal projection P with small diagonal $P_{ii} \leq \epsilon$ corresponds to a set of vectors $v_1, \dots, v_n$ whose covariance is P and norms are $\|v_i\|^2 \leq \epsilon$. The Marcus–Spielman–Srivastava theorem guarantees that we can partition the index set into two parts S_1, S_2 such that the operator norms of the submatrices are bounded by $(1 + \sqrt{\epsilon})^2/2$. For small ϵ , this is strictly less than 1, proving the projection can be paved.

Python/SageMath Starter Code

```

# Chapter 22: Expected
# characteristic polynomial
# of rank-one sum
import numpy as np

def
    mixed_characteristic_poly(
        vectors):
    # Computes expected
    # characteristic

```

```

    polynomial under rank-
    one updates
d = len(vectors[0])
# For independent random
# projections, the
# expected polynomial is
# computed
# analytically. This
# function simulates
# random projections sum
.
import scipy.linalg as
    la
eigvals = []
for _ in range(100):
    # random sign
    # partition
    signs = np.random.
        choice([-1, 1],
            len(vectors))
    matrix = sum(signs[i]
        * np.outer(
            vectors[i],
            vectors[i]) for i
            in range(len(
                vectors)))
    eigvals.append(la.
        eigvalsh(matrix))
return np.mean(eigvals,
    axis=0)

# 3 orthogonal vectors in R
# 3
vecs = [np.array([1.0, 0.0,
    0.0]), np.array([0.0, 1.0,
    0.0]), np.array([0.0,
    0.0, 1.0])]
print("Simulated expected
    eigenvalues:",
    mixed_characteristic_poly(
        vecs))

```

Chapter 23: Break-throughs in Random Matrix Theory

Exercise 24.83 (Basic). Verify Wigner's Semicircle Law for a 2×2 real symmetric matrix. Why does the semicircle shape not appear? Solution: For a 2×2 symmetric matrix with entries $H_{11} = a, H_{22} = b, H_{12} = c$, the eigenvalues are $\lambda_{\pm} = \frac{a+b \pm \sqrt{(a-b)^2 + 4c^2}}{2}$. The global semicircle law is an asymptotic limit theorem as $N \rightarrow \infty$; for finite N (especially $N = 2$), the joint eigenvalue density is highly non-semicircular and reflects the entry distribution rather than the universal $N \rightarrow \infty$ density.

Exercise 24.84 (Intermediate). Explain the Stieltjes transform and how it is used to relate the resolvent of a matrix to its eigenvalue distribution. Solution: The Stieltjes transform of a probability distribution μ is $m(z) = \int \frac{d\mu(x)}{x-z}$. For a random matrix with empirical spectral distribution $\mu_N = \frac{1}{N} \sum \delta_{\lambda_i}$, the Stieltjes transform is $m_N(z) = \frac{1}{N} \text{Tr}(H - zI)^{-1}$, which is the trace of the resolvent operator. The inversion formula $\rho(x) = \lim_{\eta \rightarrow 0^+} \frac{1}{\pi} \text{Im } m(x + i\eta)$ recovers the density $\rho(x)$ from the Stieltjes transform, enabling analytic local analysis by studying the trace of the resolvent.

Exercise 24.85 (Challenge). Show that the spacing between adjacent eigenvalues of a GUE matrix exhibits eigenvalue repulsion. Why is this different from a Poisson point process? Solution: The joint probability density of eigenvalues for GUE contains the Vandermonde determinant term $\prod_{i < j} (\lambda_i - \lambda_j)^2$. If two eigenvalues are close, this term goes to 0 quadratically, making the probability of small spacing extremely small. This is eigenvalue repulsion. In a Poisson

process, points are independent, so the probability of two points being close does not vanish, having a non-zero density at zero spacing (no repulsion).

Python/SageMath Starter Code

```
# Chapter 23: Wigner
# Semicircle Law histogram
# simulation
import numpy as np

def
    simulate_wigner_semicircle
    (N=500, trials=10):
    eigvals = []
    for _ in range(trials):
        # Generate Gaussian
        # Orthogonal
        # Ensemble (GOE)
        # matrix
        A = np.random.normal
            (0, 1, (N, N))
        H = (A + A.T) / (2 *
            np.sqrt(N))
        eigvals.extend(np.
            linalg.eigvalsh(H)
        )
    return eigvals

# Print simple statistics of
# eigenvalues (range should
# be close to [-2, 2])
evs =
    simulate_wigner_semicircle
    (N=200, trials=5)
print("Min eigenvalue:", min
    (evs), "Max eigenvalue:",
    max(evs))
```

Chapter 24: The Kakeya Conjecture

Exercise 24.86 (Basic). Show that a Kakeya set in $\mathbb{R}^2$ containing segments

in all directions can be constructed with measure 0. Solution: Besicovitch's construction works by dividing a triangle of height 1 and base 2 into 2^m thin triangles, translating them horizontally so they overlap heavily, and then gluing them back. By increasing m and arranging the overlaps, the area of the union can be made arbitrarily small ($\leq \epsilon$) while still containing unit line segments in all directions. The intersection of these sets for $\epsilon = 1/k$ as $k \rightarrow \infty$ yields a Kakeya set of measure zero.

Exercise 24.87 (Intermediate). Write down the details of Zeev Dvir's finite field Kakeya set proof. Show why the degree of the vanishing polynomial must be less than q . Solution: If $K \subset \mathbb{F}_q^n$ is Kakeya and $|K| < \binom{n+q-1}{n}$, there exists a non-zero polynomial $P(X_1, \dots, X_n)$ of degree $d < q$ vanishing on K . For any direction $y \in \mathbb{F}_q^n \setminus \{0\}$, there exists x such that $\{x + ty : t \in \mathbb{F}_q\} \subset K$. The polynomial $g(t) = P(x + ty)$ vanishes on $t \in \mathbb{F}_q$. Since $\deg(g) \leq d < q$, and it has q roots, it must be the zero polynomial. Thus the coefficient of t^d , which is the homogeneous part $P_d(y)$, must vanish. Since this holds for all y , the non-zero polynomial P_d vanishes on all vectors, a contradiction for $d < q$.

Exercise 24.88 (Challenge). Explain why Dvir's algebraic proof fails in the setting of real numbers $\mathbb{R}^n$. What is the main difference between lines in $\mathbb{F}_q^n$ and $\mathbb{R}^n$? Solution: In finite fields, a line segment is a collection of q points, which is the same as the size of the field. A polynomial of degree $< q$ vanishing on q points must vanish on the whole line. In $\mathbb{R}^n$, a line segment is a bounded interval (like $[0, 1]$), whereas the field $\mathbb{R}$ is infinite. A polynomial of degree d can easily vanish on $[0, 1]$ (meaning it is the zero polynomial), but if we restrict

the degree, there is no finite boundary bound like q . Real lines do not have the discrete finiteness that links polynomial degree to field size, requiring geometric partitioning instead.

Python/SageMath Starter Code

```
# Chapter 24: Verifying Dvir
's finite field Kakeya set
size bounds
def minimal_kakeya_bound(n,
q):
    # Dvir's theorem states
     $|K| \geq (1/n!) * q^n$  (
```

```
asymptotically)
# The exact binomial
bound is  $\text{binom}(q + n - 1, n)$ 
import math
exact_bound = math.comb(
    q + n - 1, n)
asymptotic_bound = (q**n
) / math.factorial(n)
return exact_bound,
asymptotic_bound

print("Bounds for n=3, q=11
(exact, asymptotic):",
minimal_kakeya_bound(3,
11))
```